Science Monk · Presents

NEET Physics

Chapter-wise Previous Year Questions

28 NCERT Chapters · 2016 – 2026 · 515 Questions

The important thing
is not to stop questioning

Albert Einstein

every question retyped *every diagram redrawn*



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"Appa Deepo Bhava"
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About this book. The 515 questions of NEET (UG) Physics 2016–2026 are arranged by the 28 chapters of the 2026-27 NCERT textbooks (Class XI, then Class XII). Inside a chapter the questions are grouped by topic, in the order the topics appear in the textbook, newest year first; each question shows the year and question number of the original paper (Q 136–180 in 2017 and 2020) and keeps its original option labels. Every question carries its NCERT tag (Direct or Applied, with the page and a link to the official chapter PDF). Each chapter ends with its answer key (official NTA final keys for 2021, 2022, 2023, 2025 and 2026; solved by the editor for the other years) and a chapter summary: questions per year, topic-wise counts, direct and applied split and what to focus on. Questions on deleted portions are collected in their own section. The book ends with an overall summary and a preparation plan.

Prepared by Sumeet Birajadar for Science Monk online classes.

Class XI · Chapter 1

Units and Measurements

21 questions · 2 topics · NEET 2016–2026

Dimensional analysis 9 Qs

1. NEET 2026 · Q37 Consider that σ_s, k_B, b represents Stefan–Boltzmann constant, Boltzmann constant and Wien's displacement law constant, respectively. The dimension of $\sigma_s k_B^{-1} b$ is

APPLIED Class XI Part 1 · Ch 1 · **Page 7** Open PDF ↗

(1) $[L^{-1}T^{-1}K^{-2}]$ **(2)** $[L^{-1}K^{-2}]$
(3) $[L^{-1}T^{-1}K^{-3}]$ **(4)** $[L^{-1}T^{-1}K^{-4}]$

2. NEET 2025 · Q24 A balloon is made of a material of surface tension S and its inflation outlet (from where gas is filled in it) has small area A. It is filled with a gas of density ρ and takes a spherical shape of radius R. When the gas is allowed to flow freely out of it, its radius r changes from R to 0 (zero) in time T. If the speed $v(r)$ of gas coming out of the balloon depends on r as r^a and $T \propto S^\alpha A^\beta \rho^\gamma R^\delta$ then

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(1) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -1, \gamma = +1, \delta = \frac{3}{2}$
(2) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = -\frac{1}{2}, \delta = \frac{5}{2}$
(3) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$
(4) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -\frac{1}{2}, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$

3. NEET 2024 · Q23 The quantities which have the same dimensions as those of solid angle are:

DIRECT Class XI Part 1 · Ch 1 · **Page 7** · dimensionless quantities Open PDF ↗

(1) angular speed and stress
(2) strain and angle
(3) stress and angle
(4) strain and arc

4. NEET 2024 · Q50 A force defined by $F = \alpha t^2 + \beta t$ acts on a particle at a given time t. The factor which is dimensionless, if α and β are constants, is:

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(1) $\alpha\beta/t$ **(2)** $\beta t/\alpha$ **(3)** $\alpha t/\beta$ **(4)** $\alpha\beta t$

5. NEET 2022 · Q28 Plane angle and solid angle have:

DIRECT Class XI Part 1 · Ch 1 · **Page 2** · both are dimensionless (radian, steradian) Open PDF ↗

(1) Both units and dimensions
(2) Units but no dimensions
(3) Dimensions but no units
(4) No units and no dimensions

6. NEET 2022 · Q39 Match List-I with List-II:

List-I		*List-II*	
(a)	Gravitational constant (G)	(i)	$[L^2T^{-2}]$
(b)	Gravitational potential energy	(ii)	$[M^{-1}L^3T^{-2}]$
(c)	Gravitational potential	(iii)	$[LT^{-2}]$
(d)	Gravitational intensity	(iv)	$[ML^2T^{-2}]$

Choose the correct answer from the options given below:

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(1) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
(2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(3) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
(4) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

7. NEET 2021 · Q9 If force [F], acceleration [A] and time [T] are chosen as the fundamental physical quantities. Find the dimensions of energy.

APPLIED Class XI Part 1 · Ch 1 · **Page 8** Open PDF ↗

(1) [F][A][T] **(2)** $[F][A][T^2]$
(3) $[F][A][T^{-1}]$ **(4)** $[F][A^{-1}][T]$

8. NEET 2021 · Q32 If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of:

APPLIED Class XI Part 1 · Ch 1 · **Page 8** Open PDF ↗

(1) $[M^2][L^{-1}][T^0]$ **(2)** $[M][L^{-1}][T^{-1}]$
(3) $[M][L^0][T^0]$ **(4)** $[M^2][L^{-2}][T^{-1}]$

9. NEET 2017 · Q173 A physical quantity of the dimensions of length that can be formed out of c, G and $\frac{e^2}{4\pi\varepsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and e is charge]:

APPLIED Class XI Part 1 · Ch 1 · **Page 8** Open PDF ↗

(1) $c^2\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$ **(2)** $\frac{1}{c^2}\left[\frac{e^2}{G4\pi\varepsilon_0}\right]^{1/2}$
(3) $\frac{1}{c}G\frac{e^2}{4\pi\varepsilon_0}$ **(4)** $\frac{1}{c^2}\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$

Measurement and errors 12 Qs

10. NEET 2026 · Q20 Which of the following measurements require 'index correction'?

APPLIED Class XI Part 1 · Ch 1 · **Page 6** · Lab manual: index correction on the optical bench Open PDF ↗

(1) Measurement of resistance of a wire using meter bridge
(2) Measurement of gravitational acceleration using simple pendulum
(3) Measurement of focal length of lenses using optical bench
(4) Measurement of speed of sound using resonance tube

11. NEET 2026 · Q26 One main scale division of a Vernier calliper is equal to 1 mm and the number of divisions on the Vernier scale is 10. When both the jaws touch each other, the Vernier scale shifts to the left of zero of the main scale in such a way that 4th Vernier division coincides with a division of the main scale. If this Vernier calliper measures the length of the wire to be 1 cm, the actual length of the wire is:

APPLIED Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: vernier zero error Open PDF ↗

(1) 0.60 cm **(2)** 0.96 cm
(3) 1.00 cm **(4)** 1.04 cm

12. NEET 2025 · Q25 Consider the diameter of a spherical object being measured with the help of a Vernier callipers. Suppose its 10 Vernier Scale Divisions (V.S.D.) are equal to its 9 Main Scale Divisions (M.S.D.). The least division in the M.S. is 0.1 cm and the zero of V.S. is at $x = 0.1$ cm when the jaws of Vernier callipers are closed. If the main scale reading for the diameter is $M = 5$ cm and the number of coinciding vernier division is 8, the measured diameter after zero error correction, is

APPLIED Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: vernier zero error Open PDF ↗

(1) 5.18 cm **(2)** 5.08 cm
(3) 4.98 cm **(4)** 5.00 cm

13. **NEET 2025 · Q42** A physical quantity P is related to four observations a, b, c and d as follows: $P = a^3b^2/c\sqrt{d}$. The percentage errors of measurement in a, b, c and d are 1%, 3%, 2%, and 4% respectively. The percentage error in the quantity P is

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(1) 10% **(2)** 2% **(3)** 13% **(4)** 15%

14. **NEET 2024 · Q12** In a vernier calipers, $(N + 1)$ divisions of vernier scale coincide with N divisions of main scale. If 1 MSD represents 0.1 mm, the vernier constant (in cm) is:

APPLIED — Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: vernier constant Open PDF ↗

(1) $10(N + 1)$ **(2)** $\dfrac{1}{10N}$

(3) $\dfrac{1}{100(N + 1)}$ **(4)** $100N$

15. **NEET 2023 · Q7** A metal wire has mass (0.4 ± 0.002) g, radius (0.3 ± 0.001) mm and length (5 ± 0.02) cm. The maximum possible percentage error in the measurement of density will nearly be:

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(1) 1.4% **(2)** 1.2% **(3)** 1.3% **(4)** 1.6%

16. **NEET 2022 · Q47** The area of a rectangular field (in m^2) of length 55.3 m and breadth 25 m after rounding off the value for correct significant digits is:

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(1) 14×10^2 **(2)** 138×10^1

(3) 1382 **(4)** 1382.5

17. **NEET 2021 · Q22** A screw gauge gives the following readings when used to measure the diameter of a wire: Main scale reading: 0 mm; Circular scale reading: 52 divisions. Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is:

APPLIED — Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: screw gauge Open PDF ↗

(1) 0.52 cm **(2)** 0.026 cm

(3) 0.26 cm **(4)** 0.052 cm

18. **NEET 2020 · Q146** A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale. The pitch of the screw gauge is:

APPLIED — Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: pitch = LC x divisions Open PDF ↗

(1) 0.01 mm **(2)** 0.25 mm

(3) 0.5 mm **(4)** 1.0 mm

19. **NEET 2020 · Q176** Taking into account of the significant figures, what is the value of 9.99 m – 0.0099 m?

DIRECT — Class XI Part 1 · Ch 1 · **Page 5** · rounding in subtraction Open PDF ↗

(1) 9.9801 m **(2)** 9.98 m

(3) 9.980 m **(4)** 9.9 m

20. **NEET 2019 · Q44** In an experiment, the percentage of error occurred in the measurement of physical quantities A, B, C and D are 1%, 2%, 3% and 4% respectively. Then the maximum percentage of error in the measurement X, where $X = \dfrac{A^2B^{1/2}}{C^{1/3}D^3}$, will be:

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(1) $\left(\dfrac{3}{13}\right)\%$ **(2)** 16%

(3) –10% **(4)** 10%

21. **NEET 2018 · Q37** A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above the reference level. If screw gauge has a zero error of −0.004 cm, the correct diameter of the ball is

APPLIED

Class XI Part 1 · Ch 1 · **Page 11** · Lab manual: screw gauge zero error **Open PDF** ↗

(1) 0.053 cm **(2)** 0.525 cm
(3) 0.521 cm **(4)** 0.529 cm

Units and Measurements · Answer Key

1 (1)	**2** (3)	**3** (2)	**4** (3)	**5** (2)	**6** (3)	**7** (2)	**8** (1)
9 (4)	**10** (3)	**11** (–)*	**12** (3)	**13** (3)	**14** (3)	**15** (4)	**16** (1)
17 (4)	**18** (3)	**19** (2)	**20** (2)	**21** (4)			

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor. –: question dropped by NTA. *: see note.

11. (NEET 2026 Q26) Dropped by NTA. The zero error is −0.06 cm, so the actual length is 1.06 cm, which is not among the options.

Chapter Summary · Units and Measurements

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	0	1	1	1	2	3	3	1	3	3	3	**21**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Dimensional analysis	**9**	2	6	16 **17** 18 19 20 **21** **22** 23 **24** **25** **26**
Measurement and errors	**12**	4	9	16 17 **18** **19** **20** **21** **22** **23** **24** **25** **26**

Direct vs Applied

Direct from NCERT	**6**	29%
text, table, figure	6	
solved Example	0	
Exercise	0	
Applied	**15**	71%
Easy / Mod. / Hard	9 / 10 / 2	

What to focus on

- 21 questions in 11 papers, about 1.9 per paper; rank 7 of 28 chapters by number of questions.
- Most asked topic: **Measurement and errors** (12 questions, in 9 of the 11 years).
- Asked in 8 or more of the 11 years: Measurement and errors.
- NCERT pages that gave three or more questions: page 11 (6 questions), page 8 (5 questions), page 6 (4 questions), page 7 (3 questions). Read these pages line by line.
- Difficulty: 9 easy, 10 moderate, 2 hard (hard: Q 2, 9).
- 15 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XI · Chapter 2

Motion in a Straight Line

13 questions · 1 topics · NEET 2016–2026

Kinematics in one dimension 13 Qs

1. NEET 2026 · Q6 Consider a particle moving along a straight line, whose position as a function of time is given by $s(t) = \alpha t^2 - \beta t + \gamma$, where $\alpha = 1\,\mathrm{m\,s^{-2}}$, $\beta = 6\,\mathrm{m\,s^{-1}}$ and $\gamma = 5\,\mathrm{m}$. The average speed of the particle, in $\mathrm{m\,s^{-1}}$ from $t = 0$ to $t = 6\,\mathrm{s}$ is:

APPLIED Class XI Part 1 · Ch 2 · **Page 21** Open PDF ↗

(1) 12 **(2)** 6 **(3)** 3 **(4)** 0

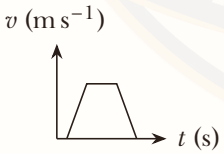
2. NEET 2025 · Q15 In some appropriate units, time (t) and position (x) relation of a moving particle is given by $t = x^2 + x$. The acceleration of the particle is

APPLIED Class XI Part 1 · Ch 2 · **Page 15** Open PDF ↗

(1) $-\dfrac{2}{(x+2)^3}$ **(2)** $-\dfrac{2}{(2x+1)^3}$

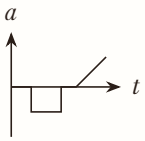
(3) $+\dfrac{2}{(x+1)^3}$ **(4)** $+\dfrac{2}{2x+1}$

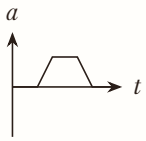
3. NEET 2024 · Q41 The velocity (v)–time (t) plot of the motion of a body is shown below:

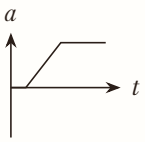


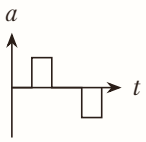
The acceleration (a)–time (t) graph that best suits this motion is:

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(1) 

(2) 

(3) 

(4) 

4. NEET 2023 · Q31 A vehicle travels half the distance with speed ϑ and the remaining distance with speed 2ϑ. Its average speed is:

APPLIED Class XI Part 1 · Ch 2 · **Page 21** Open PDF ↗

(1) $\dfrac{3\vartheta}{4}$ **(2)** $\dfrac{\vartheta}{3}$ **(3)** $\dfrac{2\vartheta}{3}$ **(4)** $\dfrac{4\vartheta}{3}$

5. NEET 2023 · Q48 A bullet from a gun is fired on a rectangular wooden block with velocity u. When bullet travels 24 cm through the block along its length horizontally, velocity of bullet becomes $\dfrac{u}{3}$. Then it further penetrates into the block in the same direction before coming to rest exactly at the other end of the block. The total length of the block is:

APPLIED Class XI Part 1 · Ch 2 · **Page 17** Open PDF ↗

(1) 30 cm **(2)** 27 cm **(3)** 24 cm **(4)** 28 cm

6. NEET 2023 · Q50 A horizontal bridge is built across a river. A student standing on the bridge throws a small ball vertically upwards with a velocity $4\,\mathrm{m\,s^{-1}}$. The ball strikes the water surface after 4 s. The height of bridge above water surface is (Take $g = 10\,\mathrm{m\,s^{-2}}$):

DIRECT Class XI Part 1 · Ch 2 · **Page 18** · Example 2.3 Open PDF ↗

(1) 68 m **(2)** 56 m **(3)** 60 m **(4)** 64 m

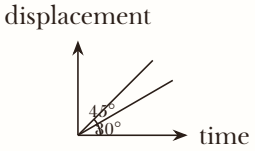
7. NEET 2022 · Q13 The ratio of the distances travelled by a freely falling body in the 1st, 2nd, 3rd and 4th second:

DIRECT Class XI Part 1 · Ch 2 · **Page 20** · Example 2.5 Open PDF ↗

(1) 1:1:1:1 **(2)** 1:2:3:4

(3) 1:4:9:16 **(4)** 1:3:5:7

8. **NEET 2022 · Q25** The displacement-time graphs of two moving particles make angles of 30° and 45° with the x-axis as shown in the figure. The ratio of their respective velocity is:



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(1) $1 : \sqrt{3}$ **(2)** $\sqrt{3} : 1$ **(3)** 1:1 **(4)** 1:2

9. **NEET 2021 · Q24** A small block slides down on a smooth inclined plane, starting from rest at time $t = 0$. Let S_n be the distance travelled by the block in the interval $t = n - 1$ to $t = n$. Then, the ratio $\dfrac{S_n}{S_{n+1}}$ is:

DIRECT Class XI Part 1 · Ch 2 · **Page 20** · Example 2.5 Open PDF ↗

(1) $\dfrac{2n-1}{2n}$ **(2)** $\dfrac{2n-1}{2n+1}$

(3) $\dfrac{2n+1}{2n-1}$ **(4)** $\dfrac{2n}{2n-1}$

10. **NEET 2020 · Q177** A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is: ($g = 10\ \text{m/s}^2$)

DIRECT Class XI Part 1 · Ch 2 · **Page 17** · v^2=v0^2+2ax Open PDF ↗

(1) 360 m **(2)** 340 m

(3) 320 m **(4)** 300 m

11. **NEET 2019 · Q30** When an object is shot from the bottom of a long smooth inclined plane kept at an angle 60° with horizontal, it can travel a distance x_1 along the plane. But when the inclination is decreased to 30° and the same object is shot with the same velocity, it can travel x_2 distance. Then $x_1 : x_2$ will be:

APPLIED Class XI Part 1 · Ch 2 · **Page 17** Open PDF ↗

(1) $1 : \sqrt{2}$ **(2)** $\sqrt{2} : 1$

(3) $1 : \sqrt{3}$ **(4)** $1 : 2\sqrt{3}$

12. **NEET 2018 · Q34** A toy car with charge q moves on a frictionless horizontal plane surface under the influence of a uniform electric field $\vec{E}$. Due to the force $q\vec{E}$, its velocity increases from 0 to 6 m/s in one second duration. At that instant the direction of the field is reversed. The car continues to move for two more seconds under the influence of this field. The average velocity and the average speed of the toy car between 0 to 3 seconds are respectively

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(1) 1 m/s, 3.5 m/s **(2)** 1 m/s, 3 m/s

(3) 2 m/s, 4 m/s **(4)** 1.5 m/s, 3 m/s

13. **NEET 2016 · Q41** If the velocity of a particle is $v = At + Bt^2$, where A and B are constants, then the distance travelled by it between 1 s and 2 s is:

APPLIED Class XI Part 1 · Ch 2 · **Page 18** · calculus method Open PDF ↗

(1) $\dfrac{3}{2}A + 4B$ **(2)** $3A + 7B$

(3) $\dfrac{3}{2}A + \dfrac{7}{3}B$ **(4)** $\dfrac{A}{2} + \dfrac{B}{3}$

Motion in a Straight Line · Answer Key

1 (3)	**2** (2)	**3** (4)	**4** (4)	**5** (2)	**6** (4)	**7** (4)	**8** (1)
9 (2)	**10** (4)	**11** (3)	**12** (2)	**13** (3)			

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Motion in a Straight Line

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	0	1	1	1	1	2	3	1	1	1	**13**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Kinematics in one dimension	**13**	4	10	**16** 17 **18 19 20 21 22 23 24 25 26**

Direct vs Applied

Direct from NCERT	**4**	31%
text, table, figure	1	
solved Example	3	
Exercise	0	
Applied	**9**	69%
Easy / Mod. / Hard	6 / 7 / 0	

What to focus on

- 13 questions in 11 papers, about 1.2 per paper; rank 19 of 28 chapters by number of questions.
- Most asked topic: **Kinematics in one dimension** (13 questions, in 10 of the 11 years).
- Asked in 8 or more of the 11 years: Kinematics in one dimension.
- 3 questions are NCERT solved Examples or Exercises: Example 2.3 (Q 6), Example 2.5 (Q 7), Example 2.5 (Q 9). Work these by hand.
- NCERT pages that gave three or more questions: page 17 (4 questions). Read these pages line by line.
- Difficulty: 6 easy, 7 moderate, 0 hard.
- 9 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XI · Chapter 3

Motion in a Plane

13 questions · 3 topics · NEET 2016–2026

Motion in a plane (vectors, relative velocity) 6 Qs

1. NEET 2025 · Q21 Two cities X and Y are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scooty with a speed of 60 km/h in the direction X to Y notices that a bus goes past her every 30 minutes in the direction of her motion, and every 10 minutes in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses.

APPLIED Class XI Part 1 · Ch 3 · **Page 46** · relative velocity (Points to ponder) Open PDF ↗

(1) 9 min, 40 km/h
(2) 25 min, 100 km/h
(3) 10 min, 90 km/h
(4) 15 min, 120 km/h

2. NEET 2021 · Q42 A car starts from rest and accelerates at 5 m/s^2. At $t = 4$ s, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at $t = 6$ s? (Take $g = 10$ m/s^2)

APPLIED Class XI Part 1 · Ch 3 · **Page 37** Open PDF ↗

(1) 20 m/s, 5 m/s^2
(2) 20 m/s, 0
(3) $20\sqrt{2}$ m/s, 0
(4) $20\sqrt{2}$ m/s, 10 m/s^2

3. NEET 2019 · Q32 The speed of a swimmer in still water is 20 m/s. The speed of river water is 10 m/s and is flowing due east. If he is standing on the south bank and wishes to cross the river along the shortest path, the angle at which he should make his strokes w.r.t. north is given by:

APPLIED Class XI Part 1 · Ch 3 · **Page 34** · Example 3.3 setting Open PDF ↗

(1) 30° west **(2)** 0°
(3) 60° west **(4)** 45° west

4. NEET 2017 · Q154 Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk up on the moving escalator will be:

APPLIED Class XI Part 1 · Ch 3 · **Page 46** · relative velocity Open PDF ↗

(1) $\dfrac{t_1 t_2}{t_2 - t_1}$ **(2)** $\dfrac{t_1 t_2}{t_2 + t_1}$
(3) $t_1 - t_2$ **(4)** $\dfrac{t_1 + t_2}{2}$

5. NEET 2017 · Q178 The x and y coordinates of the particle at any time are $x = 5t - 2t^2$ and $y = 10t$ respectively, where x and y are in meters and t in seconds. The acceleration of the particle at $t = 2$ s is:

APPLIED Class XI Part 1 · Ch 3 · **Page 37** Open PDF ↗

(1) 5 m/s^2 **(2)** −4 m/s^2
(3) −8 m/s^2 **(4)** 0

6. NEET 2016 · Q37 If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is:

APPLIED Class XI Part 1 · Ch 3 · **Page 33** Open PDF ↗

(1) 0° **(2)** 90° **(3)** 45° **(4)** 180°

Projectile motion 3 Qs

7. NEET 2023 · Q25 A bullet is fired from a gun at the speed of 280 m s^{-1} in the direction 30° above the horizontal. The maximum height attained by the bullet is (g = 9.8 m s^{-2}, sin 30° = 0.5):

DIRECT Class XI Part 1 · Ch 3 · **Page 39** · hm=(v sin theta)^2/2g Open PDF ↗

(1) 3000 m (2) 2800 m
(3) 2000 m (4) 1000 m

8. NEET 2022 · Q41 A ball is projected with a velocity, 10 m s^{-1}, at an angle of 60° with the vertical direction. Its speed at the highest point of its trajectory will be:

APPLIED Class XI Part 1 · Ch 3 · **Page 39** Open PDF ↗

(1) 10 m s^{-1} (2) Zero
(3) $5\sqrt{3}$ m s^{-1} (4) 5 m s^{-1}

9. NEET 2021 · Q47 A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution. If this particle were projected with the same speed at an angle 'θ' to the horizontal, the maximum height attained by it equals $4R$. The angle of projection, θ, is then given by:

APPLIED Class XI Part 1 · Ch 3 · **Page 39** Open PDF ↗

(1) $\theta = \cos^{-1}\left(\frac{gT^2}{\pi^2 R}\right)^{1/2}$

(2) $\theta = \cos^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{1/2}$

(3) $\theta = \sin^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{1/2}$

(4) $\theta = \sin^{-1}\left(\frac{2gT^2}{\pi^2 R}\right)^{1/2}$

Circular motion 4 Qs

10. NEET 2024 · Q7 A particle moving with uniform speed in a circular path maintains:

DIRECT Class XI Part 1 · Ch 3 · **Page 42** · acceleration of UCM is not a constant vector Open PDF ↗

(1) varying velocity and varying acceleration.
(2) constant velocity.
(3) constant acceleration.
(4) constant velocity but varying acceleration.

11. NEET 2019 · Q34 Two particles A and B are moving in uniform circular motion in concentric circles of radii r_A and r_B with speed v_A and v_B respectively. Their time period of rotation is the same. The ratio of angular speed of A to that of B will be:

DIRECT Class XI Part 1 · Ch 3 · **Page 42** · omega = 2 pi/T Open PDF ↗

(1) $r_A : r_B$ (2) $v_A : v_B$
(3) $r_B : r_A$ (4) 1:1

12. NEET 2016 · Q24 A particle of mass 10 g moves along a circle of radius 6.4 cm with a constant tangential acceleration. What is the magnitude of this acceleration if the kinetic energy of the particle becomes equal to 8×10^{-4} J by the end of the second revolution after the beginning of the motion?

APPLIED Class XI Part 1 · Ch 3 · **Page 42** Open PDF ↗

(1) 0.1 m/s^2 (2) 0.15 m/s^2
(3) 0.18 m/s^2 (4) 0.2 m/s^2

13. NEET 2016 · Q27 A particle moves so that its position vector is given by $\vec{r} = \cos\omega t\,\hat{x} + \sin\omega t\,\hat{y}$, where ω is a constant. Which of the following is true?

APPLIED Class XI Part 1 · Ch 3 · **Page 42** · uniform circular motion Open PDF ↗

(1) Velocity and acceleration both are perpendicular to $\vec{r}$.
(2) Velocity and acceleration both are parallel to $\vec{r}$.
(3) Velocity is perpendicular to $\vec{r}$ and acceleration is directed towards the origin.
(4) Velocity is perpendicular to $\vec{r}$ and acceleration is directed away from the origin.

Motion in a Plane · Answer Key

1 (1)	**2** (4)	**3** (1)	**4** (2)	**5** (2)	**6** (2)	**7** (4)	**8** (3)
9 (4)	**10** (1)	**11** (4)	**12** (1)	**13** (3)			

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Motion in a Plane

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	3	2	0	2	0	2	1	1	1	1	0	**13**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Motion in a plane (vectors, relative velocity)	**6**	0	5	**16** **17** 18 **19** 20 **21** 22 23 24 **25** 26
Projectile motion	**3**	1	3	16 17 18 19 20 **21** **22** **23** 24 25 26
Circular motion	**4**	2	3	**16** 17 18 **19** 20 21 22 23 **24** 25 26

Direct vs Applied

Direct from NCERT	**3**	23%
text, table, figure	3	
solved Example	0	
Exercise	0	
Applied	**10**	77%
Easy / Mod. / Hard	7 / 5 / 1	

What to focus on

- 13 questions in 11 papers, about 1.2 per paper; rank 20 of 28 chapters by number of questions.
- Most asked topic: **Motion in a plane (vectors, relative velocity)** (6 questions, in 5 of the 11 years).
- NCERT pages that gave three or more questions: page 42 (4 questions), page 39 (3 questions). Read these pages line by line.
- Difficulty: 7 easy, 5 moderate, 1 hard (hard: Q 9).
- 10 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

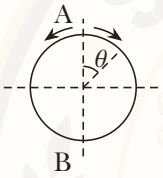
Class XI · Chapter 4

Laws of Motion

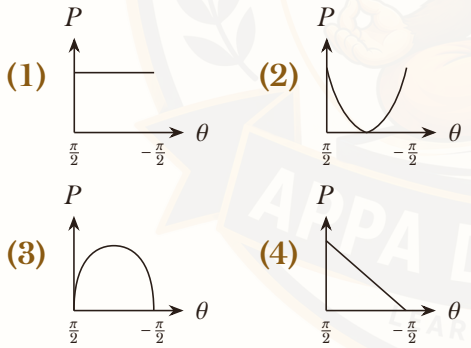
18 questions · 7 topics · NEET 2016–2026

Kinetic energy and momentum 1 Q

1. NEET 2026 · Q23 A frictionless circular wire of unit radius is fixed on the horizontal plane. Two point particles of unit mass start moving simultaneously from point A ($\theta = \frac{\pi}{2}$) with identical uniform angular speeds in opposite directions, and meet again at point B ($\theta = -\frac{\pi}{2}$). During this time, which of the following figures schematically represent the magnitude of the total linear momentum P of the system, as a function of θ?



APPLIED Class XI Part 1 · Ch 4 · **Page 53** Open PDF ↗



Impulse and momentum 3 Qs

2. NEET 2025 · Q10 A ball of mass 0.5 kg is dropped from a height of 40 m. The ball hits the ground and rises to a height of 10 m. The impulse imparted to the ball during its collision with the ground is (Take $g = 9.8\ \text{m/s}^2$)

APPLIED Class XI Part 1 · Ch 4 · **Page 55** Open PDF ↗

(1) 21 N s **(2)** 7 N s **(3)** 0 **(4)** 84 N s

3. NEET 2023 · Q13 A football player is moving southward and suddenly turns eastward with the same speed to avoid an opponent. The force that acts on the player while turning is:

APPLIED Class XI Part 1 · Ch 4 · **Page 54** Open PDF ↗

(1) along south-west **(2)** along eastward
(3) along northward **(4)** along north-east

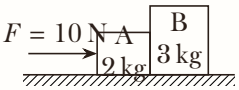
4. NEET 2021 · Q45 A ball of mass 0.15 kg is dropped from a height 10 m, strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is ($g = 10\ \text{m/s}^2$) nearly:

APPLIED Class XI Part 1 · Ch 4 · **Page 55** Open PDF ↗

(1) 0 kg m/s **(2)** 4.2 kg m/s
(3) 2.1 kg m/s **(4)** 1.4 kg m/s

Laws of motion (lift, apparent weight) 3 Qs

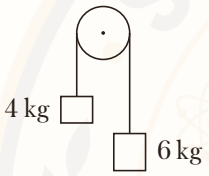
5. **NEET 2024 · Q31** A horizontal force 10 N is applied to a block A as shown in figure. The mass of blocks A and B are 2 kg and 3 kg, respectively. The blocks slide over a frictionless surface. The force exerted by block A on block B is:



APPLIED Class XI Part 1 · Ch 4 · **Page 54** Open PDF ↗

(1) 10 N **(2)** zero **(3)** 4 N **(4)** 6 N

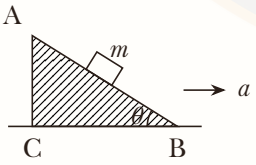
6. **NEET 2020 · Q160** Two bodies of mass 4 kg and 6 kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless (see figure). The acceleration of the system in terms of acceleration due to gravity (g) is:



DIRECT Class XI Part 1 · Ch 4 · **Page 70** · Exercise 4.16 Open PDF ↗

(1) g **(2)** $g/2$ **(3)** $g/5$ **(4)** $g/10$

7. **NEET 2018 · Q35** A block of mass m is placed on a smooth inclined wedge ABC of inclination θ as shown in the figure. The wedge is given an acceleration 'a' towards the right. The relation between a and θ for the block to remain stationary on the wedge is



APPLIED Class XI Part 1 · Ch 4 · **Page 65** Open PDF ↗

(1) $a = g\cos\theta$ **(2)** $a = \dfrac{g}{\sin\theta}$

(3) $a = \dfrac{g}{\operatorname{cosec}\theta}$ **(4)** $a = g\tan\theta$

Collisions and momentum conservation 1 Q

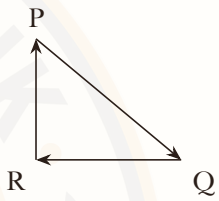
8. **NEET 2022 · Q20** A shell of mass m is at rest initially. It explodes into three fragments having mass in the ratio 2 : 2 : 1. If the fragments having equal mass fly off along mutually perpendicular directions with speed v, the speed of the third (lighter) fragment is:

APPLIED Class XI Part 1 · Ch 4 · **Page 57** Open PDF ↗

(1) $3\sqrt{2}\,v$ **(2)** v **(3)** $\sqrt{2}\,v$ **(4)** $2\sqrt{2}\,v$

Equilibrium of forces and tension 2 Qs

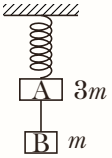
9. **NEET 2019 · Q33** A particle moving with velocity $\vec{V}$ is acted by three forces shown by the vector triangle PQR. The velocity of the particle will:



DIRECT Class XI Part 1 · Ch 4 · **Page 58** · forces forming a closed triangle Open PDF ↗

(1) increase
(2) decrease
(3) remain constant
(4) change according to the smallest force $\overrightarrow{QR}$

10. **NEET 2017 · Q167** Two blocks A and B of masses $3m$ and m respectively are connected by a massless and inextensible string. The whole system is suspended by a massless spring as shown in figure. The magnitudes of acceleration of A and B immediately after the string is cut, are respectively:



APPLIED Class XI Part 1 · Ch 4 · **Page 59** · spring force Open PDF ↗

(1) $\dfrac{g}{3}, g$ **(2)** g, g **(3)** $\dfrac{g}{3}, \dfrac{g}{3}$ **(4)** $g, \dfrac{g}{3}$

Friction 4 Qs

11. **NEET 2025 · Q4** There are two inclined surfaces of equal length (L) and same angle of inclination 45° with the horizontal. One of them is rough and the other is perfectly smooth. A given body takes 2 times as much time to slide down on rough surface than on the smooth surface. The coefficient of kinetic friction (μ_k) between the object and the rough surface is close to

APPLIED Class XI Part 1 · Ch 4 · **Page 61** Open PDF ↗

(1) 0.25 **(2)** 0.40 **(3)** 0.5 **(4)** 0.75

12. **NEET 2023 · Q41** Calculate the maximum acceleration of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is 0.15 ($g = 10\ \text{m s}^{-2}$).

DIRECT Class XI Part 1 · Ch 4 · **Page 61** · Example 4.7 Open PDF ↗

(1) $50\ \text{m s}^{-2}$ **(2)** $1.2\ \text{m s}^{-2}$
(3) $150\ \text{m s}^{-2}$ **(4)** $1.5\ \text{m s}^{-2}$

13. **NEET 2019 · Q35** A block of mass 10 kg is in contact against the inner wall of a hollow cylindrical drum of radius 1 m. The coefficient of friction between the block and the inner wall of the cylinder is 0.1. The minimum angular velocity needed for the cylinder to keep the block stationary when the cylinder is vertical and rotating about its axis, will be: ($g = 10\ \text{m/s}^2$)

APPLIED Class XI Part 1 · Ch 4 · **Page 63** Open PDF ↗

(1) $\sqrt{10}$ rad/s **(2)** $\dfrac{10}{2\pi}$ rad/s
(3) 10 rad/s **(4)** 10π rad/s

14. **NEET 2018 · Q33** Which one of the following statements is *incorrect*?

DIRECT Class XI Part 1 · Ch 4 · **Page 60** · mu is dimensionless Open PDF ↗

(1) Frictional force opposes the relative motion.
(2) Limiting value of static friction is directly proportional to normal reaction.
(3) Rolling friction is smaller than sliding friction.
(4) Coefficient of sliding friction has dimensions of length.

Circular motion 4 Qs

15. **NEET 2026 · Q41** A car travels on a circular racetrack of radius 50 m, which is banked at an angle θ. If the car travels at a speed $10\ \text{m s}^{-1}$, then the wear and tear on its tyres is minimum. Taking the acceleration due to gravity to be $10\ \text{m s}^{-2}$, the value of θ is:

DIRECT Class XI Part 1 · Ch 4 · **Page 64** · v0=sqrt(Rg tan theta) Open PDF ↗

(1) $\tan^{-1}\left(\dfrac{1}{5}\right)$ **(2)** $\tan^{-1}\left(\dfrac{2}{5}\right)$
(3) $\tan^{-1}(\sqrt{3}/2)$ **(4)** $\tan^{-1}(2\sqrt{3})$

16. **NEET 2024 · Q25** A bob is whirled in a horizontal plane by means of a string with an initial speed of ω rpm. The tension in the string is T. If speed becomes 2ω while keeping the same radius, the tension in the string becomes:

APPLIED Class XI Part 1 · Ch 4 · **Page 63** Open PDF ↗

(1) $\sqrt{2}T$ **(2)** T **(3)** $4T$ **(4)** $\dfrac{T}{4}$

17. **NEET 2017 · Q156** One end of string of length l is connected to a particle of mass 'm' and the other end is connected to a small peg on a smooth horizontal table. If the particle moves in circle with speed 'v', the net force on the particle (directed towards center) will be (T represents the tension in the string):

DIRECT Class XI Part 1 · Ch 4 · **Page 69** · Exercise 4.4 Open PDF ↗

(1) $T + \frac{mv^2}{l}$ **(2)** $T - \frac{mv^2}{l}$

(3) Zero **(4)** T

18. **NEET 2016 · Q45** A car is negotiating a curved road of radius R. The road is banked at an angle θ. The coefficient of friction between the tyres of the car and the road is μ_s. The maximum safe velocity on this road is:

DIRECT Class XI Part 1 · Ch 4 · **Page 64** · banked road Open PDF ↗

(1) $\sqrt{gR^2\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$

(2) $\sqrt{gR\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$

(3) $\sqrt{\frac{g}{R}\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$

(4) $\sqrt{\frac{g}{R^2}\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$

Laws of Motion · Answer Key

1 (3)	**2** (1)	**3** (4)	**4** (2)	**5** (4)	**6** (3)	**7** (4)	**8** (4)
9 (3)	**10** (1)	**11** (4)	**12** (4)	**13** (3)	**14** (4)	**15** (1)	**16** (3)
17 (4)	**18** (2)						

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Laws of Motion

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	2	2	2	1	1	1	2	2	2	2	**18**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Kinetic energy and momentum	**1**	0	1	16 17 18 19 20 21 22 23 24 25 **26**
Impulse and momentum	**3**	0	3	16 17 18 19 20 **21** 22 **23** 24 **25** 26
Laws of motion (lift, apparent weight)	**3**	1	3	16 17 **18** 19 **20** 21 22 23 **24** 25 26
Collisions and momentum conservation	**1**	0	1	16 17 18 19 20 21 **22** 23 24 25 26
Equilibrium of forces and tension	**2**	1	2	16 **17** 18 **19** 20 21 22 23 24 25 26
Friction	**4**	2	4	16 17 **18** **19** 20 21 22 **23** 24 **25** 26
Circular motion	**4**	3	4	**16** **17** 18 19 20 21 22 23 **24** 25 **26**

Direct vs Applied

Direct from NCERT	**7**	39%
text, table, figure	4	
solved Example	1	
Exercise	2	
Applied	**11**	61%
Easy / Mod. / Hard	11 / 7 / 0	

What to focus on

- 18 questions in 11 papers, about 1.6 per paper; rank 9 of 28 chapters by number of questions.
- Most asked topic: **Friction** (4 questions, in 4 of the 11 years).
- 3 questions are NCERT solved Examples or Exercises: Example 4.7 (Q 12), Exercise 4.16 (Q 6), Exercise 4.4 (Q 17). Work these by hand.
- Difficulty: 11 easy, 7 moderate, 0 hard.
- 11 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XI · Chapter 5

Work, Energy and Power

18 questions · 3 topics · NEET 2016–2026

Work-energy theorem and power 11 Qs

1. **NEET 2026 · Q1** A particle of mass M moves along a horizontal x axis from $x = 0$ to $x = L$. The coefficient of kinetic friction varies as a function of x as $\mu_k(x) = \mu_0 - \alpha x$, where μ_0, α are constants of appropriate dimensions, so that $\mu_k(L) = 0$. The total work done by the frictional force during the motion is $n\mu_0 MgL$, where g is the acceleration due to gravity. The value of n is:

APPLIED Class XI Part 1 · Ch 5 · **Page 75** · work by variable force (area under F-x) Open PDF ↗

(1) 3 (2) 1 (3) $\frac{1}{3}$ (4) $\frac{1}{2}$

2. **NEET 2025 · Q5** The kinetic energies of two similar cars A and B are 100 J and 225 J respectively. On applying brakes, car A stops after 1000 m and car B stops after 1500 m. If F_A and F_B are the forces applied by the brakes on cars A and B, respectively, then the ratio F_A/F_B is

APPLIED Class XI Part 1 · Ch 5 · **Page 73** Open PDF ↗

(1) $\frac{3}{2}$ (2) $\frac{2}{3}$ (3) $\frac{1}{3}$ (4) $\frac{1}{2}$

3. **NEET 2024 · Q13** At any instant of time t, the displacement of any particle is given by $2t - 1$ (SI unit) under the influence of force of 5 N. The value of instantaneous power is (in SI unit):

DIRECT Class XI Part 1 · Ch 5 · **Page 83** · P=F.v Open PDF ↗

(1) 6 (2) 10 (3) 5 (4) 7

4. **NEET 2023 · Q29** The potential energy of a long spring when stretched by 2 cm is U. If the spring is stretched by 8 cm, potential energy stored in it will be:

DIRECT Class XI Part 1 · Ch 5 · **Page 81** · U=kx^2/2 Open PDF ↗

(1) 16U (2) 2U (3) 4U (4) 8U

5. **NEET 2022 · Q15** The energy that will be ideally radiated by a 100 kW transmitter in 1 hour is:

DIRECT Class XI Part 1 · Ch 5 · **Page 83** · energy = power x time Open PDF ↗

(1) 1×10^5 J (2) 36×10^7 J
(3) 36×10^4 J (4) 36×10^5 J

6. **NEET 2022 · Q16** An electric lift with a maximum load of 2000 kg (lift + passengers) is moving up with a constant speed of 1.5 m s^{-1}. The frictional force opposing the motion is 3000 N. The minimum power delivered by the motor to the lift in watts is: ($g = 10$ m s^{-2})

DIRECT Class XI Part 1 · Ch 5 · **Page 83** · Example 5.10 Open PDF ↗

(1) 23500 (2) 23000
(3) 20000 (4) 34500

7. **NEET 2021 · Q11** A particle is released from height S from the surface of the Earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of earth and the speed of the particle at that instant are respectively:

APPLIED Class XI Part 1 · Ch 5 · **Page 78** Open PDF ↗

(1) $\frac{S}{4}, \frac{3gS}{2}$ (2) $\frac{S}{4}, \frac{\sqrt{3gS}}{2}$
(3) $\frac{S}{2}, \frac{\sqrt{3gS}}{2}$ (4) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

8. **NEET 2021 · Q23** Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine? ($g = 10\,m/s^2$)

APPLIED Class XI Part 1 · Ch 5 · **Page 83** Open PDF ↗

(1) 10.2 kW **(2)** 8.1 kW
(3) 12.3 kW **(4)** 7.0 kW

9. **NEET 2019 · Q22** A force $F = 20 + 10y$ acts on a particle in y-direction where F is in newton and y in meter. Work done by this force to move the particle from $y = 0$ to $y = 1$ m is:

DIRECT Class XI Part 1 · Ch 5 · **Page 75** · W = integral of F dx Open PDF ↗

(1) 30 J **(2)** 5 J **(3)** 25 J **(4)** 20 J

10. **NEET 2017 · Q177** Consider a drop of rain water having mass 1 g falling from a height of 1 km. It hits the ground with a speed of 50 m/s. Take 'g' constant with a value $10\,m/s^2$. The work done by the (i) gravitational force and the (ii) resistive force of air is:

DIRECT Class XI Part 1 · Ch 5 · **Page 73** · Example 5.2 Open PDF ↗

(1) (i) 1.25 J (ii) −8.25 J
(2) (i) 100 J (ii) 8.75 J
(3) (i) 10 J (ii) −8.75 J
(4) (i) −10 J (ii) −8.25 J

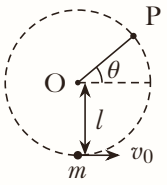
11. **NEET 2016 · Q39** A body of mass 1 kg begins to move under the action of a time dependent force $\vec{F} = (2t\hat{i} + 3t^2\hat{j})$ N, where $\hat{i}$ and $\hat{j}$ are unit vectors along x and y axis. What power will be developed by the force at the time t?

APPLIED Class XI Part 1 · Ch 5 · **Page 83** · P=F.v Open PDF ↗

(1) $(2t^2 + 3t^3)$ W **(2)** $(2t^2 + 4t^4)$ W
(3) $(2t^3 + 3t^4)$ W **(4)** $(2t^3 + 3t^5)$ W

Circular motion 4 Qs

12. **NEET 2025 · Q7** A bob of heavy mass m is suspended by a light string of length l. The bob is given a horizontal velocity v_0 as shown in figure. If the string gets slack at some point P making an angle θ from the horizontal, the ratio of the speed v of the bob at point P to its initial speed v_0 is:



APPLIED Class XI Part 1 · Ch 5 · **Page 79** · Example 5.7 setting Open PDF ↗

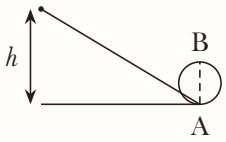
(1) $(\sin\theta)^{1/2}$ **(2)** $\left(\frac{1}{2+3\sin\theta}\right)^{1/2}$
(3) $\left(\frac{\cos\theta}{2+3\sin\theta}\right)^{1/2}$ **(4)** $\left(\frac{\sin\theta}{2+3\sin\theta}\right)^{1/2}$

13. **NEET 2019 · Q2** A mass m is attached to a thin wire and whirled in a vertical circle. The wire is most likely to break when:

APPLIED Class XI Part 1 · Ch 5 · **Page 79** · vertical circle (Example 5.7) Open PDF ↗

(1) the mass is at the highest point
(2) the wire is horizontal
(3) the mass is at the lowest point
(4) inclined at an angle of 60° from vertical

14. **NEET 2018 · Q30** A body initially at rest and sliding along a frictionless track from a height h (as shown in the figure) just completes a vertical circle of diameter $AB = D$. The height h is equal to



APPLIED Class XI Part 1 · Ch 5 · **Page 79** · Example 5.7 setting Open PDF ↗

(1) $\frac{7}{5}D$ **(2)** D **(3)** $\frac{3}{2}D$ **(4)** $\frac{5}{4}D$

15. NEET 2016 · Q28 What is the minimum velocity with which a body of mass m must enter a vertical loop of radius R so that it can complete the loop?

DIRECT Class XI Part 1 · Ch 5 · **Page 79** · Example 5.7 Open PDF ↗

(1) $\sqrt{gR}$ **(2)** $\sqrt{2gR}$ **(3)** $\sqrt{3gR}$ **(4)** $\sqrt{5gR}$

Collisions and momentum conservation — 3 Qs

16. NEET 2026 · Q44 Bob B of mass m at rest is hanging vertically from the ceiling via a massless string of length 10 m, as shown in the figure. Point mass A of mass m travelling horizontally with speed $10\ \text{m s}^{-1}$ hits bob B elastically. The bob B rises h metre after the collision. Taking the acceleration due to gravity $g = 10\ \text{m s}^{-2}$ and neglecting the size of the bob, the value of h is:



APPLIED Class XI Part 1 · Ch 5 · **Page 84** Open PDF ↗

(1) 8 **(2)** 7 **(3)** 5 **(4)** 2.5

17. NEET 2024 · Q29 Two bodies A and B of same mass undergo completely inelastic one dimensional collision. The body A moves with velocity v_1 while body B is at rest before collision. The velocity of the system after collision is v_2. The ratio $v_1 : v_2$ is:

DIRECT Class XI Part 1 · Ch 5 · **Page 84** · completely inelastic collision Open PDF ↗

(1) 1:4 **(2)** 1:2 **(3)** 2:1 **(4)** 4:1

18. NEET 2019 · Q4 Body A of mass $4m$ moving with speed u collides with another body B of mass $2m$, at rest. The collision is head on and elastic in nature. After the collision the fraction of energy lost by the colliding body A is:

APPLIED Class XI Part 1 · Ch 5 · **Page 84** Open PDF ↗

(1) $\frac{1}{9}$ **(2)** $\frac{8}{9}$ **(3)** $\frac{4}{9}$ **(4)** $\frac{5}{9}$

Work, Energy and Power · Answer Key

1 (4)	**2** (2)	**3** (2)	**4** (1)	**5** (2)	**6** (4)	**7** (4)	**8** (2)
9 (3)	**10** (3)	**11** (4)	**12** (4)	**13** (3)	**14** (4)	**15** (4)	**16** (3)
17 (3)	**18** (2)						

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Work, Energy and Power

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	2	1	1	3	0	2	2	1	2	2	2	**18**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Work-energy theorem and power	**11**	6	9	**16 17** 18 **19** 20 **21 22 23 24 25 26**
Circular motion	**4**	1	4	**16** 17 **18 19** 20 21 22 23 24 **25** 26
Collisions and momentum conservation	**3**	1	3	16 17 18 **19** 20 21 22 23 **24** 25 **26**

Direct vs Applied

Direct from NCERT	**8**	44%
text, table, figure	5	
solved Example	3	
Exercise	0	
Applied	**10**	56%
Easy / Mod. / Hard	9 / 8 / 1	

What to focus on

- 18 questions in 11 papers, about 1.6 per paper; rank 10 of 28 chapters by number of questions.
- Most asked topic: **Work-energy theorem and power** (11 questions, in 9 of the 11 years).
- Asked in 8 or more of the 11 years: Work-energy theorem and power.
- 3 questions are NCERT solved Examples or Exercises: Example 5.10 (Q 6), Example 5.2 (Q 10), Example 5.7 (Q 15). Work these by hand.
- NCERT pages that gave three or more questions: page 83 (5 questions), page 79 (4 questions), page 84 (3 questions). Read these pages line by line.
- Difficulty: 9 easy, 8 moderate, 1 hard (hard: Q 12).
- 10 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XI · Chapter 6

System of Particles and Rotational Motion

22 questions · 3 topics · NEET 2016–2026

Rotational dynamics (torque, moment of inertia)

15 Qs

1. NEET 2025 · Q22 A uniform rod of mass 20 kg and length 5 m leans against a smooth vertical wall making an angle of 60° with it. The other end rests on a rough horizontal floor. The friction force that the floor exerts on the rod is (take $g = 10\ \text{m/s}^2$)

APPLIED Class XI Part 1 · Ch 6 · **Page 113** Open PDF ↗

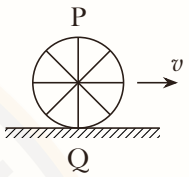
(1) 100 N (2) $100\sqrt{3}$ N
(3) 200 N (4) $200\sqrt{3}$ N

2. NEET 2024 · Q10 The moment of inertia of a thin rod about an axis passing through its mid point and perpendicular to the rod is $2400\ \text{g cm}^2$. The length of the 400 g rod is nearly:

DIRECT Class XI Part 1 · Ch 6 · **Page 116** · Table 6.1 Open PDF ↗

(1) 72.0 cm (2) 8.5 cm
(3) 17.5 cm (4) 20.7 cm

3. NEET 2024 · Q32 A wheel of a bullock cart is rolling on a level road as shown in the figure below. If its linear speed is v in the direction shown, which one of the following options is correct (P and Q are any highest and lowest points on the wheel, respectively)?



APPLIED Class XI Part 1 · Ch 6 · **Page 94** · rolling (qualitative) Open PDF ↗

(1) Point P has zero speed.
(2) Point P moves slower than point Q.
(3) Point P moves faster than point Q.
(4) Both the points P and Q move with equal speed.

4. NEET 2023 · Q15 The angular acceleration of a body, moving along the circumference of a circle, is:

DIRECT Class XI Part 1 · Ch 6 · **Page 105** · alpha along the axis Open PDF ↗

(1) along the axis of rotation
(2) along the radius, away from centre
(3) along the radius towards the centre
(4) along the tangent to its position

5. NEET 2023 · Q32 The ratio of radius of gyration of a solid sphere of mass M and radius R about its own axis to the radius of gyration of the thin hollow sphere of same mass and radius about its axis is:

APPLIED Class XI Part 1 · Ch 6 · **Page 115** Open PDF ↗

(1) 5:2 (2) 3:5 (3) 5:3 (4) 2:5

6. **NEET 2022 · Q7** The ratio of the radius of gyration of a thin uniform disc about an axis passing through its centre and normal to its plane to the radius of gyration of the disc about its diameter is:

DIRECT — Class XI Part 1 · Ch 6 · **Page 116** · Table 6.1 (k=R/sqrt2, R/2) Open PDF ↗

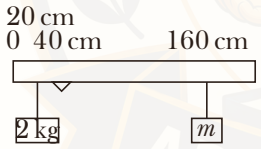
(1) 1 : $\sqrt{2}$ **(2)** 2:1 **(3)** $\sqrt{2}$: 1 **(4)** 4:1

7. **NEET 2022 · Q22** The angular speed of a fly wheel moving with uniform angular acceleration changes from 1200 rpm to 3120 rpm in 16 seconds. The angular acceleration in rad/s^2 is:

DIRECT — Class XI Part 1 · Ch 6 · **Page 117** · Example 6.11 Open PDF ↗

(1) 104π **(2)** 2π **(3)** 4π **(4)** 12π

8. **NEET 2021 · Q40** A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass 'm' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of 'm' such that the rod is in equilibrium. (g = 10 m/s^2)



APPLIED — Class XI Part 1 · Ch 6 · **Page 111** Open PDF ↗

(1) $\frac{1}{2}$ kg **(2)** $\frac{1}{3}$ kg **(3)** $\frac{1}{6}$ kg **(4)** $\frac{1}{12}$ kg

9. **NEET 2021 · Q50** From a circular ring of mass 'M' and radius 'R' an arc corresponding to a 90° sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to the plane of the ring is 'K' times 'MR^2'. Then the value of 'K' is:

APPLIED — Class XI Part 1 · Ch 6 · **Page 115** Open PDF ↗

(1) $\frac{3}{4}$ **(2)** $\frac{7}{8}$ **(3)** $\frac{1}{4}$ **(4)** $\frac{1}{8}$

10. **NEET 2020 · Q149** Find the torque about the origin when a force of $3\hat{j}$ N acts on a particle whose position vector is $2\hat{k}$ m.

DIRECT — Class XI Part 1 · Ch 6 · **Page 103** · vector product Open PDF ↗

(1) $6\hat{i}$ N m **(2)** $6\hat{j}$ N m
(3) $-6\hat{i}$ N m **(4)** $6\hat{k}$ N m

11. **NEET 2019 · Q21** A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis at the rate of 3 rpm. The torque required to stop after 2π revolutions is:

APPLIED — Class XI Part 1 · Ch 6 · **Page 117** Open PDF ↗

(1) 2×10^{-6} N m **(2)** 2×10^{-3} N m
(3) 12×10^{-4} N m **(4)** 2×10^{6} N m

12. **NEET 2018 · Q31** Three objects, A: (a solid sphere), B: (a thin circular disk) and C: (a circular ring), each have the same mass M and radius R. They all spin with the same angular speed ω about their own symmetry axes. The amounts of work (W) required to bring them to rest, would satisfy the relation

APPLIED — Class XI Part 1 · Ch 6 · **Page 116** Open PDF ↗

(1) $W_B > W_A > W_C$ **(2)** $W_A > W_B > W_C$
(3) $W_C > W_B > W_A$ **(4)** $W_A > W_C > W_B$

13. **NEET 2018 · Q36** The moment of the force, $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at (2, 0, −3), about the point (2, −2, −2), is given by

APPLIED — Class XI Part 1 · Ch 6 · **Page 106** Open PDF ↗

(1) $-7\hat{i} - 8\hat{j} - 4\hat{k}$ **(2)** $-4\hat{i} - \hat{j} - 8\hat{k}$
(3) $-8\hat{i} - 4\hat{j} - 7\hat{k}$ **(4)** $-7\hat{i} - 4\hat{j} - 8\hat{k}$

14. **NEET 2017 · Q161** A rope is wound around a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder if the rope is pulled with a force of 30 N?

DIRECT — Class XI Part 1 · Ch 6 · **Page 126** · Exercise 6.13 Open PDF ↗

(1) 0.25 rad/s^2 **(2)** 25 rad/s^2
(3) 5 m/s^2 **(4)** 25 m/s^2

15. NEET 2016 · Q19 A uniform circular disc of radius 50 cm at rest is free to turn about an axis which is perpendicular to its plane and passes through its centre. It is subjected to a torque which produces a constant angular acceleration of 2.0 rad s^{-2}. Its net acceleration in m s^{-2} at the end of 2.0 s is approximately:

APPLIED — Class XI Part 1 · Ch 6 · **Page 117** · rotational kinematics, a_t and a_c Open PDF ↗

(1) 8.0 **(2)** 7.0 **(3)** 6.0 **(4)** 3.0

Centre of mass — 3 Qs

16. NEET 2022 · Q14 Two objects of mass 10 kg and 20 kg respectively are connected to the two ends of a rigid rod of length 10 m with negligible mass. The distance of the center of mass of the system from the 10 kg mass is:

DIRECT — Class XI Part 1 · Ch 6 · **Page 96** · two-particle CM Open PDF ↗

(1) 5 m **(2)** $\frac{10}{3}$ m **(3)** $\frac{20}{3}$ m **(4)** 10 m

17. NEET 2020 · Q148 Two particles of mass 5 kg and 10 kg respectively are attached to the two ends of a rigid rod of length 1 m with negligible mass. The centre of mass of the system from the 5 kg particle is nearly at a distance of:

DIRECT — Class XI Part 1 · Ch 6 · **Page 96** · two-particle CM Open PDF ↗

(1) 33 cm **(2)** 50 cm **(3)** 67 cm **(4)** 80 cm

18. NEET 2017 · Q165 Which of the following statements are correct?

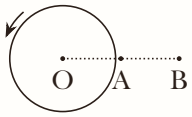
(a) Centre of mass of a body always coincides with the centre of gravity of the body.
(b) Centre of mass of a body is the point at which the total gravitational torque on the body is zero.
(c) A couple on a body produce both translational and rotational motion in a body.
(d) Mechanical advantage greater than one means that small effort can be used to lift a large load.

DIRECT — Class XI Part 1 · Ch 6 · **Page 112** · centre of gravity and mechanical advantage Open PDF ↗

(1) (a) and (b) **(2)** (b) and (c)
(3) (c) and (d) **(4)** (b) and (d)

Angular momentum and its conservation — 4 Qs

19. NEET 2026 · Q31 A thin horizontal disc is rotating about a vertical axis passing through its fixed centre O. Its angular momentum is L_A and L_B computed about points A and B, respectively, with $OB = 2 \times OA$. The value of $\frac{L_A}{L_B}$ is:



APPLIED — Class XI Part 1 · Ch 6 · **Page 108** Open PDF ↗

(1) $\frac{1}{4}$ **(2)** $\frac{1}{2}$ **(3)** 1 **(4)** 2

20. NEET 2025 · Q17 The Sun rotates around its centre once in 27 days. What will be the period of revolution if the Sun were to expand to twice its present radius without any external influence? Assume the Sun to be a sphere of uniform density.

APPLIED — Class XI Part 1 · Ch 6 · **Page 122** Open PDF ↗

(1) 100 days **(2)** 105 days
(3) 115 days **(4)** 108 days

21. NEET 2018 · Q38 A solid sphere is rotating freely about its symmetry axis in free space. The radius of the sphere is increased keeping its mass same. Which of the following physical quantities would remain constant for the sphere?

DIRECT — Class XI Part 1 · Ch 6 · **Page 108** · L constant when torque is zero Open PDF ↗

(1) Rotational kinetic energy
(2) Moment of inertia
(3) Angular velocity
(4) Angular momentum

22. **NEET 2017 · Q162** Two discs of same moment of inertia rotating about their regular axis passing through centre and perpendicular to the plane of disc with angular velocities ω_1 and ω_2. They are brought into contact face to face coinciding the axis of rotation. The expression for loss of energy during this process is:

APPLIED Class XI Part 1 · Ch 6 · **Page 122** Open PDF ↗

(1) $\frac{1}{4}I(\omega_1-\omega_2)^2$ **(2)** $I(\omega_1-\omega_2)^2$

(3) $\frac{1}{8}(\omega_1-\omega_2)^2$ **(4)** $\frac{1}{2}I(\omega_1+\omega_2)^2$

System of Particles and Rotational Motion · Answer Key

1 (2)	**2** (2)	**3** (3)	**4** (1)	**5** (–)*	**6** (3)	**7** (3)	**8** (4)
9 (1)	**10** (3)	**11** (1)	**12** (3)	**13** (4)	**14** (2)	**15** (1)	**16** (3)
17 (3)	**18** (4)	**19** (3)	**20** (4)	**21** (4)	**22** (1)		

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor. –: question dropped by NTA. *: see note.

5. (NEET 2023 Q32) Dropped by NTA. The ratio is $\sqrt{2/5} : \sqrt{2/3} = \sqrt{3} : \sqrt{5}$, which is not among the options.

Chapter Summary · System of Particles and Rotational Motion

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	3	3	1	2	2	3	2	2	2	1	**22**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Rotational dynamics (torque, moment of inertia)	**15**	6	10	**16 17 18 19 20 21 22 23 24 25** 26
Centre of mass	**3**	3	3	16 **17** 18 19 **20** 21 **22** 23 24 25 26
Angular momentum and its conservation	**4**	1	4	16 **17 18** 19 20 21 22 23 24 **25 26**

Direct vs Applied

Direct from NCERT	**10**	45%
text, table, figure	8	
solved Example	1	
Exercise	1	
Applied	**12**	55%
Easy / Mod. / Hard	12 / 10 / 0	

What to focus on

- 22 questions in 11 papers, about 2.0 per paper; rank 5 of 28 chapters by number of questions.
- Most asked topic: **Rotational dynamics (torque, moment of inertia)** (15 questions, in 10 of the 11 years).
- Asked in 8 or more of the 11 years: Rotational dynamics (torque, moment of inertia).
- 2 questions are NCERT solved Examples or Exercises: Example 6.11 (Q 7), Exercise 6.13 (Q 14). Work these by hand.
- NCERT pages that gave three or more questions: page 116 (3 questions), page 117 (3 questions). Read these pages line by line.
- Difficulty: 12 easy, 10 moderate, 0 hard.
- 12 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XI · Chapter 7

Gravitation

20 questions · 4 topics · NEET 2016–2026

Gravitation (force, field, energy) 5 Qs

1. NEET 2023 · Q33 Two bodies of mass m and $9m$ are placed at a distance R. The gravitational potential on the line joining the bodies where the gravitational field equals zero, will be (G = gravitational constant):

APPLIED — Class XI Part 1 · Ch 7 · **Page 136** · Example 7.4 setting Open PDF ↗

(1) $-\dfrac{20Gm}{R}$ (2) $-\dfrac{8Gm}{R}$

(3) $-\dfrac{12Gm}{R}$ (4) $-\dfrac{16Gm}{R}$

2. NEET 2022 · Q32 A body of mass 60 g experiences a gravitational force of 3.0 N, when placed at a particular point. The magnitude of the gravitational field intensity at that point is:

DIRECT — Class XI Part 1 · Ch 7 · **Page 134** · field = force per unit mass Open PDF ↗

(1) 180 N/kg (2) 0.05 N/kg

(3) 50 N/kg (4) 20 N/kg

3. NEET 2019 · Q29 The work done to raise a mass m from the surface of the earth to a height h, which is equal to the radius of the earth, is:

APPLIED — Class XI Part 1 · Ch 7 · **Page 134** Open PDF ↗

(1) mgR (2) $2mgR$

(3) $\dfrac{1}{2}mgR$ (4) $\dfrac{3}{2}mgR$

4. NEET 2017 · Q136 Two astronauts are floating in gravitational free space after having lost contact with their spaceship. The two will:

APPLIED — Class XI Part 1 · Ch 7 · **Page 129** · law of gravitation Open PDF ↗

(1) move towards each other.

(2) move away from each other.

(3) will become stationary.

(4) keep floating at the same distance between them.

5. NEET 2016 · Q7 At what height from the surface of earth the gravitation potential and the value of g are -5.4×10^7 J kg^{-2} and 6.0 m s^{-2} respectively? Take the radius of earth as 6400 km:

APPLIED — Class XI Part 1 · Ch 7 · **Page 134** · V=-GM/r, g=GM/r^2 Open PDF ↗

(1) 2600 km (2) 1600 km

(3) 1400 km (4) 2000 km

Kepler's laws and planetary motion 3 Qs

6. NEET 2026 · Q14 In a solar system, the time-period of revolution of a planet tracing a circular orbit of radius R is proportional to:

DIRECT Class XI Part 1 · Ch 7 · **Page 129** · T^2 proportional to a^3 Open PDF ↗

(1) $R^{1/2}$ **(2)** $R^{3/2}$ **(3)** R^2 **(4)** R^3

7. NEET 2025 · Q31 The radius of Martian orbit around the Sun is about 4 times the radius of the orbit of Mercury. The Martian year is 687 Earth days. Then which of the following is the length of 1 year on Mercury?

DIRECT Class XI Part 1 · Ch 7 · **Page 129** · T^2 proportional to a^3 Open PDF ↗

(1) 88 earth days **(2)** 225 earth days
(3) 172 earth days **(4)** 124 earth days

8. NEET 2018 · Q39 The kinetic energies of a planet in an elliptical orbit about the Sun, at positions A, B and C are K_A, K_B and K_C, respectively. AC is the major axis and SB is perpendicular to AC at the position of the Sun S as shown in the figure. Then



DIRECT Class XI Part 1 · Ch 7 · **Page 129** · Example 7.1 Open PDF ↗

(1) $K_B < K_A < K_C$ **(2)** $K_A > K_B > K_C$
(3) $K_A < K_B < K_C$ **(4)** $K_B > K_A > K_C$

Acceleration due to gravity (g, variation) 6 Qs

9. NEET 2025 · Q32 A body weighs 48 N on the surface of the earth. The gravitational force experienced by the body due to the earth at a height equal to one-third the radius of the earth from its surface is:

DIRECT Class XI Part 1 · Ch 7 · **Page 133** · g(h)=GM/(R+h)^2 Open PDF ↗

(1) 16 N **(2)** 27 N **(3)** 32 N **(4)** 36 N

10. NEET 2024 · Q33 The mass of a planet is $\frac{1}{10}$th that of the earth and its diameter is half that of the earth. The acceleration due to gravity on that planet is:

APPLIED Class XI Part 1 · Ch 7 · **Page 133** Open PDF ↗

(1) 3.92 m s^{-2} **(2)** 19.6 m s^{-2}
(3) 9.8 m s^{-2} **(4)** 4.9 m s^{-2}

11. NEET 2020 · Q143 A body weighs 72 N on the surface of the earth. What is the gravitational force on it, at a height equal to half the radius of the earth?

DIRECT Class XI Part 1 · Ch 7 · **Page 143** · Exercise 7.15 Open PDF ↗

(1) 48 N **(2)** 32 N **(3)** 30 N **(4)** 24 N

12. NEET 2019 · Q16 A body weighs 200 N on the surface of the earth. How much will it weigh half way down to the centre of the earth?

DIRECT Class XI Part 1 · Ch 7 · **Page 143** · Exercise 7.16 Open PDF ↗

(1) 150 N **(2)** 200 N **(3)** 250 N **(4)** 100 N

13. NEET 2018 · Q40 If the mass of the Sun were ten times smaller and the universal gravitational constant were ten times larger in magnitude, which of the following is *not* correct?

APPLIED Class XI Part 1 · Ch 7 · **Page 133** Open PDF ↗

(1) Time period of a simple pendulum on the Earth would decrease.
(2) Walking on the ground would become more difficult.
(3) Raindrops will fall faster.
(4) 'g' on the Earth will not change.

14. NEET 2017 · Q166 The acceleration due to gravity at a height 1 km above the earth is the same as at a depth d below the surface of earth. Then:

DIRECT Class XI Part 1 · Ch 7 · **Page 133** · g(h) and g(d) Open PDF ↗

(1) $d = 1\,\text{km}$ (2) $d = \frac{3}{2}\,\text{km}$

(3) $d = 2\,\text{km}$ (4) $d = \frac{1}{2}\,\text{km}$

Satellites, orbital and escape speed 6 Qs

15. NEET 2026 · Q13 Two planets P_1 and P_2 with equal mass have radii R_1 and R_2, respectively, where $R_2 = \frac{R_1}{2}$. The escape speeds of P_1 and P_2 are v_1 and v_2, respectively. Then $\frac{v_2}{v_1}$ is:

DIRECT Class XI Part 1 · Ch 7 · **Page 136** · v=sqrt(2GM/R) Open PDF ↗

(1) $\frac{1}{\sqrt{2}}$ (2) 1 (3) $\sqrt{2}$ (4) 2

16. NEET 2024 · Q48 The minimum energy required to launch a satellite of mass m from the surface of earth of mass M and radius R in a circular orbit at an altitude of $2R$ from the surface of the earth is:

APPLIED Class XI Part 1 · Ch 7 · **Page 138** Open PDF ↗

(1) $\frac{GmM}{3R}$ (2) $\frac{5GmM}{6R}$

(3) $\frac{2GmM}{3R}$ (4) $\frac{GmM}{2R}$

17. NEET 2023 · Q43 A satellite is orbiting just above the surface of the earth with period T. If d is the density of the earth and G is the universal constant of gravitation, the quantity $\frac{3\pi}{Gd}$ represents:

APPLIED Class XI Part 1 · Ch 7 · **Page 137** Open PDF ↗

(1) $\sqrt{T}$ (2) T (3) T^2 (4) T^3

18. NEET 2021 · Q21 The escape velocity from the Earth's surface is v. The escape velocity from the surface of another planet having a radius, four times that of Earth and same mass density is:

APPLIED Class XI Part 1 · Ch 7 · **Page 136** Open PDF ↗

(1) v (2) $2v$ (3) $3v$ (4) $4v$

19. NEET 2021 · Q48 A particle of mass 'm' is projected with a velocity $v = kV_e$ ($k < 1$) from the surface of the earth. (V_e = escape velocity). The maximum height above the surface reached by the particle is:

APPLIED Class XI Part 1 · Ch 7 · **Page 136** Open PDF ↗

(1) $R\left(\frac{k}{1-k}\right)^2$ (2) $R\left(\frac{k}{1+k}\right)^2$

(3) $\frac{R^2k}{1+k}$ (4) $\frac{Rk^2}{1-k^2}$

20. NEET 2016 · Q36 The ratio of escape velocity at earth (v_e) to the escape velocity at a planet (v_p) whose radius and mean density are twice as that of earth is:

APPLIED Class XI Part 1 · Ch 7 · **Page 136** · v_e=sqrt(2GM/R) Open PDF ↗

(1) 1:2 (2) $1 : 2\sqrt{2}$

(3) 1:4 (4) $1 : \sqrt{2}$

Gravitation · Answer Key

1 (4)	**2** (3)	**3** (3)	**4** (1)	**5** (1)	**6** (2)	**7** (1)	**8** (2)
9 (2)	**10** (1)	**11** (2)	**12** (4)	**13** (4)	**14** (3)	**15** (3)	**16** (2)
17 (3)	**18** (4)	**19** (4)	**20** (2)				

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Gravitation

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	2	2	2	2	1	2	1	2	2	2	2	**20**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Gravitation (force, field, energy)	**5**	1	5	**16 17** 18 **19** 20 21 **22 23** 24 25 26
Kepler's laws and planetary motion	**3**	3	3	16 17 **18** 19 20 21 22 23 24 **25 26**
Acceleration due to gravity (g, variation)	**6**	4	6	16 **17 18 19 20** 21 22 23 **24 25** 26
Satellites, orbital and escape speed	**6**	1	5	**16** 17 18 19 20 **21** 22 **23 24** 25 **26**

Direct vs Applied

Direct from NCERT	**9**	45%
text, table, figure	6	
solved Example	1	
Exercise	2	
Applied	**11**	55%
Easy / Mod. / Hard	12 / 8 / 0	

What to focus on

- 20 questions in 11 papers, about 1.8 per paper; rank 8 of 28 chapters by number of questions.
- Most asked topic: **Acceleration due to gravity (g, variation)** (6 questions, in 6 of the 11 years).
- 3 questions are NCERT solved Examples or Exercises: Example 7.1 (Q 8), Exercise 7.15 (Q 11), Exercise 7.16 (Q 12). Work these by hand.
- NCERT pages that gave three or more questions: page 136 (5 questions), page 129 (4 questions), page 133 (4 questions), page 134 (3 questions). Read these pages line by line.
- Difficulty: 12 easy, 8 moderate, 0 hard.
- 11 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XI · Chapter 8

Mechanical Properties of Solids

8 questions · 1 topics · NEET 2016–2026

Elasticity (Young's modulus) 8 Qs

1. NEET 2024 · Q34 The maximum elongation of a steel wire of 1 m length if the elastic limit of steel and its Young's modulus, respectively, are 8×10^8 N m^{-2} and 2×10^{11} N m^{-2}, is:

DIRECT Class XI Part 2 · Ch 8 · **Page 170** · Y=FL/(A dL) Open PDF ↗

(1) 8 mm (2) 4 mm
(3) 0.4 mm (4) 40 mm

2. NEET 2023 · Q10 Let a wire be suspended from the ceiling (rigid support) and stretched by a weight W attached at its free end. The longitudinal stress at any point of cross-sectional area A of the wire is:

DIRECT Class XI Part 2 · Ch 8 · **Page 168** · stress=F/A Open PDF ↗

(1) Zero (2) $2W/A$
(3) W/A (4) $W/2A$

3. NEET 2022 · Q43 Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): The stretching of a spring is determined by the shear modulus of the material of the spring.
Reason (R): A coil spring of copper has more tensile strength than a steel spring of same dimensions.
In the light of the above statements, choose the most appropriate answer from the options given below:

APPLIED Class XI Part 2 · Ch 8 · **Page 172** Open PDF ↗

(1) (A) is false but (R) is true
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(4) (A) is true but (R) is false

4. NEET 2020 · Q161 A wire of length L, area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is:

DIRECT Class XI Part 2 · Ch 8 · **Page 170** · Y=FL/(A dL) Open PDF ↗

(1) $\dfrac{MgL_1}{AL}$ (2) $\dfrac{Mg(L_1 - L)}{AL}$
(3) $\dfrac{MgL}{AL_1}$ (4) $\dfrac{MgL}{A(L_1 - L)}$

5. NEET 2020 · Q167 Dimensions of stress are:

DIRECT Class XI Part 2 · Ch 8 · **Page 168** · stress unit Open PDF ↗

(1) $[MLT^{-2}]$ (2) $[ML^2T^{-2}]$
(3) $[ML^0T^{-2}]$ (4) $[ML^{-1}T^{-2}]$

6. NEET 2019 · Q1 When a block of mass M is suspended by a long wire of length L, the length of the wire becomes $(L + l)$. The elastic potential energy stored in the extended wire is:

DIRECT Class XI Part 2 · Ch 8 · **Page 174** · elastic PE = (1/2) F x extension Open PDF ↗

(1) Mgl (2) MgL
(3) $\frac{1}{2}Mgl$ (4) $\frac{1}{2}MgL$

7. NEET 2018 · Q44 Two wires are made of the same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area 3A. If the length of the first wire is increased by Δl on applying a force F, how much force is needed to stretch the second wire by the same amount?

APPLIED Class XI Part 2 · Ch 8 · **Page 170** Open PDF ↗

(1) 4F (2) 6F (3) 9F (4) F

8. **NEET 2017 · Q164** The bulk modulus of a spherical object is 'B'. If it is subjected to uniform pressure 'p', the fractional decrease in radius is:

APPLIED

Class XI Part 2 · Ch 8 · **Page 173** · bulk modulus
Open PDF ↗

(1) $\frac{B}{3p}$ **(2)** $\frac{3p}{B}$ **(3)** $\frac{p}{3B}$ **(4)** $\frac{p}{B}$

Mechanical Properties of Solids · Answer Key

1 (2)	**2** (3)	**3** (4)	**4** (4)	**5** (4)	**6** (3)	**7** (3)	**8** (3)

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Mechanical Properties of Solids

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	0	1	1	1	2	0	1	1	1	0	0	**8**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Elasticity (Young's modulus)	**8**	5	7	16 **17 18 19 20** 21 **22 23 24** 25 26

Direct vs Applied

Direct from NCERT	5	62%
text, table, figure	5	
solved Example	0	
Exercise	0	
Applied	**3**	38%
Easy / Mod. / Hard	6 / 2 / 0	

What to focus on

- 8 questions in 11 papers, about 0.7 per paper; rank 26 of 28 chapters by number of questions.
- Most asked topic: **Elasticity (Young's modulus)** (8 questions, in 7 of the 11 years).
- NCERT pages that gave three or more questions: page 170 (3 questions). Read these pages line by line.
- Difficulty: 6 easy, 2 moderate, 0 hard.
- 3 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

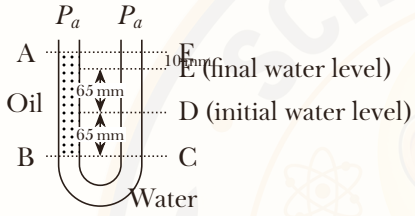
Class XI · Chapter 9

Mechanical Properties of Fluids

15 questions · 4 topics · NEET 2016–2026

Pressure, Pascal's law and hydrostatics 2 Qs

1. **NEET 2017 · Q176** A U tube with both ends open to the atmosphere, is partially filled with water. Oil, which is immiscible with water, is poured into one side until it stands at a distance of 10 mm above the water level on the other side. Meanwhile the water rises by 65 mm from its original level (see diagram). The density of the oil is:



APPLIED Class XI Part 2 · Ch 9 · **Page 184** · equal pressure at the same level Open PDF ↗

(1) 425 kg m^{-3} **(2)** 800 kg m^{-3}
(3) 928 kg m^{-3} **(4)** 650 kg m^{-3}

2. **NEET 2016 · Q33** Two non-mixing liquids of densities ρ and $n\rho$ ($n > 1$) are put in a container. The height of each liquid is h. A solid cylinder of length L and density d is put in this container. The cylinder floats with its axis vertical and length pL ($p < 1$) in the denser liquid. The density d is equal to:

APPLIED Class XI Part 2 · Ch 9 · **Page 184** · upthrust from both liquids Open PDF ↗

(1) $\{1 + (n + 1)p\}\rho$ **(2)** $\{2 + (n + 1)p\}\rho$
(3) $\{2 + (n - 1)p\}\rho$ **(4)** $\{1 + (n - 1)p\}\rho$

Bernoulli's principle and fluid flow 3 Qs

3. **NEET 2026 · Q10** Water flows in a streamline motion through a horizontal pipe of circular cross-section as shown in the figure. The pressure difference of water between P and Q is 15 N m^{-2}. The area of cross-section at P and Q are 40 cm^2 and 20 cm^2, respectively. The rate of flow of water through the pipe, in cm^3 s^{-1}, is: [Take density of water = 1000 kg m^{-3}]



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(1) 100 **(2)** 200 **(3)** 300 **(4)** 400

4. **NEET 2023 · Q34** The venturi-meter works on:

APPLIED Class XI Part 2 · Ch 9 · **Page 188** · venturimeter (Bernoulli) Open PDF ↗

(1) The principle of perpendicular axes
(2) Huygen's principle
(3) Bernoulli's principle
(4) The principle of parallel axes

5. **NEET 2019 · Q38** A small hole of area of cross-section 2 mm^2 is present near the bottom of a fully filled open tank of height 2 m. Taking $g = 10$ m/s^2, the rate of flow of water through the open hole would be nearly:

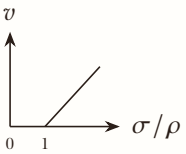
DIRECT Class XI Part 2 · Ch 9 · **Page 189** · v=sqrt(2gh) Open PDF ↗

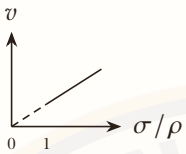
(1) 12.6×10^{-6} m^3/s
(2) 8.9×10^{-6} m^3/s
(3) 2.23×10^{-6} m^3/s
(4) 6.4×10^{-6} m^3/s

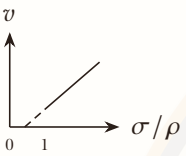
Viscosity and Stokes' law 4 Qs

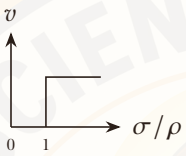
6. NEET 2026 · Q12 In the measurement of viscosity of liquids using terminal velocity experiment, spherical balls of same radius but having different densities are used. The variation of the terminal velocity (v) with the ratio of density of spherical ball (σ) to density of the liquid (ρ), is best represented by:

DIRECT Class XI Part 2 · Ch 9 · **Page 192** · vt=2a^2(rho-sigma)g/9eta Open PDF ↗

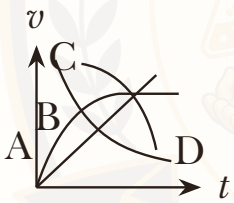
(1) 

(2) 

(3) 

(4) 

7. NEET 2022 · Q21 A spherical ball is dropped in a long column of a highly viscous liquid. The curve in the graph shown, which represents the speed of the ball (v) as a function of time (t) is:



DIRECT Class XI Part 2 · Ch 9 · **Page 192** · terminal velocity Open PDF ↗

(1) D (2) A (3) B (4) C

8. NEET 2021 · Q12 The velocity of a small ball of mass M and density d, when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $\frac{d}{2}$, then the viscous force acting on the ball will be:

APPLIED Class XI Part 2 · Ch 9 · **Page 192** Open PDF ↗

(1) $\frac{Mg}{2}$ (2) Mg (3) $\frac{3}{2}Mg$ (4) $2Mg$

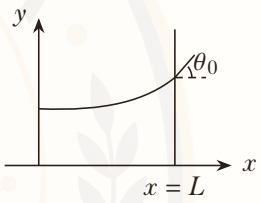
9. NEET 2018 · Q42 A small sphere of radius 'r' falls from rest in a viscous liquid. As a result, heat is produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity, is proportional to

APPLIED Class XI Part 2 · Ch 9 · **Page 192** Open PDF ↗

(1) r^5 (2) r^2 (3) r^3 (4) r^4

Surface tension and capillarity 6 Qs

10. NEET 2025 · Q1 Consider a water tank shown in the figure. It has one wall at $x = L$ and can be taken to be very wide in the z direction. When filled with a liquid of surface tension S and density ρ, the liquid surface makes angle θ_0 ($\theta_0 \ll 1$) with the x-axis at $x = L$. If $y(x)$ is the height of the surface then the equation for $y(x)$ is: (take $\theta(x) = \sin\theta(x) = \tan\theta(x) = \frac{dy}{dx}$, g is the acceleration due to gravity)



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(1) $\frac{d^2y}{dx^2} = \frac{\rho g}{S}x$ (2) $\frac{d^2y}{dx^2} = \frac{\rho g}{S}y$

(3) $\frac{d^2y}{dx^2} = \sqrt{\frac{\rho g}{S}}$ (4) $\frac{dy}{dx} = \sqrt{\frac{\rho g}{S}}\,x$

11. NEET 2024 · Q15 A thin flat circular disc of radius 4.5 cm is placed gently over the surface of water. If surface tension of water is 0.07 N m^{-1}, then the excess force required to take it away from the surface is:

APPLIED Class XI Part 2 · Ch 9 · **Page 195** Open PDF ↗

(1) 99 N (2) 19.8 mN
(3) 198 N (4) 1.98 mN

12. **NEET 2023 · Q27** The amount of energy required to form a soap bubble of radius 2 cm from a soap solution is nearly: (surface tension of soap solution = 0.03 N m^{-1})

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(1) 50.1×10^{-4} J **(2)** 30.16×10^{-4} J
(3) 5.06×10^{-4} J **(4)** 3.01×10^{-4} J

13. **NEET 2022 · Q18** If a soap bubble expands, the pressure inside the bubble:

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(1) is equal to the atmospheric pressure
(2) decreases
(3) increases
(4) remains the same

14. **NEET 2020 · Q178** A capillary tube of radius r is immersed in water and water rises in it to a height h. The mass of the water in the capillary is 5 g. Another capillary tube of radius $2r$ is immersed in water. The mass of water that will rise in this tube is:

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(1) 2.5 g **(2)** 5.0 g **(3)** 10.0 g **(4)** 20.0 g

15. **NEET 2019 · Q7** A soap bubble, having radius of 1 mm, is blown from a detergent solution having a surface tension of 2.5×10^{-2} N/m. The pressure inside the bubble equals at a point Z_0 below the free surface of water in a container. Taking $g = 10$ m/s^2, density of water = 10^3 kg/m^3, the value of Z_0 is:

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(1) 100 cm **(2)** 10 cm
(3) 1 cm **(4)** 0.5 cm

Mechanical Properties of Fluids · Answer Key

1 (3)	**2** (4)	**3** (4)	**4** (3)	**5** (1)	**6** (1)	**7** (3)	**8** (1)
9 (1)	**10** (2)	**11** (2)	**12** (4)	**13** (2)	**14** (3)	**15** (3)	

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Mechanical Properties of Fluids

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	1	1	2	1	1	2	2	1	1	2	**15**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Pressure, Pascal's law and hydrostatics	**2**	0	2	**16 17** 18 19 20 21 22 23 24 25 26
Bernoulli's principle and fluid flow	**3**	1	3	16 17 18 **19** 20 21 22 **23** 24 25 **26**
Viscosity and Stokes' law	**4**	2	4	16 17 **18** 19 20 **21 22** 23 24 25 **26**
Surface tension and capillarity	**6**	0	6	16 17 18 **19 20** 21 **22 23 24 25** 26

Direct vs Applied

Direct from NCERT	**3**	20%
text, table, figure	3	
solved Example	0	
Exercise	0	
Applied	**12**	80%
Easy / Mod. / Hard	5 / 9 / 1	

What to focus on

- 15 questions in 11 papers, about 1.4 per paper; rank 15 of 28 chapters by number of questions.
- Most asked topic: **Surface tension and capillarity** (6 questions, in 6 of the 11 years).
- NCERT pages that gave three or more questions: page 192 (4 questions), page 196 (4 questions). Read these pages line by line.
- Difficulty: 5 easy, 9 moderate, 1 hard (hard: Q 10).
- 12 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XI · Chapter 10

Thermal Properties of Matter

13 questions · 3 topics · NEET 2016–2026

Thermal expansion 4 Qs

1. NEET 2026 · Q24 The temperature of a metallic sphere of radius R is increased by a small amount ΔT. If the linear coefficient of thermal expansion of the metal is α, the approximate increase in the volume of the sphere is:

DIRECT Class XI Part 2 · Ch 10 · **Page 207** · alpha_V=3 alpha_1 Open PDF ↗

(1) $2\pi R^3\alpha\Delta T$ (2) $3\pi R^3\alpha\Delta T$
(3) $4\pi R^3\alpha\Delta T$ (4) $6\pi R^3\alpha\Delta T$

2. NEET 2024 · Q43 A metallic bar of Young's modulus, 0.5×10^{11} N m^{-2} and coefficient of linear thermal expansion 10^{-5} °C^{-1}, length 1 m and area of cross-section 10^{-3} m^2 is heated from 0°C to 100°C without expansion or bending. The compressive force developed in it is:

DIRECT Class XI Part 2 · Ch 10 · **Page 207** · thermal stress Open PDF ↗

(1) 2×10^3 N (2) 5×10^3 N
(3) 50×10^3 N (4) 100×10^3 N

3. NEET 2019 · Q14 A copper rod of 88 cm and an aluminium rod of unknown length have their increase in length independent of increase in temperature. The length of aluminium rod is: ($\alpha_{Cu} = 1.7 \times 10^{-5}$ K^{-1} and $\alpha_{Al} = 2.2 \times 10^{-5}$ K^{-1})

APPLIED Class XI Part 2 · Ch 10 · **Page 205** Open PDF ↗

(1) 6.8 cm (2) 113.9 cm
(3) 88 cm (4) 68 cm

4. NEET 2016 · Q16 Coefficient of linear expansion of brass and steel rods are α_1 and α_2. Lengths of brass and steel rods are l_1 and l_2 respectively. If $(l_2 - l_1)$ is maintained same at all temperatures, which one of the following relations holds good?

APPLIED Class XI Part 2 · Ch 10 · **Page 207** · linear expansion Open PDF ↗

(1) $\alpha_1 l_2 = \alpha_2 l_1$ (2) $\alpha_1 l_2^2 = \alpha_2 l_1^2$
(3) $\alpha_1^2 l_2 = \alpha_2^2 l_1$ (4) $\alpha_1 l_1 = \alpha_2 l_2$

Calorimetry and latent heat 2 Qs

5. NEET 2020 · Q174 The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 ($r_1 = 1.5r_2$) through 1 K are in the ratio:

APPLIED Class XI Part 2 · Ch 10 · **Page 208** Open PDF ↗

(1) $\frac{27}{8}$ (2) $\frac{9}{4}$ (3) $\frac{3}{2}$ (4) $\frac{5}{3}$

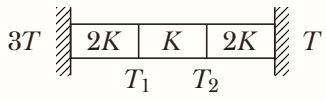
6. NEET 2016 · Q35 A piece of ice falls from a height h so that it melts completely. Only one-quarter of the heat produced is absorbed by the ice and all energy of ice gets converted into heat during its fall. The value of h is: [Latent heat of ice is 3.4×10^5 J/kg and $g = 10$ N/kg]

APPLIED Class XI Part 2 · Ch 10 · **Page 212** · latent heat Open PDF ↗

(1) 34 km (2) 544 km
(3) 136 km (4) 68 km

Heat transfer (conduction, radiation, cooling) 7 Qs

7. **NEET 2025 · Q19** Three identical heat conducting rods are connected in series as shown in the figure. The rods on the sides have thermal conductivity $2K$ while that in the middle has thermal conductivity K. The left end of the combination is maintained at temperature $3T$ and the right end at T. The rods are thermally insulated from outside. In steady state, temperature at the left junction is T_1 and that at the right junction is T_2. The ratio T_1/T_2 is



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(1) $\frac{3}{2}$ (2) $\frac{4}{3}$ (3) $\frac{5}{3}$ (4) $\frac{5}{4}$

8. **NEET 2021 · Q10** A cup of coffee cools from 90°C to 80°C in t minutes, when the room temperature is 20°C. The time taken by a similar cup of coffee to cool from 80°C to 60°C at a room temperature same at 20°C is:

DIRECT Class XI Part 2 · Ch 10 · **Page 221** · Example 10.8 Open PDF ↗

(1) $\frac{13}{10}t$ (2) $\frac{13}{5}t$ (3) $\frac{10}{13}t$ (4) $\frac{5}{13}t$

9. **NEET 2019 · Q20** The unit of thermal conductivity is:

DIRECT Class XI Part 2 · Ch 10 · **Page 214** · unit of K Open PDF ↗

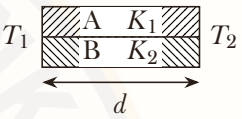
(1) $\mathrm{J\,m\,K^{-1}}$ (2) $\mathrm{J\,m^{-1}K^{-1}}$
(3) $\mathrm{W\,m\,K^{-1}}$ (4) $\mathrm{W\,m^{-1}K^{-1}}$

10. **NEET 2018 · Q43** The power radiated by a black body is P and it radiates maximum energy at wavelength, λ_0. If the temperature of the black body is now changed so that it radiates maximum energy at wavelength $\frac{3}{4}\lambda_0$, the power radiated by it becomes nP. The value of n is

APPLIED Class XI Part 2 · Ch 10 · **Page 219** Open PDF ↗

(1) $\frac{256}{81}$ (2) $\frac{4}{3}$ (3) $\frac{3}{4}$ (4) $\frac{81}{256}$

11. **NEET 2017 · Q146** Two rods A and B of different materials are welded together as shown in figure. Their thermal conductivities are K_1 and K_2. The thermal conductivity of the composite rod will be:



APPLIED Class XI Part 2 · Ch 10 · **Page 216** · rods side by side (parallel) Open PDF ↗

(1) $\frac{3(K_1+K_2)}{2}$ (2) K_1+K_2
(3) $2(K_1+K_2)$ (4) $\frac{K_1+K_2}{2}$

12. **NEET 2017 · Q160** A spherical black body with a radius of 12 cm radiates 450 watt power at 500 K. If the radius were halved and the temperature doubled, the power radiated in watt would be:

APPLIED Class XI Part 2 · Ch 10 · **Page 219** · Stefan-Boltzmann law Open PDF ↗

(1) 450 (2) 1000 (3) 1800 (4) 225

13. **NEET 2016 · Q15** A black body is at a temperature of 5760 K. The energy of radiation emitted by the body at wavelength 250 nm is U_1, at wavelength 500 nm is U_2 and that at 1000 nm is U_3. Wien's constant, $b = 2.88 \times 10^6$ nm K. Which of the following is correct?

APPLIED Class XI Part 2 · Ch 10 · **Page 218** · Wien's law Open PDF ↗

(1) $U_1 = 0$ (2) $U_3 = 0$
(3) $U_1 > U_2$ (4) $U_2 > U_1$

Thermal Properties of Matter · Answer Key

1 (3)	**2** (3)	**3** (4)	**4** (4)	**5** (1)	**6** (3)	**7** (3)	**8** (2)
9 (4)	**10** (1)	**11** (4)	**12** (3)	**13** (4)			

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Thermal Properties of Matter

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	3	2	1	2	1	1	0	0	1	1	1	**13**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Thermal expansion	**4**	2	4	**16** 17 18 **19** 20 21 22 23 **24** 25 **26**
Calorimetry and latent heat	**2**	0	2	**16** 17 18 19 **20** 21 22 23 24 25 26
Heat transfer (conduction, radiation, cooling)	**7**	2	6	**16** **17** **18** **19** 20 **21** 22 23 24 **25** 26

Direct vs Applied

Direct from NCERT	**4**	31%
text, table, figure	3	
solved Example	1	
Exercise	0	
Applied	**9**	69%
Easy / Mod. / Hard	6 / 7 / 0	

What to focus on

- 13 questions in 11 papers, about 1.2 per paper; rank 21 of 28 chapters by number of questions.
- Most asked topic: **Heat transfer (conduction, radiation, cooling)** (7 questions, in 6 of the 11 years).
- 1 questions are NCERT solved Examples or Exercises: Example 10.8 (Q 8). Work these by hand.
- NCERT pages that gave three or more questions: page 207 (3 questions). Read these pages line by line.
- Difficulty: 6 easy, 7 moderate, 0 hard.
- 9 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XI · Chapter 11

Thermodynamics

15 questions · 3 topics · NEET 2016–2026

First law of thermodynamics 2 Qs

1. NEET 2025 · Q41 Two gases A and B are filled at the same pressure in separate cylinders with movable pistons of radius r_A and r_B, respectively. On supplying an equal amount of heat to both the systems reversibly under constant pressure, the pistons of gas A and B are displaced by 16 cm and 9 cm, respectively. If the change in their internal energy is the same, then the ratio r_A/r_B is equal to

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(1) $\frac{4}{3}$ (2) $\frac{3}{4}$ (3) $\frac{2}{\sqrt{3}}$ (4) $\frac{\sqrt{3}}{2}$

2. NEET 2018 · Q45 A sample of 0.1 g of water at 100°C and normal pressure (1.013×10^5 N m^{-2}) requires 54 cal of heat energy to convert to steam at 100°C. If the volume of the steam produced is 167.1 cc, the change in internal energy of the sample, is

APPLIED Class XI Part 2 · Ch 11 · **Page 231** Open PDF ↗

(1) 42.2 J (2) 208.7 J
(3) 104.3 J (4) 84.5 J

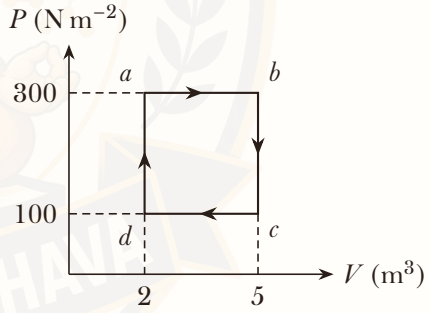
Thermodynamic processes 9 Qs

3. NEET 2026 · Q17 In an adiabatic expansion, the temperature of one mole of an ideal monatomic gas ($\gamma = 5/3$) decreases from 60 K to 50 K. The work done by the gas in the process is: (Take the universal gas constant as $R = 8.3$ J mol^{-1}K^{-1})

DIRECT Class XI Part 2 · Ch 11 · **Page 235** · W=muR(T1-T2)/(gamma-1) Open PDF ↗

(1) 41.5 J (2) 83 J
(3) 124.5 J (4) 166 J

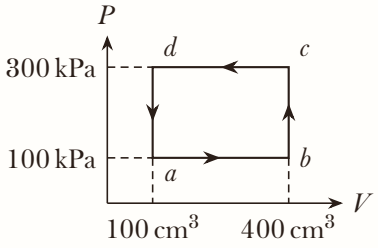
4. NEET 2026 · Q39 One mole of an ideal monatomic gas undergoes a cyclic process as shown in the figure. The total heat supplied to the gas is:



DIRECT Class XI Part 2 · Ch 11 · **Page 236** · cyclic process: Q=W=area Open PDF ↗

(1) 400 J (2) 500 J (3) 600 J (4) 800 J

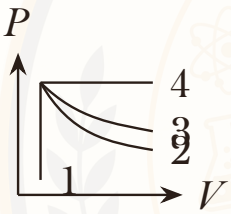
5. **NEET 2024 · Q27** A thermodynamic system is taken through the cycle *abcda*. The work done by the gas along the path *bc* is:



DIRECT Class XI Part 2 · Ch 11 · **Page 235** · isochoric: W=0 Open PDF ↗

(1) −60 J **(2)** zero **(3)** 30 J **(4)** −90 J

6. **NEET 2022 · Q12** An ideal gas undergoes four different processes from the same initial state as shown in the figure below. Those processes are adiabatic, isothermal, isobaric and isochoric. The curve which represents the adiabatic process among 1, 2, 3 and 4 is:



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(1) 4 **(2)** 1 **(3)** 2 **(4)** 3

7. **NEET 2020 · Q141** Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is:

APPLIED Class XI Part 2 · Ch 11 · **Page 236** · free expansion Open PDF ↗

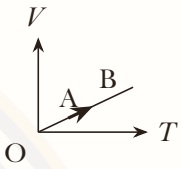
(1) isothermal **(2)** adiabatic
(3) isochoric **(4)** isobaric

8. **NEET 2019 · Q40** In which of the following processes, heat is neither absorbed nor released by a system?

DIRECT Class XI Part 2 · Ch 11 · **Page 234** · Table 11.2 Open PDF ↗

(1) isothermal **(2)** adiabatic
(3) isobaric **(4)** isochoric

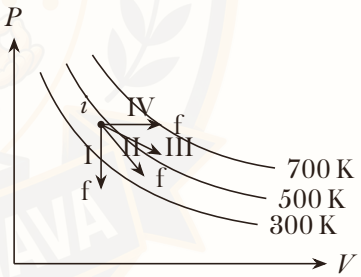
9. **NEET 2018 · Q1** The volume (V) of a monatomic gas varies with its temperature (T), as shown in the graph. The ratio of work done by the gas, to the heat absorbed by it, when it undergoes a change from state A to state B, is



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(1) $\frac{1}{3}$ **(2)** $\frac{2}{3}$ **(3)** $\frac{2}{5}$ **(4)** $\frac{2}{7}$

10. **NEET 2017 · Q155** Thermodynamic processes are indicated in the following diagram.



Match the following:

P.	Process I	a.	Adiabatic
Q.	Process II	b.	Isobaric
R.	Process III	c.	Isochoric
S.	Process IV	d.	Isothermal

APPLIED Class XI Part 2 · Ch 11 · **Page 232** Open PDF ↗

(1) P→c, Q→a, R→d, S→b
(2) P→c, Q→d, R→b, S→a
(3) P→d, Q→b, R→a, S→c
(4) P→a, Q→c, R→d, S→b

11. **NEET 2016 · Q30** A gas is compressed isothermally to half its initial volume. The same gas is compressed separately through an adiabatic process until its volume is again reduced to half. Then:

APPLIED Class XI Part 2 · Ch 11 · **Page 233** · adiabatic curve is steeper Open PDF ↗

(1) Compressing the gas isothermally will require more work to be done.
(2) Compressing the gas through adiabatic process will require more work to be done.
(3) Compressing the gas isothermally or adiabatically will require the same amount of work.
(4) Which of the case (whether compression through isothermal or through adiabatic process) requires more work will depend upon the atomicity of the gas.

Second law and heat engines — 4 Qs

12. **NEET 2023 · Q26** A Carnot engine has an efficiency of 50% when its source is at a temperature 327°C. The temperature of the sink is:

DIRECT Class XI Part 2 · Ch 11 · **Page 238** · eta=1-T2/T1 Open PDF ↗

(1) 200°C **(2)** 27°C
(3) 15°C **(4)** 100°C

13. **NEET 2018 · Q4** The efficiency of an ideal heat engine working between the freezing point and boiling point of water, is

DIRECT Class XI Part 2 · Ch 11 · **Page 238** · eta=1-T2/T1 Open PDF ↗

(1) 6.25% **(2)** 20% **(3)** 26.8% **(4)** 12.5%

14. **NEET 2017 · Q180** A carnot engine having an efficiency of $\frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is:

APPLIED Class XI Part 2 · Ch 11 · **Page 236** · refrigerator Open PDF ↗

(1) 90 J **(2)** 99 J **(3)** 100 J **(4)** 1 J

15. **NEET 2016 · Q11** A refrigerator works between 4°C and 30°C. It is required to remove 600 calories of heat every second in order to keep the temperature of the refrigerated space constant. The power required is: (Take 1 cal = 4.2 Joules)

APPLIED Class XI Part 2 · Ch 11 · **Page 236** · refrigerator Open PDF ↗

(1) 2.365 W **(2)** 23.65 W
(3) 236.5 W **(4)** 2365 W

Thermodynamics · Answer Key

1 (2)	**2** (2)	**3** (3)	**4** (3)	**5** (2)	**6** (3)	**7** (2)	**8** (2)
9 (3)	**10** (1)	**11** (2)	**12** (2)	**13** (3)	**14** (1)	**15** (3)	

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Thermodynamics

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	2	2	3	1	1	0	1	1	1	1	2	**15**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
First law of thermodynamics	**2**	0	2	16 17 **18** 19 20 21 22 23 24 **25** 26
Thermodynamic processes	**9**	4	8	**16 17 18 19 20** 21 **22** 23 **24** 25 **26**
Second law and heat engines	**4**	2	4	**16 17 18** 19 20 21 22 **23** 24 25 26

Direct vs Applied

Direct from NCERT	**6**	40%
text, table, figure	6	
solved Example	0	
Exercise	0	
Applied	**9**	60%
Easy / Mod. / Hard	7 / 8 / 0	

What to focus on

- 15 questions in 11 papers, about 1.4 per paper; rank 14 of 28 chapters by number of questions.
- Most asked topic: **Thermodynamic processes** (9 questions, in 8 of the 11 years).
- Asked in 8 or more of the 11 years: Thermodynamic processes.
- NCERT pages that gave three or more questions: page 235 (4 questions), page 236 (4 questions). Read these pages line by line.
- Difficulty: 7 easy, 8 moderate, 0 hard.
- 9 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XI · Chapter 12

Kinetic Theory

15 questions · 1 topics · NEET 2016–2026

Kinetic theory of gases 15 Qs

1. NEET 2026 · Q2 The mean free path of molecules in an ideal gas A is half that of another ideal gas B. The diameter of the spherical molecules of gas A is twice the diameter of the molecules of B. If number densities of the gases A and B are n_A and n_B, respectively, the correct option is:

DIRECT Class XI Part 2 · Ch 12 · **Page 254** · l=1/(sqrt2 n pi d^2) Open PDF ↗

(1) $n_A = n_B$ **(2)** $n_A = 2n_B$

(3) $n_A = \frac{1}{4}n_B$ **(4)** $n_A = \frac{1}{2}n_B$

2. NEET 2026 · Q45 An ideal gas is made of polyatomic molecules. Each of the molecules has three translational, three rotational and f number of vibrational modes. If the ratio of heat capacities C_P/C_V of the gas is 8/7, then the value of f is:

DIRECT Class XI Part 2 · Ch 12 · **Page 253** · Cv=(3+f)R Open PDF ↗

(1) 4 **(2)** 3 **(3)** 2 **(4)** 1

3. NEET 2025 · Q14 An oxygen cylinder of volume 30 litre has 18.20 moles of oxygen. After some oxygen is withdrawn from the cylinder, its gauge pressure drops to 11 atmospheric pressure at temperature 27°C. The mass of the oxygen withdrawn from the cylinder is nearly equal to: [Given, $R = \frac{100}{12}\,\text{J mol}^{-1}\text{K}^{-1}$, and molecular mass of $O_2 = 32$, 1 atm pressure $= 1.01 \times 10^5\ \text{N/m}^2$]

APPLIED Class XI Part 2 · Ch 12 · **Page 246** Open PDF ↗

(1) 0.125 kg **(2)** 0.144 kg

(3) 0.116 kg **(4)** 0.156 kg

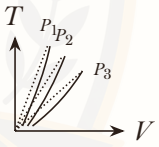
4. NEET 2025 · Q29 A container has two chambers of volumes $V_1 = 2$ litres and $V_2 = 3$ litres separated by a partition made of a thermal insulator. The chambers contain $n_1 = 5$ and $n_2 = 4$ moles of ideal gas at pressures $p_1 = 1$ atm and $p_2 = 2$ atm, respectively. When the partition is removed, the mixture attains an equilibrium pressure of:

APPLIED Class XI Part 2 · Ch 12 · **Page 247** Open PDF ↗

(1) 1.3 atm **(2)** 1.6 atm

(3) 1.4 atm **(4)** 1.8 atm

5. NEET 2024 · Q49 The following graph represents the T-V curves of an ideal gas (where T is the temperature and V the volume) at three pressures P_1, P_2 and P_3 compared with those of Charles's law represented as dotted lines.



Then the correct relation is:

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(1) $P_1 > P_2 > P_3$ **(2)** $P_3 > P_2 > P_1$

(3) $P_1 > P_3 > P_2$ **(4)** $P_2 > P_1 > P_3$

6. NEET 2023 · Q12 The temperature of a gas is −50°C. To what temperature the gas should be heated so that the rms speed is increased by 3 times?

APPLIED Class XI Part 2 · Ch 12 · **Page 250** Open PDF ↗

(1) 223 K **(2)** 669°C

(3) 3295°C **(4)** 3097 K

7. **NEET 2022 · Q37** The volume occupied by the molecules contained in 4.5 kg water at STP, if the intermolecular forces vanish away is:

APPLIED Class XI Part 2 · Ch 12 · **Page 246** Open PDF ↗

(1) $5.6\,\text{m}^3$ **(2)** $5.6 \times 10^6\,\text{m}^3$
(3) $5.6 \times 10^3\,\text{m}^3$ **(4)** $5.6 \times 10^{-3}\,\text{m}^3$

8. **NEET 2021 · Q7** Match Column-I and Column-II and choose the correct match from the given choices.

Column-I		*Column-II*	
(A)	Root mean square speed of gas molecules	(P)	$\frac{1}{3}nm\bar{v}^2$
(B)	Pressure exerted by ideal gas	(Q)	$\sqrt{\frac{3RT}{M}}$
(C)	Average kinetic energy of a molecule	(R)	$\frac{5}{2}RT$
(D)	Total internal energy of 1 mole of a diatomic gas	(S)	$\frac{3}{2}k_BT$

DIRECT Class XI Part 2 · Ch 12 · **Page 250** · vrms, P=(1/3)nm<v^2>, (3/2)kBT Open PDF ↗

(1) (A)-(R), (B)-(P), (C)-(S), (D)-(Q)
(2) (A)-(Q), (B)-(R), (C)-(S), (D)-(P)
(3) (A)-(Q), (B)-(P), (C)-(S), (D)-(R)
(4) (A)-(R), (B)-(Q), (C)-(P), (D)-(S)

9. **NEET 2020 · Q139** A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C. Its density is: ($R = 8.3\,\text{J mol}^{-1}\text{K}^{-1}$)

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(1) $0.5\,\text{kg/m}^3$ **(2)** $0.2\,\text{kg/m}^3$
(3) $0.1\,\text{kg/m}^3$ **(4)** $0.02\,\text{kg/m}^3$

10. **NEET 2020 · Q155** The mean free path for a gas, with molecular diameter d and number density n can be expressed as:

DIRECT Class XI Part 2 · Ch 12 · **Page 254** · l=1/(sqrt2 n pi d^2) Open PDF ↗

(1) $\frac{1}{\sqrt{2}\,n\pi d}$ **(2)** $\frac{1}{\sqrt{2}\,n\pi d^2}$
(3) $\frac{1}{\sqrt{2}\,n^2\pi d^2}$ **(4)** $\frac{1}{\sqrt{2}\,n^2\pi^2 d^2}$

11. **NEET 2020 · Q162** The average thermal energy for a mono-atomic gas is: (k_B is Boltzmann constant and T, absolute temperature)

DIRECT Class XI Part 2 · Ch 12 · **Page 250** · (3/2)kBT Open PDF ↗

(1) $\frac{1}{2}k_BT$ **(2)** $\frac{3}{2}k_BT$ **(3)** $\frac{5}{2}k_BT$ **(4)** $\frac{7}{2}k_BT$

12. **NEET 2019 · Q12** Increase in temperature of a gas filled in a container would lead to:

DIRECT Class XI Part 2 · Ch 12 · **Page 250** · KE proportional to T Open PDF ↗

(1) increase in its mass
(2) increase in its kinetic energy
(3) decrease in its pressure
(4) decrease in intermolecular distance

13. **NEET 2018 · Q3** At what temperature will the rms speed of oxygen molecules become just sufficient for escaping from the Earth's atmosphere? (Given: Mass of oxygen molecule $m = 2.76 \times 10^{-26}$ kg, Boltzmann's constant $k_B = 1.38 \times 10^{-23}\,\text{J K}^{-1}$)

APPLIED Class XI Part 2 · Ch 12 · **Page 250** Open PDF ↗

(1) 5.016×10^4 K **(2)** 8.360×10^4 K
(3) 2.508×10^4 K **(4)** 1.254×10^4 K

14. **NEET 2017 · Q163** A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T. Neglecting all vibrational modes, the total internal energy of the system is:

APPLIED Class XI Part 2 · Ch 12 · **Page 252** · equipartition Open PDF ↗

(1) $15RT$ **(2)** $9RT$ **(3)** $11RT$ **(4)** $4RT$

15. **NEET 2016 · Q25** The molecules of a given mass of a gas have r.m.s. velocity of $200\,\text{m s}^{-1}$ at 27°C and $1.0 \times 10^5\,\text{N m}^{-2}$ pressure. When the temperature and pressure of the gas are respectively, 127°C and $0.05 \times 10^5\,\text{N m}^{-2}$, the r.m.s. velocity of its molecules in m s^{-1} is:

APPLIED Class XI Part 2 · Ch 12 · **Page 252** · v_rms proportional to sqrt(T) Open PDF ↗

(1) $100\sqrt{2}$ **(2)** $\frac{400}{\sqrt{3}}$
(3) $\frac{100\sqrt{2}}{3}$ **(4)** $\frac{100}{3}$

Kinetic Theory · Answer Key

1 (4)	**2** (1)	**3** (3)	**4** (2)	**5** (1)	**6** (3)	**7** (1)	**8** (3)
9 (2)	**10** (2)	**11** (2)	**12** (2)	**13** (2)	**14** (3)	**15** (2)	

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Kinetic Theory

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	1	1	1	3	1	1	1	1	2	2	**15**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Kinetic theory of gases	**15**	6	11	**16 17 18 19 20 21 22 23 24 25 26**

Direct vs Applied

Direct from NCERT	**6**	40%
text, table, figure	6	
solved Example	0	
Exercise	0	
Applied	**9**	60%
Easy / Mod. / Hard	8 / 7 / 0	

What to focus on

- 15 questions in 11 papers, about 1.4 per paper; rank 16 of 28 chapters by number of questions.
- Most asked topic: **Kinetic theory of gases** (15 questions, in 11 of the 11 years).
- Asked in 8 or more of the 11 years: Kinetic theory of gases.
- NCERT pages that gave three or more questions: page 250 (5 questions), page 246 (4 questions). Read these pages line by line.
- Difficulty: 8 easy, 7 moderate, 0 hard.
- 9 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XI · Chapter 13

Oscillations

17 questions · 1 topics · NEET 2016–2026

Simple harmonic motion 17 Qs

1. NEET 2026 · Q25 A cylindrical cork of uniform density floats in a liquid of density ρ_1. If the cork is depressed slightly and released, it oscillates harmonically with time period T. If the same cork floats in another liquid of density ρ_2, then the similar oscillation has time period $2T$. The value of ρ_2/ρ_1 is:

DIRECT Class XI Part 2 · Ch 13 · **Page 277** · Exercise 13.17 Open PDF ↗

(1) 4 **(2)** 2 **(3)** 1/2 **(4)** 1/4

2. NEET 2026 · Q28 Consider a spring-mass simple harmonic oscillator in one dimension. The mass of the particle is m kg and the spring constant is k N m^{-1}. At a given instant, the extension of the spring is x metre and the speed of the particle is v m s^{-1}. On the x–v plane, if the graph of v as a function of x is a circle, then the correct option is:

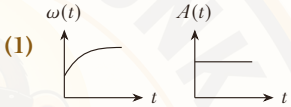
APPLIED Class XI Part 2 · Ch 13 · **Page 266** Open PDF ↗

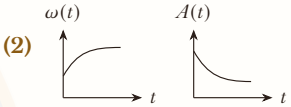
(1) $k = \dfrac{1}{m}$ **(2)** $k = m$

(3) $k = m^2$ **(4)** $k = \sqrt{m}$

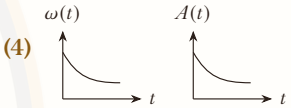
3. NEET 2025 · Q23 In an oscillating spring mass system, a spring is connected to a box filled with sand. As the box oscillates, sand leaks slowly out of the box vertically so that the average frequency $\omega(t)$ and average amplitude $A(t)$ of the system change with time t. Which one of the following options schematically depicts these changes correctly?

APPLIED Class XI Part 2 · Ch 13 · **Page 267** Open PDF ↗

(1) 

(2) 

(3) 

(4) 

4. NEET 2025 · Q44 Two identical point masses P and Q, suspended from two separate massless springs of spring constants k_1 and k_2, respectively, oscillate vertically. If their maximum speeds are the same, the ratio (A_Q/A_P) of the amplitude A_Q of mass Q to the amplitude A_P of mass P is:

APPLIED Class XI Part 2 · Ch 13 · **Page 266** Open PDF ↗

(1) $\dfrac{k_2}{k_1}$ **(2)** $\dfrac{k_1}{k_2}$ **(3)** $\sqrt{\dfrac{k_2}{k_1}}$ **(4)** $\sqrt{\dfrac{k_1}{k_2}}$

5. NEET 2024 · Q26 If $x = 5\sin\left(\pi t + \dfrac{\pi}{3}\right)$ m represents the motion of a particle executing simple harmonic motion, the amplitude and time period of motion, respectively, are:

DIRECT Class XI Part 2 · Ch 13 · **Page 264** · T=2 pi/omega Open PDF ↗

(1) 5 m, 1 s **(2)** 5 cm, 2 s

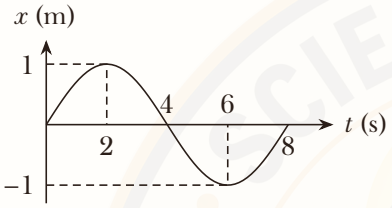
(3) 5 m, 2 s **(4)** 5 cm, 1 s

6. **NEET 2024 · Q42** If the mass of the bob in a simple pendulum is increased to thrice its original mass and its length is made half its original length, then the new time period of oscillation is $\frac{x}{2}$ times its original time period. Then the value of x is:

DIRECT Class XI Part 2 · Ch 13 · **Page 271** · T=2 pi sqrt(L/g) Open PDF ↗

(1) 4 **(2)** $\sqrt{3}$ **(3)** $\sqrt{2}$ **(4)** $2\sqrt{3}$

7. **NEET 2023 · Q42** The x-t graph of a particle performing simple harmonic motion is shown in the figure. The acceleration of the particle at $t = 2$ s is:



APPLIED Class XI Part 2 · Ch 13 · **Page 266** Open PDF ↗

(1) $-\frac{\pi^2}{16}$ m s^{-2} **(2)** $\frac{\pi^2}{8}$ m s^{-2}
(3) $-\frac{\pi^2}{8}$ m s^{-2} **(4)** $\frac{\pi^2}{16}$ m s^{-2}

8. **NEET 2022 · Q36** Two pendulums of length 121 cm and 100 cm start vibrating in phase. At some instant, the two are at their mean position in the same phase. The minimum number of vibrations of the shorter pendulum after which the two are again in phase at the mean position is:

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(1) 8 **(2)** 11 **(3)** 9 **(4)** 10

9. **NEET 2021 · Q3** A body is executing simple harmonic motion with frequency 'n', the frequency of its potential energy is:

DIRECT Class XI Part 2 · Ch 13 · **Page 269** · K and U have period T/2 Open PDF ↗

(1) n **(2)** $2n$ **(3)** $3n$ **(4)** $4n$

10. **NEET 2021 · Q15** A spring is stretched by 5 cm by a force 10 N. The time period of the oscillations when a mass of 2 kg is suspended by it is:

DIRECT Class XI Part 2 · Ch 13 · **Page 267** · omega=sqrt(k/m) Open PDF ↗

(1) 0.0628 s **(2)** 6.28 s
(3) 3.14 s **(4)** 0.628 s

11. **NEET 2020 · Q145** The phase difference between displacement and acceleration of a particle in a simple harmonic motion is:

DIRECT Class XI Part 2 · Ch 13 · **Page 266** · a=-omega^2 x Open PDF ↗

(1) π rad **(2)** $\frac{3\pi}{2}$ rad
(3) $\frac{\pi}{2}$ rad **(4)** zero

12. **NEET 2019 · Q10** The displacement of a particle executing simple harmonic motion is given by $y = A_0 + A\sin\omega t + B\cos\omega t$. Then the amplitude of its oscillation is given by:

DIRECT Class XI Part 2 · Ch 13 · **Page 262** · A sin wt + B cos wt = D sin(wt+phi) Open PDF ↗

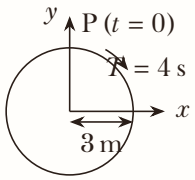
(1) $A_0 + \sqrt{A^2 + B^2}$ **(2)** $\sqrt{A^2 + B^2}$
(3) $\sqrt{A_0^2 + (A + B)^2}$ **(4)** $A + B$

13. **NEET 2019 · Q19** Average velocity of a particle executing SHM in one complete vibration is:

APPLIED Class XI Part 2 · Ch 13 · **Page 266** Open PDF ↗

(1) $\frac{A\omega}{2}$ **(2)** $A\omega$ **(3)** $\frac{A\omega^2}{2}$ **(4)** zero

14. **NEET 2019 · Q42** The radius of circle, the period of revolution, initial position and sense of revolution are indicated in the fig.



y-projection of the radius vector of rotating particle P is:

DIRECT — Class XI Part 2 · Ch 13 · **Page 265** · reference circle Open PDF ↗

(1) $y(t) = -3\cos 2\pi t$, where y in m

(2) $y(t) = 4\sin\left(\frac{\pi t}{2}\right)$, where y in m

(3) $y(t) = 3\cos\left(\frac{3\pi t}{2}\right)$, where y in m

(4) $y(t) = 3\cos\left(\frac{\pi t}{2}\right)$, where y in m

15. **NEET 2018 · Q25** A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m/s^2 at a distance of 5 m from the mean position. The time period of oscillation is

DIRECT — Class XI Part 2 · Ch 13 · **Page 266** · a=-omega^2 x Open PDF ↗

(1) 2 s **(2)** π s **(3)** 2π s **(4)** 1 s

16. **NEET 2017 · Q170** A spring of force constant k is cut into lengths of ratio 1:2:3. They are connected in series and the new force constant is k'. Then they are connected in parallel and force constant is k''. Then $k' : k''$ is:

APPLIED — Class XI Part 2 · Ch 13 · **Page 267** · Example 13.6 setting Open PDF ↗

(1) 1:9 **(2)** 1:11 **(3)** 1:14 **(4)** 1:6

17. **NEET 2017 · Q175** A particle executes linear simple harmonic motion with an amplitude of 3 cm. When the particle is at 2 cm from the mean position, the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is:

APPLIED — Class XI Part 2 · Ch 13 · **Page 266** Open PDF ↗

(1) $\frac{\sqrt{5}}{2\pi}$ **(2)** $\frac{4\pi}{\sqrt{5}}$ **(3)** $\frac{2\pi}{\sqrt{3}}$ **(4)** $\frac{\sqrt{5}}{\pi}$

Oscillations · Answer Key

1 (4)	**2** (2)	**3** (2)	**4** (4)	**5** (3)	**6** (3)	**7** (1)	**8** (2)
9 (2)	**10** (4)	**11** (1)	**12** (2)	**13** (4)	**14** (4)	**15** (2)	**16** (2)
17 (2)							

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Oscillations

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	0	2	1	3	1	2	1	1	2	2	2	**17**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Simple harmonic motion	**17**	9	10	16 **17 18 19 20 21 22 23 24 25 26**

Direct vs Applied

Direct from NCERT	**9**	53%
text, table, figure	8	
solved Example	0	
Exercise	1	
Applied	**8**	47%
Easy / Mod. / Hard	8 / 9 / 0	

What to focus on

- 17 questions in 11 papers, about 1.5 per paper; rank 13 of 28 chapters by number of questions.
- Most asked topic: **Simple harmonic motion** (17 questions, in 10 of the 11 years).
- Asked in 8 or more of the 11 years: Simple harmonic motion.
- 1 questions are NCERT solved Examples or Exercises: Exercise 13.17 (Q 1). Work these by hand.
- NCERT pages that gave three or more questions: page 266 (7 questions), page 267 (3 questions). Read these pages line by line.
- Difficulty: 8 easy, 9 moderate, 0 hard.
- 8 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XI · Chapter 14

Waves

10 questions · 2 topics · NEET 2016–2026

Wave motion and speed of waves — 2 Qs

1. NEET 2022 · Q5 If the initial tension on a stretched string is doubled, then the ratio of the initial and final speeds of a transverse wave along the string is:

DIRECT — Class XI Part 2 · Ch 14 · **Page 285** · v=sqrt(T/mu) Open PDF ↗

(1) 1:2 **(2)** 1:1 **(3)** $\sqrt{2} : 1$ **(4)** $1 : \sqrt{2}$

2. NEET 2016 · Q10 A uniform rope of length L and mass m_1 hangs vertically from a rigid support. A block of mass m_2 is attached to the free end of the rope. A transverse pulse of wavelength λ_1 is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope is λ_2. The ratio λ_2/λ_1 is:

APPLIED — Class XI Part 2 · Ch 14 · **Page 285** · v=sqrt(T/mu) Open PDF ↗

(1) $\sqrt{\frac{m_1}{m_2}}$ **(2)** $\sqrt{\frac{m_1+m_2}{m_2}}$

(3) $\sqrt{\frac{m_2}{m_1}}$ **(4)** $\sqrt{\frac{m_1+m_2}{m_1}}$

Beats, organ pipes and strings — 8 Qs

3. NEET 2026 · Q4 For sound waves, if the number of nodes for the 5th harmonic of an open-ended pipe is n and that for the 9th harmonic of the same pipe with one of its ends closed is m, the ratio $\frac{n}{m}$ is

APPLIED — Class XI Part 2 · Ch 14 · **Page 292** Open PDF ↗

(1) $\frac{5}{9}$ **(2)** $\frac{9}{5}$ **(3)** 1 **(4)** $\frac{3}{5}$

4. NEET 2025 · Q45 A pipe open at both ends has a fundamental frequency f in air. The pipe is now dipped vertically in a water drum to half of its length. The fundamental frequency of the air column is now equal to:

APPLIED — Class XI Part 2 · Ch 14 · **Page 292** Open PDF ↗

(1) $\frac{f}{2}$ **(2)** f **(3)** $\frac{3f}{2}$ **(4)** $2f$

5. NEET 2023 · Q14 The ratio of frequencies of fundamental harmonic produced by an open pipe to that of closed pipe having the same length is:

DIRECT — Class XI Part 2 · Ch 14 · **Page 292** · open v/2L, closed v/4L Open PDF ↗

(1) 3:1 **(2)** 1:2 **(3)** 2:1 **(4)** 1:3

6. NEET 2020 · Q147 In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be:

DIRECT — Class XI Part 2 · Ch 14 · **Page 294** · Example 14.6 Open PDF ↗

(1) 523 Hz **(2)** 524 Hz

(3) 536 Hz **(4)** 537 Hz

7. **NEET 2018 · Q2** The fundamental frequency in an open organ pipe is equal to the third harmonic of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is

APPLIED Class XI Part 2 · Ch 14 · **Page 292** Open PDF ↗

(1) 12.5 cm **(2)** 8 cm
(3) 13.2 cm **(4)** 16 cm

8. **NEET 2018 · Q24** A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27°C two successive resonances are produced at 20 cm and 73 cm of column length. If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27°C is

DIRECT Class XI Part 2 · Ch 14 · **Page 299** · Exercise 14.15 Open PDF ↗

(1) 350 m/s **(2)** 339 m/s
(3) 330 m/s **(4)** 300 m/s

9. **NEET 2017 · Q157** The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system?

APPLIED Class XI Part 2 · Ch 14 · **Page 293** · odd harmonics Open PDF ↗

(1) 20 Hz **(2)** 30 Hz **(3)** 40 Hz **(4)** 10 Hz

10. **NEET 2016 · Q12** An air column, closed at one end and open at the other, resonates with a tuning fork when the smallest length of the column is 50 cm. The next larger length of the column resonating with the same tuning fork is:

APPLIED Class XI Part 2 · Ch 14 · **Page 293** · odd harmonics Open PDF ↗

(1) 66.7 cm **(2)** 100 cm
(3) 150 cm **(4)** 200 cm

Waves · Answer Key

1 (4)	**2** (2)	**3** (3)	**4** (2)	**5** (3)	**6** (2)	**7** (3)	**8** (2)
9 (1)	**10** (3)						

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Waves

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	2	1	2	0	1	0	1	1	0	1	1	**10**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Wave motion and speed of waves	**2**	1	2	**16** 17 18 19 20 21 **22** 23 24 25 26
Beats, organ pipes and strings	**8**	3	7	**16** **17** **18** 19 **20** 21 22 **23** 24 **25** **26**

Direct vs Applied

Direct from NCERT	**4**	40%
text, table, figure	2	
solved Example	1	
Exercise	1	
Applied	**6**	60%
Easy / Mod. / Hard	4 / 6 / 0	

What to focus on

- 10 questions in 11 papers, about 0.9 per paper; rank 25 of 28 chapters by number of questions.
- Most asked topic: **Beats, organ pipes and strings** (8 questions, in 7 of the 11 years).
- 2 questions are NCERT solved Examples or Exercises: Example 14.6 (Q 6), Exercise 14.15 (Q 8). Work these by hand.
- NCERT pages that gave three or more questions: page 292 (4 questions). Read these pages line by line.
- Difficulty: 4 easy, 6 moderate, 0 hard.
- 6 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 1

Electric Charges and Fields

13 questions · 4 topics · NEET 2016–2026

Electric charge and Coulomb's law 4 Qs

1. NEET 2025 · Q28 Two identical charged conducting spheres A and B have their centres separated by a certain distance. Charge on each sphere is q and the force of repulsion between them is F. A third identical uncharged conducting sphere is brought in contact with sphere A first and then with B and finally removed from both. New force of repulsion between spheres A and B (radii of A and B are negligible compared to the distance of separation so that for calculating force between them they can be considered as point charges) is best given as:

APPLIED Class XII Part 1 · Ch 1 · **Page 10** Open PDF ↗

(1) $\frac{3F}{5}$ (2) $\frac{2F}{3}$ (3) $\frac{F}{2}$ (4) $\frac{3F}{8}$

2. NEET 2019 · Q37 Two point charges A and B, having charges $+Q$ and $-Q$ respectively, are placed at certain distance apart and force acting between them is F. If 25% charge of A is transferred to B, then force between the charges becomes:

APPLIED Class XII Part 1 · Ch 1 · **Page 8** Open PDF ↗

(1) F (2) $\frac{9F}{16}$ (3) $\frac{16F}{9}$ (4) $\frac{4F}{3}$

3. NEET 2017 · Q172 Suppose the charge of a proton and an electron differ slightly. One of them is $-e$, the other is $(e + \Delta e)$. If the net of electrostatic force and gravitational force between two hydrogen atoms placed at a distance d (much greater than atomic size) apart is zero, then Δe is of the order of [Given mass of hydrogen $m_h = 1.67 \times 10^{-27}$ kg]

APPLIED Class XII Part 1 · Ch 1 · **Page 9** Open PDF ↗

(1) 10^{-23} C (2) 10^{-37} C
(3) 10^{-47} C (4) 10^{-20} C

4. NEET 2016 · Q9 Two identical charged spheres suspended from a common point by two massless strings of lengths l, are initially at a distance d ($d \ll l$) apart because of their mutual repulsion. The charges begin to leak from both the spheres at a constant rate. As a result, the spheres approach each other with a velocity v. Then v varies as a function of the distance x between the spheres, as:

APPLIED Class XII Part 1 · Ch 1 · **Page 9** Open PDF ↗

(1) $v \propto x^{1/2}$ (2) $v \propto x$
(3) $v \propto x^{-1/2}$ (4) $v \propto x^{-1}$

Coulomb force and electric field 1 Q

5. NEET 2018 · Q22 An electron falls from rest through a vertical distance h in a uniform and vertically upward directed electric field E. The direction of electric field is now reversed, keeping its magnitude the same. A proton is allowed to fall from rest in it through the same vertical distance h. The time of fall of the electron, in comparison to the time of fall of the proton is

DIRECT Class XII Part 1 · Ch 1 · **Page 17** · Example 1.7 Open PDF ↗

(1) 10 times greater (2) 5 times greater
(3) smaller (4) equal

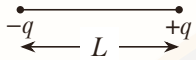
Electric dipole 4 Qs

6. NEET 2023 · Q9 An electric dipole is placed at an angle of 30° with an electric field of intensity 2×10^5 N C^{-1}. It experiences a torque equal to 4 N m. Calculate the magnitude of charge on the dipole, if the dipole length is 2 cm.

DIRECT Class XII Part 1 · Ch 1 · **Page 27** · tau=pE sin theta Open PDF ↗

(1) 2 mC **(2)** 8 mC **(3)** 6 mC **(4)** 4 mC

7. NEET 2022 · Q49 Two point charges $-q$ and $+q$ are placed at a distance of L, as shown in the figure.

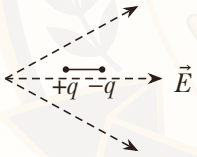


The magnitude of electric field intensity at a distance R ($R \gg L$) varies as:

DIRECT Class XII Part 1 · Ch 1 · **Page 25** · dipole field falls as 1/r^3 Open PDF ↗

(1) $\frac{1}{R^6}$ **(2)** $\frac{1}{R^2}$ **(3)** $\frac{1}{R^3}$ **(4)** $\frac{1}{R^4}$

8. NEET 2021 · Q6 A dipole is placed in an electric field as shown. In which direction will it move?



APPLIED Class XII Part 1 · Ch 1 · **Page 27** Open PDF ↗

(1) towards the left as its potential energy will increase.
(2) towards the right as its potential energy will decrease.
(3) towards the left as its potential energy will decrease.
(4) towards the right as its potential energy will increase.

9. NEET 2021 · Q25 Polar molecules are the molecules:

DIRECT Class XII Part 1 · Ch 1 · **Page 25** · polar molecules Open PDF ↗

(1) having zero dipole moment.
(2) acquire a dipole moment only in the presence of electric field due to displacement of charges.
(3) acquire a dipole moment only when magnetic field is absent.
(4) having a permanent electric dipole moment.

Gauss's law and electric flux 4 Qs

10. NEET 2023 · Q17 If $\oint_S \vec{E} \cdot d\vec{S} = 0$ over a surface, then:

DIRECT Class XII Part 1 · Ch 1 · **Page 31** · zero net flux Open PDF ↗

(1) the electric field inside the surface is necessarily uniform.
(2) the number of flux lines entering the surface must be equal to the number of flux lines leaving it.
(3) the magnitude of electric field on the surface is constant.
(4) all the charges must necessarily be inside the surface.

11. NEET 2020 · Q151 A spherical conductor of radius 10 cm has a charge of 3.2×10^{-7} C distributed uniformly. What is the magnitude of electric field at a point 15 cm from the centre of the sphere? $\left(\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \text{ N m}^2/\text{C}^2\right)$

DIRECT Class XII Part 1 · Ch 1 · **Page 35** · shell: field outside q/4 pi eps0 r^2 Open PDF ↗

(1) 1.28×10^4 N/C **(2)** 1.28×10^5 N/C
(3) 1.28×10^6 N/C **(4)** 1.28×10^7 N/C

12. **NEET 2019 · Q25** A hollow metal sphere of radius R is uniformly charged. The electric field due to the sphere at a distance r from the centre:

DIRECT

Class XII Part 1 · Ch 1 · **Page 36** · shell: zero inside, kq/r^2 outside **Open PDF ↗**

(1) increases as r increases for $r < R$ and for $r > R$

(2) zero as r increases for $r < R$, decreases as r increases for $r > R$

(3) zero as r increases for $r < R$, increases as r increases for $r > R$

(4) decreases as r increases for $r < R$ and for $r > R$

13. **NEET 2019 · Q36** Two parallel infinite line charges with linear charge densities $+\lambda$ C/m and $-\lambda$ C/m are placed at a distance of $2R$ in free space. What is the electric field mid-way between the two line charges?

APPLIED

Class XII Part 1 · Ch 1 · **Page 34** **Open PDF ↗**

(1) zero

(2) $\dfrac{2\lambda}{\pi\varepsilon_0 R}$ N/C

(3) $\dfrac{\lambda}{\pi\varepsilon_0 R}$ N/C

(4) $\dfrac{\lambda}{2\pi\varepsilon_0 R}$ N/C

Electric Charges and Fields · Answer Key

1 (4)	**2** (2)	**3** (2)	**4** (3)	**5** (3)	**6** (1)	**7** (3)	**8** (3)
9 (4)	**10** (2)	**11** (2)	**12** (2)	**13** (3)			

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Electric Charges and Fields

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	1	1	3	1	2	1	2	0	1	0	**13**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Electric charge and Coulomb's law	**4**	0	4	**16 17** 18 **19** 20 21 22 23 24 **25** 26
Coulomb force and electric field	**1**	1	1	16 17 **18** 19 20 21 22 23 24 25 26
Electric dipole	**4**	3	3	16 17 18 19 20 **21 22 23** 24 25 26
Gauss's law and electric flux	**4**	3	3	16 17 18 **19 20** 21 22 **23** 24 25 26

Direct vs Applied

Direct from NCERT	**7**	54%
text, table, figure	6	
solved Example	1	
Exercise	0	
Applied	**6**	46%
Easy / Mod. / Hard	7 / 4 / 2	

What to focus on

- 13 questions in 11 papers, about 1.2 per paper; rank 22 of 28 chapters by number of questions.
- Most asked topic: **Electric charge and Coulomb's law** (4 questions, in 4 of the 11 years).
- 1 questions are NCERT solved Examples or Exercises: Example 1.7 (Q 5). Work these by hand.
- Difficulty: 7 easy, 4 moderate, 2 hard (hard: Q 3, 4).
- 6 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

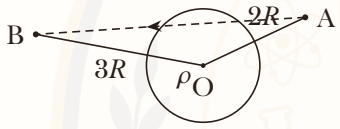
Class XII · Chapter 2

Electrostatic Potential and Capacitance

26 questions · 6 topics · NEET 2016–2026

Electric potential and potential energy 7 Qs

1. NEET 2026 · Q21 A unit positive point charge is taken slowly through an infinitesimally thin tube that is inside a charged dielectric sphere of radius R, having uniform positive charge density ρ, as shown in the figure. The initial and final positions of the charge are marked by A and B at distance $2R$ and $3R$ respectively, from the centre of the sphere. In this process, the magnitude of the total work done on the point charge is $\frac{\rho R^2}{n\varepsilon_0}$. The value of n is: (ε_0 is the permittivity of vacuum)



APPLIED Class XII Part 1 · Ch 2 · Page 52 Open PDF ↗

(1) 2 (2) 6 (3) 9 (4) 18

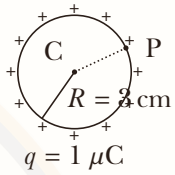
2. NEET 2026 · Q34 Consider a fixed uniformly charged insulating sphere with radius R and total charge $+Q$. A point charge $-q$ ($q \ll Q$) with mass m is released from rest at a distance of $3R$ from the centre of the charged sphere. When the point charge reaches the surface of the sphere, its speed is: (ε_0 is the permittivity of vacuum, neglect gravitational forces).

APPLIED Class XII Part 1 · Ch 2 · Page 55 Open PDF ↗

(1) $\sqrt{\frac{3Qq}{4\pi\varepsilon_0 mR}}$ (2) $\sqrt{\frac{2Qq}{3\pi\varepsilon_0 mR}}$

(3) $\sqrt{\frac{Qq}{3\pi\varepsilon_0 mR}}$ (4) $\sqrt{\frac{Qq}{4\pi\varepsilon_0 mR}}$

3. NEET 2024 · Q14 A thin spherical shell is charged by some source. The potential difference between the two points C and P (in V) shown in the figure is: (Take $\frac{1}{4\pi\varepsilon_0} = 9\times10^9$ SI units)

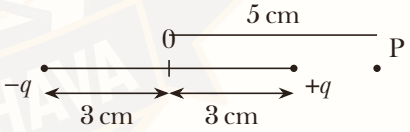


DIRECT Class XII Part 1 · Ch 2 · Page 52 · V inside a shell is constant Open PDF ↗

(1) zero (2) 3×10^5

(3) 1×10^5 (4) 0.5×10^5

4. NEET 2023 · Q49 An electric dipole is placed as shown in the figure. The electric potential (in 10^2 V) at point P due to the dipole is (ε_0 = permittivity of free space and $\frac{1}{4\pi\varepsilon_0} = K$):



APPLIED Class XII Part 1 · Ch 2 · Page 51 Open PDF ↗

(1) $\left(\frac{8}{3}\right)qK$ (2) $\left(\frac{3}{8}\right)qK$

(3) $\left(\frac{5}{8}\right)qK$ (4) $\left(\frac{8}{5}\right)qK$

5. NEET 2022 · Q2 Two hollow conducting spheres of radii R_1 and R_2 ($R_1 \gg R_2$) have equal charges. The potential would be:

DIRECT Class XII Part 1 · Ch 2 · **Page 52** · V=kQ/R Open PDF ↗

(1) dependent on the material property of the sphere
(2) more on bigger sphere
(3) more on smaller sphere
(4) equal on both the spheres

6. NEET 2021 · Q43 Twenty seven drops of same size are charged at 220 V each. They combine to form a bigger drop. Calculate the potential of the bigger drop.

APPLIED Class XII Part 1 · Ch 2 · **Page 48** Open PDF ↗

(1) 660 V **(2)** 1320 V
(3) 1520 V **(4)** 1980 V

7. NEET 2020 · Q159 The energy required to break one bond in DNA is 10^{-20} J. This value in eV is nearly:

DIRECT Class XII Part 1 · Ch 2 · **Page 59** · 1 eV = 1.6e-19 J Open PDF ↗

(1) 6 **(2)** 0.6 **(3)** 0.06 **(4)** 0.006

Equipotential surfaces and potential gradient 3 Qs

8. NEET 2022 · Q27 The angle between the electric lines of force and the equipotential surface is:

DIRECT Class XII Part 1 · Ch 2 · **Page 54** · E perpendicular to equipotential Open PDF ↗

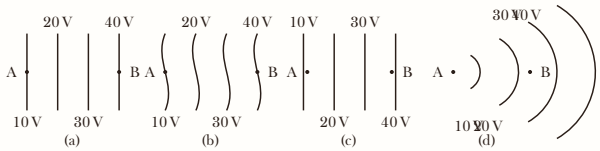
(1) 180° **(2)** 0° **(3)** 45° **(4)** 90°

9. NEET 2020 · Q152 In a certain region of space with volume 0.2 m^3, the electric potential is found to be 5 V throughout. The magnitude of electric field in this region is:

DIRECT Class XII Part 1 · Ch 2 · **Page 55** · E=-dV/dl Open PDF ↗

(1) zero **(2)** 0.5 N/C
(3) 1 N/C **(4)** 5 N/C

10. NEET 2017 · Q171 The diagrams below show regions of equipotentials.



A positive charge is moved from A to B in each diagram.

APPLIED Class XII Part 1 · Ch 2 · **Page 48** · work independent of path Open PDF ↗

(1) In all the four cases the work done is the same.
(2) Minimum work is required to move q in figure (a).
(3) Maximum work is required to move q in figure (b).
(4) Maximum work is required to move q in figure (c).

Electric dipole 3 Qs

11. NEET 2025 · Q35 An electric dipole with dipole moment 5×10^{-6} C m is aligned with the direction of a uniform electric field of magnitude 4×10^{5} N/C. The dipole is then rotated through an angle of 60° with respect to the electric field. The change in the potential energy of the dipole is:

DIRECT Class XII Part 1 · Ch 2 · **Page 61** · W=pE(cos theta0 - cos theta1) Open PDF ↗

(1) 0.8 J **(2)** 1.0 J **(3)** 1.2 J **(4)** 1.5 J

12. NEET 2024 · Q17 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: The potential (V) at any axial point, at 2 m distance (r) from the centre of the dipole of dipole moment vector $\vec{P}$ of magnitude 4×10^{-6} C m, is $\pm 9 \times 10^{3}$ V. (Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^{9}$ SI units)

Reason R: $V = \pm\frac{2P}{4\pi\varepsilon_0 r^2}$, where r is the distance of any axial point, situated at 2 m from the centre of the dipole.

In the light of the above statements, choose the *correct* answer from the options given below:

APPLIED Class XII Part 1 · Ch 2 · **Page 50** Open PDF ↗

(1) A is false but R is true.

(2) Both A and R are true and R is the correct explanation of A.

(3) Both A and R are true and R is NOT the correct explanation of A.

(4) A is true but R is false.

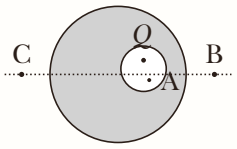
13. NEET 2020 · Q170 A short electric dipole has a dipole moment of 16×10^{-9} C m. The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole, situated on a line making an angle of 60° with the dipole axis is: $\left(\frac{1}{4\pi\varepsilon_0} = 9 \times 10^{9}\ \text{N m}^2/\text{C}^2\right)$

DIRECT Class XII Part 1 · Ch 2 · **Page 50** · V=p cos theta/(4 pi eps0 r^2) Open PDF ↗

(1) 50 V **(2)** 200 V **(3)** 400 V **(4)** zero

Gauss's law and electric flux 2 Qs

14. NEET 2026 · Q33 A point charge Q is placed inside a cavity within a solid isolated conducting sphere. Consider points A, B and C as shown in the figure, where the magnitudes of the electric fields are E_A, E_B, E_C, respectively. The points B and C are at the same distance from the center of the solid sphere. The correct option is:



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(1) $E_A = 0, E_B = E_C$ **(2)** $E_A \neq 0, E_B = E_C$

(3) $E_A = 0, E_B > E_C$ **(4)** $E_A \neq 0, E_B < E_C$

15. NEET 2021 · Q28 Two charged spherical conductors of radius R_1 and R_2 are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is:

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(1) $\frac{R_1}{R_2}$ **(2)** $\frac{R_2}{R_1}$ **(3)** $\sqrt{\frac{R_1}{R_2}}$ **(4)** $\frac{R_1^2}{R_2^2}$

Capacitors (energy, dielectrics) 7 Qs

16. NEET 2025 · Q20 The plates of a parallel plate capacitor are separated by d. Two slabs of different dielectric constant K_1 and K_2 with thickness $\frac{3}{8}d$ and $\frac{d}{2}$, respectively are inserted in the capacitor. Due to this, the capacitance becomes two times larger than when there is nothing between the plates. If $K_1 = 1.25K_2$, the value of K_1 is:

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(1) 2.66 **(2)** 2.33 **(3)** 1.60 **(4)** 1.33

17. NEET 2024 · Q37 If the plates of a parallel plate capacitor connected to a battery are moved close to each other, then

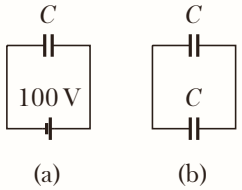
A. the charge stored in it, increases.
B. the energy stored in it, decreases.
C. its capacitance increases.
D. the ratio of charge to its potential remains the same.
E. the product of charge and voltage increases.

Choose the most appropriate answer from the options given below:

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(1) A, B and C only **(2)** A, B and E only
(3) A, C and E only **(4)** B, D and E only

18. NEET 2022 · Q50 A capacitor of capacitance $C = 900$ pF is charged fully by 100 V battery B as shown in figure (a). Then it is disconnected from the battery and connected to another uncharged capacitor of capacitance $C = 900$ pF as shown in figure (b). The electrostatic energy stored by the system (b) is:



(a) (b)

DIRECT Class XII Part 1 · Ch 2 · **Page 75** · Example 2.10 Open PDF ↗

(1) 1.5×10^{-6} J **(2)** 4.5×10^{-6} J
(3) 3.25×10^{-6} J **(4)** 2.25×10^{-6} J

19. NEET 2021 · Q16 A parallel plate capacitor has a uniform electric field '$\vec{E}$' in the space between the plates. If the distance between the plates is 'd' and the area of each plate is 'A', the energy stored in the capacitor is: (ε_0 = permittivity of free space)

DIRECT Class XII Part 1 · Ch 2 · **Page 75** · energy density (1/2)eps0 E^2 Open PDF ↗

(1) $\frac{1}{2}\varepsilon_0 E^2$ **(2)** $\varepsilon_0 EAd$
(3) $\frac{1}{2}\varepsilon_0 E^2 Ad$ **(4)** $\frac{E^2 Ad}{\varepsilon_0}$

20. NEET 2020 · Q166 The capacitance of a parallel plate capacitor with air as medium is 6 μF. With the introduction of a dielectric medium, the capacitance becomes 30 μF. The permittivity of the medium is: ($\varepsilon_0 = 8.85 \times 10^{-12}$ $C^2N^{-1}m^{-2}$)

DIRECT Class XII Part 1 · Ch 2 · **Page 70** · eps=K eps0 Open PDF ↗

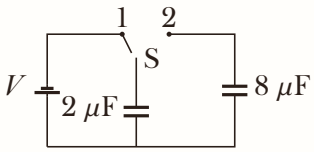
(1) 0.44×10^{-13} $C^2N^{-1}m^{-2}$
(2) 1.77×10^{-12} $C^2N^{-1}m^{-2}$
(3) 0.44×10^{-10} $C^2N^{-1}m^{-2}$
(4) 5.00 $C^2N^{-1}m^{-2}$

21. NEET 2017 · Q153 A capacitor is charged by a battery. The battery is removed and another identical uncharged capacitor is connected in parallel. The total electrostatic energy of resulting system:

APPLIED Class XII Part 1 · Ch 2 · **Page 76** · energy stored in a capacitor Open PDF ↗

(1) decreases by a factor of 2
(2) remains the same
(3) increases by a factor of 2
(4) increases by a factor of 4

22. NEET 2016 · Q5



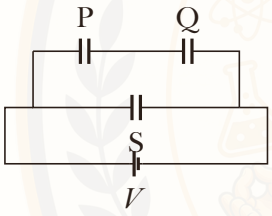
A capacitor of 2 μF is charged as shown in the diagram. When the switch S is turned to position 2, the percentage of its stored energy dissipated is:

APPLIED Class XII Part 1 · Ch 2 · **Page 76** · energy shared between capacitors Open PDF ↗

(1) 0% **(2)** 20% **(3)** 75% **(4)** 80%

Capacitors (combination) 4 Qs

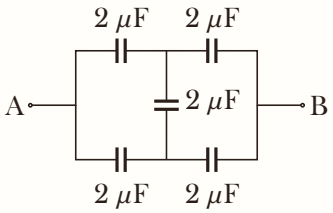
23. NEET 2026 · Q9 Three identical capacitors, P, Q and S, each of the capacitance C, are connected to a battery of voltage V, as shown in the figure. If the energy stored in the capacitor P and total energy stored in the system are U_P and U_T, respectively, then the ratio $\frac{U_P}{U_T}$ is:



APPLIED Class XII Part 1 · Ch 2 · **Page 72** Open PDF ↗

(1) 2/3 **(2)** 1/3 **(3)** 1/2 **(4)** 1/6

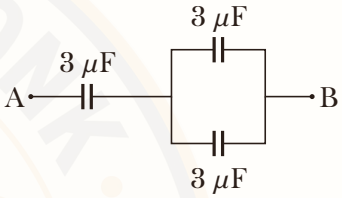
24. NEET 2024 · Q30 In the following circuit, the equivalent capacitance between terminal A and terminal B is:



APPLIED Class XII Part 1 · Ch 2 · **Page 72** Open PDF ↗

(1) 4 μF **(2)** 2 μF
(3) 1 μF **(4)** 0.5 μF

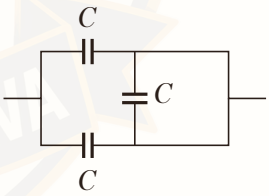
25. NEET 2023 · Q30 The equivalent capacitance of the system shown in the following circuit is:



DIRECT Class XII Part 1 · Ch 2 · **Page 72** · series and parallel Open PDF ↗

(1) 9 μF **(2)** 2 μF **(3)** 3 μF **(4)** 6 μF

26. NEET 2021 · Q4 The equivalent capacitance of the combination shown in the figure is:



APPLIED Class XII Part 1 · Ch 2 · **Page 72** Open PDF ↗

(1) $3C$ **(2)** $2C$ **(3)** $C/2$ **(4)** $3C/2$

Electrostatic Potential and Capacitance · Answer Key

1 (4)	**2** (3)	**3** (1)	**4** (2)	**5** (3)	**6** (4)	**7** (3)	**8** (4)
9 (1)	**10** (1)	**11** (2)	**12** (4)	**13** (2)	**14** (2)	**15** (2)	**16** (1)
17 (3)	**18** (4)	**19** (3)	**20** (3)	**21** (1)	**22** (4)	**23** (4)	**24** (2)
25 (2)	**26** (2)						

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Electrostatic Potential and Capacitance

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	2	0	0	4	4	3	2	4	2	4	**26**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Electric potential and potential energy	**7**	3	6	16 17 18 19 **20 21 22 23 24** 25 **26**
Equipotential surfaces and potential gradient	**3**	2	3	16 **17** 18 19 **20** 21 **22** 23 24 25 26
Electric dipole	**3**	2	3	16 17 18 19 **20** 21 22 23 **24 25** 26
Gauss's law and electric flux	**2**	0	2	16 17 18 19 20 **21** 22 23 24 25 **26**
Capacitors (energy, dielectrics)	**7**	3	7	**16 17** 18 19 **20 21 22** 23 **24 25** 26
Capacitors (combination)	**4**	1	4	16 17 18 19 20 **21** 22 **23 24** 25 **26**

Direct vs Applied

Direct from NCERT	**11**	42%
text, table, figure	10	
solved Example	1	
Exercise	0	
Applied	**15**	58%
Easy / Mod. / Hard	12 / 14 / 0	

What to focus on

- 26 questions in 11 papers, about 2.4 per paper; rank 3 of 28 chapters by number of questions.
- Most asked topic: **Capacitors (energy, dielectrics)** (7 questions, in 7 of the 11 years).
- 1 questions are NCERT solved Examples or Exercises: Example 2.10 (Q 18). Work these by hand.
- NCERT pages that gave three or more questions: page 72 (4 questions), page 52 (3 questions). Read these pages line by line.
- Difficulty: 12 easy, 14 moderate, 0 hard.
- 15 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 3

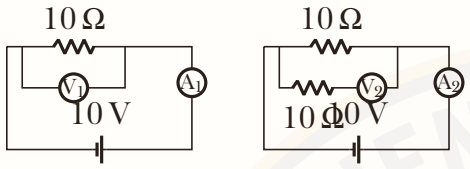
Current Electricity

30 questions · 8 topics · NEET 2016–2026

Ohm's law and measuring instruments 1 Q

1.

NEET 2019 · Q24 In the circuits shown below, the readings of the voltmeters and the ammeters will be:



Circuit 1 Circuit 2

APPLIED Class XII Part 1 · Ch 3 · **Page 83** Open PDF ↗

(1) $V_2 > V_1$ and $i_1 = i_2$
(2) $V_1 = V_2$ and $i_1 > i_2$
(3) $V_1 = V_2$ and $i_1 = i_2$
(4) $V_2 > V_1$ and $i_1 > i_2$

Resistance and resistivity 2 Qs

2.

NEET 2022 · Q17 A copper wire of length 10 m and radius $(10^{-2}/\sqrt{\pi})$ m has electrical resistance of 10 Ω. The current density in the wire for an electric field strength of 10 V/m is:

APPLIED Class XII Part 1 · Ch 3 · **Page 85** Open PDF ↗

(1) 10^5 A/m^2 **(2)** 10^4 A/m^2
(3) 10^6 A/m^2 **(4)** 10^{-5} A/m^2

3.

NEET 2017 · Q143 The resistance of a wire is 'R' ohm. If it is melted and stretched to 'n' times its original length, its new resistance will be:

APPLIED Class XII Part 1 · Ch 3 · **Page 84** · R=rho l/A, volume constant Open PDF ↗

(1) $\frac{R}{n}$ **(2)** n^2R **(3)** $\frac{R}{n^2}$ **(4)** nR

Current, drift velocity and mobility 2 Qs

4.

NEET 2021 · Q14 Column-I gives certain physical terms associated with flow of current through a metallic conductor. Column-II gives some mathematical relations involving electrical quantities. Match Column-I and Column-II with appropriate relations.

(A)	Drift Velocity	(P)	$\frac{m}{ne^2\rho}$
(B)	Electrical Resistivity	(Q)	nev_d
(C)	Relaxation Period	(R)	$\frac{eE}{m}\tau$
(D)	Current Density	(S)	$\frac{E}{J}$

DIRECT Class XII Part 1 · Ch 3 · **Page 86** · vd, tau, rho, j relations Open PDF ↗

(1) (A)-(R), (B)-(S), (C)-(P), (D)-(Q)
(2) (A)-(R), (B)-(S), (C)-(Q), (D)-(P)
(3) (A)-(R), (B)-(P), (C)-(S), (D)-(Q)
(4) (A)-(R), (B)-(Q), (C)-(S), (D)-(P)

5.

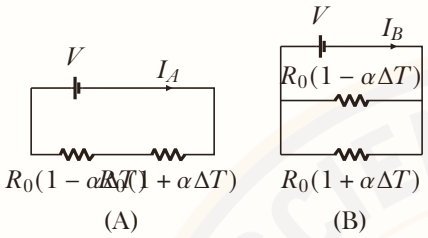
NEET 2020 · Q173 A charged particle having drift velocity of 7.5×10^{-4} m s^{-1} in an electric field of 3×10^{-10} V m^{-1}, has a mobility in m^{2}V^{-1}s^{-1} of:

DIRECT Class XII Part 1 · Ch 3 · **Page 88** · mu=vd/E Open PDF ↗

(1) 2.25×10^{15} **(2)** 2.5×10^{6}
(3) 2.5×10^{-6} **(4)** 2.25×10^{-15}

Resistivity and temperature 5 Qs

6. **NEET 2026 · Q36** Consider two circuits, (A) and (B), each having two resistors. One of them has a positive temperature coefficient of resistance, $+\alpha$, while the other one has a negative temperature coefficient, $-\alpha$, as shown in the figure. The current through these circuits are denoted by I_A and I_B. At initial temperature, the resistance of the two resistors is R_0. As the temperature is increased, the correct option that describes the variation of current in these circuits is:



APPLIED Class XII Part 1 · Ch 3 · **Page 90** Open PDF ↗

(1) I_A remains constant while I_B increases
(2) I_A decreases while I_B increases
(3) I_A increases while I_B decreases
(4) Both I_A and I_B remain constant

7. **NEET 2023 · Q37** The resistance of platinum wire at 0°C is 2 Ω and 6.8 Ω at 80°C. The temperature coefficient of resistance of the wire is:

DIRECT Class XII Part 1 · Ch 3 · **Page 90** · rho=rho0[1+alpha(T-T0)] Open PDF ↗

(1) $3 \times 10^{-1}\ °C^{-1}$ **(2)** $3 \times 10^{-4}\ °C^{-1}$
(3) $3 \times 10^{-3}\ °C^{-1}$ **(4)** $3 \times 10^{-2}\ °C^{-1}$

8. **NEET 2022 · Q4** As the temperature increases, the electrical resistance:

DIRECT Class XII Part 1 · Ch 3 · **Page 91** · conductors rise, semiconductors fall Open PDF ↗

(1) decreases for conductors but increases for semiconductors
(2) increases for both conductors and semiconductors
(3) decreases for both conductors and semiconductors
(4) increases for conductors but decreases for semiconductors

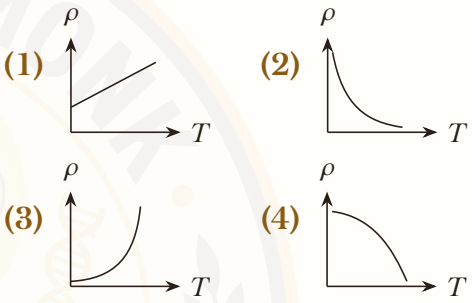
9. **NEET 2020 · Q144** The solids which have the negative temperature coefficient of resistance are:

APPLIED Class XII Part 1 · Ch 3 · **Page 91** Open PDF ↗

(1) metals
(2) insulators only
(3) semiconductors only
(4) insulators and semiconductors

10. **NEET 2020 · Q163** Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper?

DIRECT Class XII Part 1 · Ch 3 · **Page 90** · resistivity of copper vs T (Fig 3.9) Open PDF ↗



Electric power and heating 5 Qs

11. **NEET 2024 · Q40** Two heaters A and B have power rating of 1 kW and 2 kW, respectively. Those two are first connected in series and then in parallel to a fixed power source. The ratio of power outputs for these two cases is:

APPLIED Class XII Part 1 · Ch 3 · **Page 92** Open PDF ↗

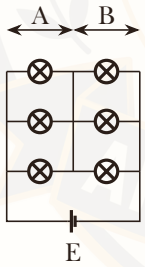
(1) 2:3 **(2)** 1:1 **(3)** 2:9 **(4)** 1:2

12. **NEET 2022 · Q1** Two resistors of resistance, 100 Ω and 200 Ω are connected in parallel in an electrical circuit. The ratio of the thermal energy developed in 100 Ω to that in 200 Ω in a given time is:

DIRECT Class XII Part 1 · Ch 3 · **Page 92** · P=V^2/R Open PDF ↗

(1) 4:1 **(2)** 1:2 **(3)** 2:1 **(4)** 1:4

13. **NEET 2019 · Q17** Six similar bulbs are connected as shown in the figure with a DC source of emf E, and zero internal resistance. The ratio of power consumption by the bulbs when (i) all are glowing and (ii) in the situation when two from section A and one from section B are glowing, will be:



APPLIED Class XII Part 1 · Ch 3 · **Page 92** Open PDF ↗

(1) 4:9 **(2)** 9:4 **(3)** 1:2 **(4)** 2:1

14. **NEET 2019 · Q23** Which of the following acts as a circuit protection device?

APPLIED Class XII Part 1 · Ch 3 · **Page 92** · fuse (heating effect) Open PDF ↗

(1) conductor **(2)** inductor
(3) switch **(4)** fuse

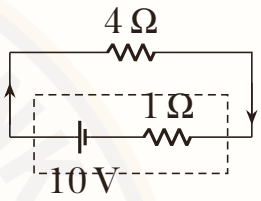
15. **NEET 2016 · Q14** The charge flowing through a resistance R varies with time t as $Q = at - bt^2$, where a and b are positive constants. The total heat produced in R is:

APPLIED Class XII Part 1 · Ch 3 · **Page 91** · P=I^2R Open PDF ↗

(1) $\frac{a^3R}{6b}$ **(2)** $\frac{a^3R}{3b}$ **(3)** $\frac{a^3R}{2b}$ **(4)** $\frac{a^3R}{b}$

Cells, emf and internal resistance 4 Qs

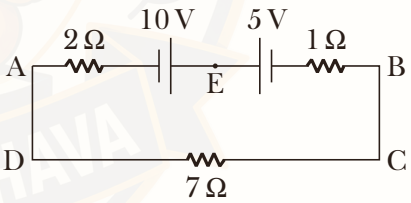
16. **NEET 2024 · Q18** The terminal voltage of the battery, whose emf is 10 V and internal resistance 1 Ω, when connected through an external resistance of 4 Ω as shown in the figure is:



DIRECT Class XII Part 1 · Ch 3 · **Page 94** · V=E-Ir Open PDF ↗

(1) 10 V **(2)** 4 V **(3)** 6 V **(4)** 8 V

17. **NEET 2023 · Q21** The magnitude and direction of the current in the following circuit is



APPLIED Class XII Part 1 · Ch 3 · **Page 95** Open PDF ↗

(1) 1.5 A from B to A through E
(2) 0.2 A from B to A through E
(3) 0.5 A from A to B through E
(4) $\frac{5}{9}$ A from A to B through E

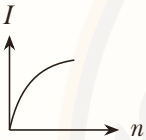
18. **NEET 2018 · Q6** A set of 'n' equal resistors, of value 'R' each, are connected in series to a battery of emf 'E' and internal resistance 'R'. The current drawn is I. Now, the 'n' resistors are connected in parallel to the same battery. Then the current drawn from battery becomes $10I$. The value of 'n' is

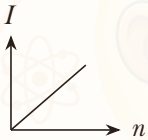
APPLIED Class XII Part 1 · Ch 3 · **Page 95** Open PDF ↗

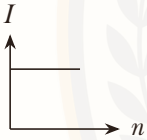
(1) 20 **(2)** 11 **(3)** 10 **(4)** 9

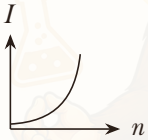
19. **NEET 2018 · Q7** A battery consists of a variable number 'n' of identical cells (having internal resistance 'r' each) which are connected in series. The terminals of the battery are short-circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n?

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(1) 

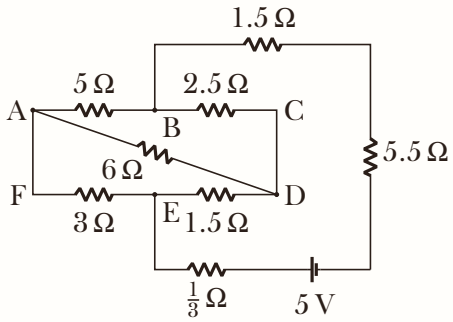
(2) 

(3) 

(4) 

Kirchhoff's rules and DC circuits 8 Qs

20. **NEET 2025 · Q6** The current passing through the battery in the given circuit, is:



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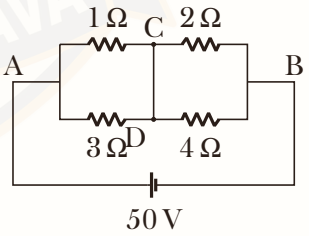
(1) 2.0 A **(2)** 0.5 A **(3)** 2.5 A **(4)** 1.5 A

21. **NEET 2025 · Q33** A wire of resistance R is cut into 8 equal pieces. From these pieces two equivalent resistances are made by adding four of these together in parallel. Then these two sets are added in series. The net effective resistance of the combination is:

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(1) $\frac{R}{64}$ **(2)** $\frac{R}{32}$ **(3)** $\frac{R}{16}$ **(4)** $\frac{R}{8}$

22. **NEET 2025 · Q36** A constant voltage of 50 V is maintained between the points A and B of the circuit shown in the figure. The current through the branch CD of the circuit is:



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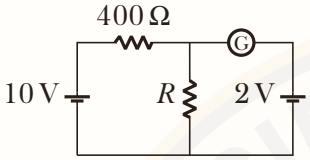
(1) 1.5 A **(2)** 2.0 A **(3)** 2.5 A **(4)** 3.0 A

23. **NEET 2024 · Q9** A wire of length 'l' and resistance 100 Ω is divided into 10 equal parts. The first 5 parts are connected in series while the next 5 parts are connected in parallel. The two combinations are again connected in series. The resistance of this final combination is:

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(1) 60 Ω **(2)** 26 Ω **(3)** 52 Ω **(4)** 55 Ω

24. **NEET 2023 · Q3** If the galvanometer G does not show any deflection in the circuit shown, the value of R is given by:



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(1) 400 Ω **(2)** 200 Ω **(3)** 50 Ω **(4)** 100 Ω

25. **NEET 2023 · Q40** 10 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. Then those are connected in parallel to the same battery, the current is increased n times. The value of n is:

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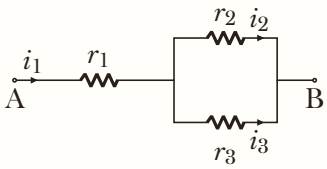
(1) 1000 **(2)** 10 **(3)** 100 **(4)** 1

26. **NEET 2021 · Q33** The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is 0.25 Ω. What will be the effective resistance if they are connected in series?

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(1) 0.25 Ω **(2)** 0.5 Ω
(3) 1 Ω **(4)** 4 Ω

27. **NEET 2021 · Q49** Three resistors having resistances r_1, r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is:



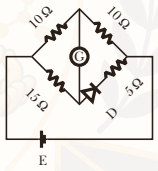
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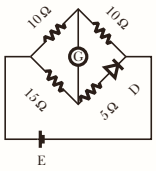
(1) $\frac{r_1}{r_2+r_3}$ **(2)** $\frac{r_2}{r_2+r_3}$
(3) $\frac{r_1}{r_1+r_2}$ **(4)** $\frac{r_2}{r_1+r_3}$

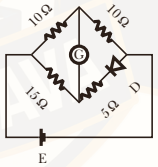
Wheatstone bridge and meter bridge 3 Qs

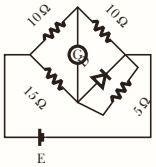
28. **NEET 2024 · Q39** Choose the correct circuit which can achieve the bridge balance. (Each figure: arms 10 Ω and 10 Ω above, 15 Ω and 5 Ω below, galvanometer G across the bridge, diode D in the 5 Ω arm, cell E and key K.)

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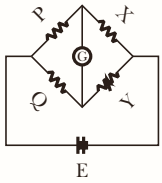
(1)


(2)


(3)


(4)


29. **NEET 2022 · Q46** A wheatstone bridge is used to determine the value of unknown resistance X by adjusting the variable resistance Y as shown in the figure. For the most precise measurement of X, the resistances P and Q:



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(1) do not play any significant role
(2) should be approximately equal to 2X
(3) should be approximately equal and are small
(4) should be very large and unequal

30. **NEET 2020 · Q179** A resistance wire connected in the left gap of a metre bridge balances a 10 Ω resistance in the right gap at a point which divides the bridge wire in the ratio 3:2. If the length of the resistance wire is 1.5 m, then the length of 1 Ω of the resistance wire is:

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(1) 1.0×10^{-2} m **(2)** 1.0×10^{-1} m
(3) 1.5×10^{-1} m **(4)** 1.5×10^{-2} m

Current Electricity · Answer Key

1 (3)	**2** (1)	**3** (2)	**4** (1)	**5** (2)	**6** (1)	**7** (4)	**8** (4)
9 (4)	**10** (3)	**11** (3)	**12** (3)	**13** (2)	**14** (4)	**15** (1)	**16** (4)
17 (3)	**18** (3)	**19** (3)	**20** (2)	**21** (3)	**22** (2)	**23** (3)	**24** (4)
25 (3)	**26** (4)	**27** (2)	**28** (3)*	**29** (3)	**30** (2)		

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor. *: see note.

28. (NEET 2024 Q39) In (3) the diode is reverse-biased, so its arm is open and no current flows through G.

Chapter Summary · Current Electricity

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	1	2	3	4	3	4	4	4	3	1	**30**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Ohm's law and measuring instruments	**1**	0	1	16 17 18 **19** 20 21 22 23 24 25 26
Resistance and resistivity	**2**	0	2	16 **17** 18 19 20 21 **22** 23 24 25 26
Current, drift velocity and mobility	**2**	2	2	16 17 18 19 **20** **21** 22 23 24 25 26
Resistivity and temperature	**5**	3	4	16 17 18 19 **20** 21 **22** **23** 24 25 **26**
Electric power and heating	**5**	1	4	**16** 17 18 **19** 20 21 **22** 23 **24** 25 26
Cells, emf and internal resistance	**4**	1	3	16 17 **18** 19 20 21 22 **23** **24** 25 26
Kirchhoff's rules and DC circuits	**8**	0	4	16 17 18 19 20 **21** 22 **23** **24** **25** 26
Wheatstone bridge and meter bridge	**3**	0	3	16 17 18 19 **20** 21 **22** 23 **24** 25 26

Direct vs Applied

Direct from NCERT	**7**	23%
text, table, figure	7	
solved Example	0	
Exercise	0	
Applied	**23**	77%
Easy / Mod. / Hard	15 / 14 / 1	

What to focus on

- 30 questions in 11 papers, about 2.7 per paper; rank 1 of 28 chapters by number of questions.
- Most asked topic: **Kirchhoff's rules and DC circuits** (8 questions, in 4 of the 11 years).
- NCERT pages that gave three or more questions: page 98 (8 questions), page 92 (4 questions), page 90 (3 questions), page 91 (3 questions), page 101 (3 questions). Read these pages line by line.
- Difficulty: 15 easy, 14 moderate, 1 hard (hard: Q 15).
- 23 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 4

Moving Charges and Magnetism

22 questions · 8 topics · NEET 2016–2026

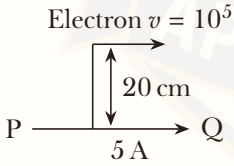
Charged particle in a magnetic field 4 Qs

1. NEET 2025 · Q18 A model for quantized motion of an electron in a uniform magnetic field B states that the flux passing through the orbit of the electron is $n(h/e)$ where n is an integer, h is Planck's constant and e is the magnitude of electron's charge. According to the model, the magnetic moment of an electron in its lowest energy state will be (m is the mass of the electron)

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(1) $\frac{he}{\pi m}$ **(2)** $\frac{he}{2\pi m}$ **(3)** $\frac{heB}{\pi m}$ **(4)** $\frac{heB}{2\pi m}$

2. NEET 2021 · Q17 An infinitely long straight conductor carries a current of 5 A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant.



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(1) 4×10^{-20} N **(2)** $8\pi \times 10^{-20}$ N
(3) $4\pi \times 10^{-20}$ N **(4)** 8×10^{-20} N

3. NEET 2021 · Q37 In the product $\vec{F} = q(\vec{v} \times \vec{B}) = q\vec{v} \times (B\hat{i} + B\hat{j} + B_0\hat{k})$. For $q = 1$ and $\vec{v} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ and $\vec{F} = 4\hat{i} - 20\hat{j} + 12\hat{k}$. What will be the complete expression for $\vec{B}$?

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(1) $-8\hat{i} - 8\hat{j} - 6\hat{k}$ **(2)** $-6\hat{i} - 6\hat{j} - 8\hat{k}$
(3) $8\hat{i} + 8\hat{j} - 6\hat{k}$ **(4)** $6\hat{i} + 6\hat{j} - 8\hat{k}$

4. NEET 2019 · Q3 Ionized hydrogen atoms and α-particles with same momenta enters perpendicular to a constant magnetic field, B. The ratio of their radii of their paths $r_H : r_\alpha$ will be:

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(1) 2:1 **(2)** 1:2 **(3)** 4:1 **(4)** 1:4

Force on a current-carrying conductor 2 Qs

5. NEET 2023 · Q47 A wire carrying a current I along the positive x-axis has length L. It is kept in a magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 4\hat{k})$ T. The magnitude of the magnetic force acting on the wire is:

DIRECT Class XII Part 1 · Ch 4 · **Page 110** · F=I l x B Open PDF ↗

(1) $\sqrt{3}\,IL$ **(2)** $3IL$ **(3)** $\sqrt{5}\,IL$ **(4)** $5IL$

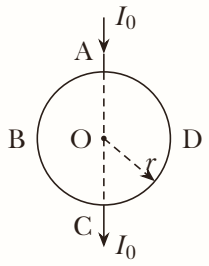
6. NEET 2018 · Q26 A metallic rod of mass per unit length 0.5 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 30° with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of induction 0.25 T is acting on it in the vertical direction. The current flowing in the rod to keep it stationary is

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(1) 14.76 A **(2)** 5.98 A
(3) 7.14 A **(4)** 11.32 A

Biot-Savart law and magnetic fields 3 Qs

7. **NEET 2026 · Q11** A current I_0 flows through a metallic circular loop of radius r as shown in the figure. Resistance of the segment ABC is half that of ADC. Magnitude of magnetic field at the centre O of the loop is:



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(1) $\frac{\mu_0 I_0}{12r}$ **(2)** $\frac{\mu_0 I_0}{4r}$ **(3)** $\frac{\mu_0 I_0}{2r}$ **(4)** $\frac{\mu_0 I_0}{2\pi r}$

8. **NEET 2024 · Q21** A tightly wound 100 turns coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (Take permeability of free space as $4\pi \times 10^{-7}$ SI units):

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(1) 44 T **(2)** 44 mT
(3) 4.4 T **(4)** 4.4 mT

9. **NEET 2022 · Q19** Given below are two statements:
Statement I: Biot-Savart's law gives us the expression for the magnetic field strength of an infinitesimal current element (Idl) of a current carrying conductor only.
Statement II: Biot-Savart's law is analogous to Coulomb's inverse square law of charge q, with the former being related to the field produced by a scalar source, Idl while the latter being produced by a vector source, q.
In light of above statements choose the most appropriate answer from the options given below:

DIRECT Class XII Part 1 · Ch 4 · **Page 114** · Biot-Savart law Open PDF ↗

(1) Statement I is incorrect and Statement II is correct
(2) Both Statement I and Statement II are correct
(3) Both Statement I and Statement II are incorrect
(4) Statement I is correct and Statement II is incorrect

Ampere's law and solenoid 6 Qs

10. **NEET 2022 · Q35** A long solenoid of radius 1 mm has 100 turns per mm. If 1 A current flows in the solenoid, the magnetic field strength at the centre of the solenoid is:

DIRECT Class XII Part 1 · Ch 4 · **Page 121** · B=mu0 n I Open PDF ↗

(1) 6.28×10^{-4} T **(2)** 6.28×10^{-2} T
(3) 12.56×10^{-2} T **(4)** 12.56×10^{-4} T

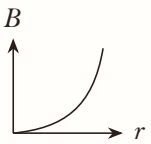
11. **NEET 2022 · Q42** From Ampere's circuital law for a long straight wire of circular cross-section carrying a steady current, the variation of magnetic field in the inside and outside region of the wire is:

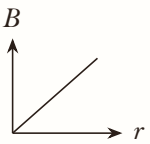
DIRECT Class XII Part 1 · Ch 4 · **Page 119** · Example 4.7 Open PDF ↗

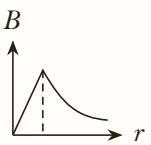
(1) a linearly decreasing function of distance upto the boundary of the wire and then a linearly increasing one for the outside region.
(2) uniform and remains constant for both the regions.
(3) a linearly increasing function of distance upto the boundary of the wire and then linearly decreasing for the outside region.
(4) a linearly increasing function of distance r upto the boundary of the wire and then decreasing one with $1/r$ dependence for the outside region.

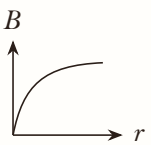
12. **NEET 2021 · Q26** A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field $B(r)$ due to the cable with the distance 'r' from the axis of the cable is represented by:

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(1) 

(2) 

(3) 

(4) 

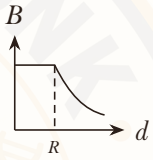
13. **NEET 2020 · Q172** A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is: ($\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1})

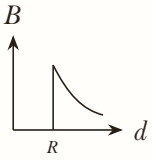
DIRECT Class XII Part 1 · Ch 4 · **Page 121** · B=mu0 n I Open PDF ↗

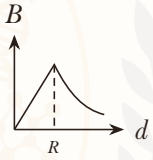
(1) 6.28×10^{-4} T **(2)** 3.14×10^{-4} T
(3) 6.28×10^{-5} T **(4)** 3.14×10^{-5} T

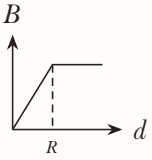
14. **NEET 2019 · Q45** A cylindrical conductor of radius R is carrying a constant current. The plot of the magnitude of the magnetic field, B with the distance, d, from the centre of the conductor, is **correctly** represented by the figure:

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(1) 

(2) 

(3) 

(4) 

15. **NEET 2016 · Q26** A long straight wire of radius a carries a steady current I. The current is uniformly distributed over its cross-section. The ratio of the magnetic fields B and B', at radial distances $\frac{a}{2}$ and $2a$ respectively, from the axis of the wire is:

DIRECT Class XII Part 1 · Ch 4 · **Page 119** · Example 4.7 Open PDF ↗

(1) $\frac{1}{4}$ **(2)** $\frac{1}{2}$ **(3)** 1 **(4)** 4

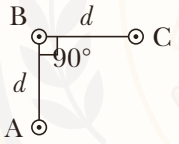
Force between parallel currents 3 Qs

16. NEET 2026 · Q15 Two infinitely long parallel conducting wires A and B carry currents I and $2I$, respectively, in the same direction. The wire A has uniform mass per unit length λ and lies on an insulated floor. The wire B is kept fixed at a height h above the floor. The minimum magnitude of h so that the wire A does not rise from the floor is: [g is the acceleration due to gravity and μ_0 is the permeability of free space.]

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(1) $\frac{\mu_0 I^2}{2\pi\lambda g}$ (2) $\frac{\mu_0 I^2}{\pi\lambda g}$
(3) $\frac{2\mu_0 I^2}{\pi\lambda g}$ (4) $\frac{4\mu_0 I^2}{\pi\lambda g}$

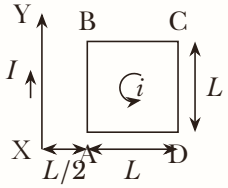
17. NEET 2017 · Q139 An arrangement of three parallel straight wires placed perpendicular to plane of paper carrying same current 'I' along the same direction is shown in Fig. Magnitude of force per unit length on the middle wire 'B' is given by:



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(1) $\frac{2\mu_0 i^2}{\pi d}$ (2) $\frac{\sqrt{2}\mu_0 i^2}{\pi d}$
(3) $\frac{\mu_0 i^2}{\sqrt{2}\pi d}$ (4) $\frac{\mu_0 i^2}{2\pi d}$

18. NEET 2016 · Q2 A square loop ABCD carrying a current i, is placed near and coplanar with a long straight conductor XY carrying a current I, the net force on the loop will be:



APPLIED Class XII Part 1 · Ch 4 · **Page 123** · force between parallel currents Open PDF ↗

(1) $\frac{2\mu_0 Ii}{3\pi}$ (2) $\frac{\mu_0 Ii}{2\pi}$
(3) $\frac{2\mu_0 IiL}{3\pi}$ (4) $\frac{\mu_0 IiL}{2\pi}$

Magnetic dipole and bar magnet 2 Qs

19. NEET 2025 · Q12 A 2 A current is flowing through two different small circular copper coils having radii ratio 1 : 2. The ratio of their respective magnetic moments will be

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(1) 1:4 (2) 1:2 (3) 2:1 (4) 4:1

20. NEET 2021 · Q44 A uniform conducting wire of length $12a$ and resistance 'R' is wound up as a current carrying coil in the shape of, (i) an equilateral triangle of side 'a', (ii) a square of side 'a'. The magnetic dipole moments of the coil in each case respectively are:

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(1) $\sqrt{3}Ia^2$ and $3Ia^2$ (2) $3Ia^2$ and Ia^2
(3) $3Ia^2$ and $4Ia^2$ (4) $4Ia^2$ and $3Ia^2$

Moving coil galvanometer, ammeter, voltmeter 1 Q

21. NEET 2018 · Q29 Current sensitivity of a moving coil galvanometer is 5 div/mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div/V. The resistance of the galvanometer is

DIRECT Class XII Part 1 · Ch 4 · **Page 130** · current and voltage sensitivity Open PDF ↗

(1) 250 Ω (2) 25 Ω (3) 40 Ω (4) 500 Ω

Dimensional analysis 1 Q

22. NEET 2022 · Q24 The dimensions $[MLT^{-2}A^{-2}]$ belong to the:

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(1) electric permittivity
(2) magnetic flux
(3) self inductance
(4) magnetic permeability

Moving Charges and Magnetism · Answer Key

1 (2)	**2** (4)	**3** (2)	**4** (1)	**5** (4)	**6** (4)	**7** (1)	**8** (4)
9 (4)	**10** (3)	**11** (4)	**12** (3)	**13** (1)	**14** (3)	**15** (3)	**16** (2)
17 (3)	**18** (1)	**19** (1)	**20** (1)	**21** (1)	**22** (4)		

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Moving Charges and Magnetism

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	2	1	2	2	1	4	4	1	1	2	2	**22**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Charged particle in a magnetic field	**4**	0	3	16 17 18 **19** 20 **21** 22 23 24 **25** 26
Force on a current-carrying conductor	**2**	1	2	16 17 **18** 19 20 21 22 **23** 24 25 26
Biot-Savart law and magnetic fields	**3**	2	3	16 17 18 19 20 21 **22** 23 **24** 25 **26**
Ampere's law and solenoid	**6**	6	5	**16** 17 18 **19** **20** **21** **22** 23 24 25 26
Force between parallel currents	**3**	0	3	**16** **17** 18 19 20 21 22 23 24 25 **26**
Magnetic dipole and bar magnet	**2**	1	2	16 17 18 19 20 **21** 22 23 24 **25** 26
Moving coil galvanometer, ammeter, voltmeter	**1**	1	1	16 17 **18** 19 20 21 22 23 24 25 26
Dimensional analysis	**1**	1	1	16 17 18 19 20 21 **22** 23 24 25 26

Direct vs Applied

Direct from NCERT	**12**	55%
text, table, figure	7	
solved Example	5	
Exercise	0	
Applied	**10**	45%
Easy / Mod. / Hard	11 / 10 / 1	

What to focus on

- 22 questions in 11 papers, about 2.0 per paper; rank 6 of 28 chapters by number of questions.
- Most asked topic: **Ampere's law and solenoid** (6 questions, in 5 of the 11 years).
- 5 questions are NCERT solved Examples or Exercises: Example 4.6 (Q 8), Example 4.7 (Q 11), Example 4.7 (Q 12), Example 4.7 (Q 14), Example 4.7 (Q 15). Work these by hand.
- NCERT pages that gave three or more questions: page 119 (4 questions), page 123 (3 questions). Read these pages line by line.
- Difficulty: 11 easy, 10 moderate, 1 hard (hard: Q 1).
- 10 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 5

Magnetism and Matter

7 questions · 3 topics · NEET 2016–2026

Magnetic dipole and bar magnet 2 Qs

1. NEET 2024 · Q47 An iron bar of length L has magnetic moment M. It is bent at the middle of its length such that the two arms make an angle 60° with each other. The magnetic moment of this new magnet is:

APPLIED Class XII Part 1 · Ch 5 · **Page 138** Open PDF ↗

(1) $\frac{M}{\sqrt{3}}$ **(2)** M **(3)** $\frac{M}{2}$ **(4)** 2M

2. NEET 2017 · Q148 A 250-turn rectangular coil of length 2.1 cm and width 1.25 cm carries a current of 85 μA and subjected to a magnetic field of strength 0.85 T. Work done for rotating the coil by 180° against the torque is:

APPLIED Class XII Part 1 · Ch 5 · **Page 139** · U=-m.B, W=2mB Open PDF ↗

(1) 4.55 μJ **(2)** 2.3 μJ
(3) 1.15 μJ **(4)** 9.1 μJ

Magnetic field lines and Gauss's law for magnetism 1 Q

3. NEET 2023 · Q2 The net magnetic flux through any closed surface is:

DIRECT Class XII Part 1 · Ch 5 · **Page 142** · net flux through a closed surface is zero Open PDF ↗

(1) Negative **(2)** Zero
(3) Positive **(4)** Infinity

Magnetic materials 4 Qs

4. NEET 2024 · Q2 Match List-I with List-II.

List-I (Material)		*List-II* (Susceptibility χ)
A. Diamagnetic	I.	$\chi = 0$
B. Ferromagnetic	II.	$0 > \chi \geq -1$
C. Paramagnetic	III.	$\chi \gg 1$
D. Non-magnetic	IV.	$0 < \chi < \varepsilon$ (a small positive number)

Choose the correct answer from the options given below:

DIRECT Class XII Part 1 · Ch 5 · **Page 147** · Table 5.2 Open PDF ↗

(1) A-IV, B-III, C-II, D-I
(2) A-II, B-III, C-IV, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-II, C-I, D-IV

5. NEET 2020 · Q171 An iron rod of susceptibility 599 is subjected to a magnetising field of 1200 A m^{-1}. The permeability of the material of the rod is: ($\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1})

DIRECT Class XII Part 1 · Ch 5 · **Page 146** · mu=mu0(1+chi) Open PDF ↗

(1) $2.4\pi \times 10^{-4}$ T m A^{-1}
(2) 8.0×10^{-5} T m A^{-1}
(3) $2.4\pi \times 10^{-5}$ T m A^{-1}
(4) $2.4\pi \times 10^{-7}$ T m A^{-1}

6. **NEET 2018 · Q27** A thin diamagnetic rod is placed vertically between the poles of an electromagnet. When the current in the electromagnet is switched on, then the diamagnetic rod is pushed up, out of the horizontal magnetic field. Hence the rod gains gravitational potential energy. The work required to do this comes from

APPLIED Class XII Part 1 · Ch 5 · **Page 147** Open PDF ↗

(1) the lattice structure of the material of the rod
(2) the magnetic field
(3) the current source
(4) the induced electric field due to the changing magnetic field

7. **NEET 2016 · Q3** The magnetic susceptibility is negative for:

DIRECT Class XII Part 1 · Ch 5 · **Page 147** · diamagnetic: chi negative Open PDF ↗

(1) diamagnetic material only
(2) paramagnetic material only
(3) ferromagnetic material only
(4) paramagnetic and ferromagnetic materials

Magnetism and Matter · Answer Key

1 (3)	**2** (4)	**3** (2)	**4** (2)	**5** (1)	**6** (3)	**7** (1)

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Magnetism and Matter

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	1	1	0	1	0	0	1	2	0	0	**7**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Magnetic dipole and bar magnet	**2**	0	2	16 **17** 18 19 20 21 22 23 **24** 25 26
Magnetic field lines and Gauss's law for magnetism	**1**	1	1	16 17 18 19 20 21 22 **23** 24 25 26
Magnetic materials	**4**	3	4	**16** 17 **18** 19 **20** 21 22 23 **24** 25 26

Direct vs Applied

Direct from NCERT	**4**	57%
text, table, figure	4	
solved Example	0	
Exercise	0	
Applied	**3**	43%
Easy / Mod. / Hard	4 / 3 / 0	

What to focus on

- 7 questions in 11 papers, about 0.6 per paper; rank 27 of 28 chapters by number of questions.
- Most asked topic: **Magnetic materials** (4 questions, in 4 of the 11 years).
- NCERT pages that gave three or more questions: page 147 (3 questions). Read these pages line by line.
- Difficulty: 4 easy, 3 moderate, 0 hard.
- 3 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 6

Electromagnetic Induction

15 questions · 3 topics · NEET 2016–2026

Magnetic flux 1 Q

1. NEET 2022 · Q33 A square loop of side 1 m and resistance 1 Ω is placed in a magnetic field of 0.5 T. If the plane of loop is perpendicular to the direction of magnetic field, the magnetic flux through the loop is:

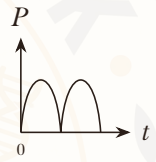
DIRECT Class XII Part 1 · Ch 6 · **Page 156** · Phi=BA Open PDF ↗

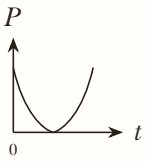
(1) zero weber **(2)** 2 weber
(3) 0.5 weber **(4)** 1 weber

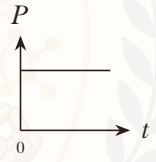
Faraday's law and Lenz's law 6 Qs

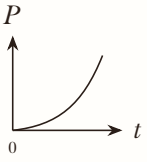
2. NEET 2026 · Q32 A conducting loop of finite resistance lies on the x–y plane. There is a constant magnetic field in the z direction. The area of the loop varies with time t, as $A = A_0(1 + \sin t)$ in appropriate units. The figure that correctly indicates the qualitative behaviour of the power P dissipated in the loop as a function of time is:

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(1)


(2)


(3)


(4)


3. NEET 2024 · Q19



In the above diagram, a strong bar magnet is moving towards solenoid-2 from solenoid-1. The direction of induced current in solenoid-1 and that in solenoid-2, respectively, are through the directions:

APPLIED Class XII Part 1 · Ch 6 · **Page 160** Open PDF ↗

(1) BA and DC **(2)** AB and DC
(3) BA and CD **(4)** AB and CD

4. **NEET 2024 · Q44** A sheet is placed on a horizontal surface in front of a strong magnetic pole. A force is needed to:

A. hold the sheet there if it is magnetic.
B. hold the sheet there if it is non-magnetic.
C. move the sheet away from the pole with uniform velocity if it is conducting.
D. move the sheet away from the pole with uniform velocity if it is both, non-conducting and non-polar.

Choose the correct statement(s) from the options given below:

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(1) C only **(2)** B and D only
(3) A and C only **(4)** A, C and D only

5. **NEET 2022 · Q48** A big circular coil of 1000 turns and average radius 10 m is rotating about its horizontal diameter at 2 rad s^{-1}. If the vertical component of earth's magnetic field at that place is 2×10^{-5} T and electrical resistance of the coil is 12.56 Ω, then the maximum induced current in the coil will be:

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(1) 2 A **(2)** 0.25 A
(3) 1.5 A **(4)** 1 A

6. **NEET 2019 · Q41** A 800 turn coil of effective area 0.05 m^2 is kept perpendicular to a magnetic field 5×10^{-5} T. When the plane of the coil is rotated by 90° around any of its coplanar axis in 0.1 s, the emf induced in the coil will be:

DIRECT Class XII Part 1 · Ch 6 · **Page 158** · emf = -N dPhi/dt Open PDF ↗

(1) 2 V **(2)** 0.2 V
(3) 2×10^{-3} V **(4)** 0.02 V

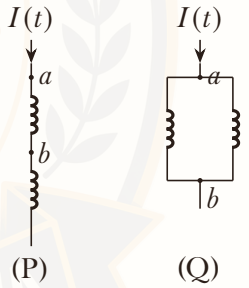
7. **NEET 2017 · Q179** A long solenoid of diameter 0.1 m has 2×10^4 turns per meter. At the centre of the solenoid, a coil of 100 turns and radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid reduces at a constant rate to 0 A from 4 A in 0.05 s. If the resistance of the coil is $10\pi^2$ Ω, the total charge flowing through the coil during this time is:

APPLIED Class XII Part 1 · Ch 6 · **Page 157** · Q = N delta-phi / R Open PDF ↗

(1) 16 μC **(2)** 32 μC
(3) 16π μC **(4)** 32π μC

Self and mutual inductance 8 Qs

8. **NEET 2026 · Q3** Two identical inductors are connected in two different configurations P and Q, where a time varying current $I(t)$ is flowing, as shown in the figure. The induced emf between points a and b for configuration P is E_P and that for configuration Q is E_Q. The ratio E_P/E_Q is: [Neglect the effect of mutual inductance.]



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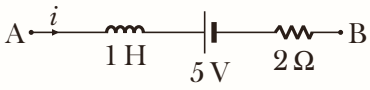
(1) $\frac{1}{4}$ **(2)** $\frac{1}{2}$ **(3)** 1 **(4)** 2

9. **NEET 2026 · Q5** Consider a long solenoid of length l and radius r. If n is the number of turns per unit length and μ_0 is the permeability of free space, the inductance of the solenoid is:

DIRECT Class XII Part 1 · Ch 6 · **Page 168** · L=mu0 n^2 A l Open PDF ↗

(1) $\mu_0 \pi n^2 r^2 l$ **(2)** $\mu_0 n^2 r^2 l$
(3) $(\mu_0/2\pi) n^2 r^2 l$ **(4)** $2\mu_0 \pi n^2 r^2 l$

10. **NEET 2025 · Q11** AB is a part of an electrical circuit (see figure). The potential difference $V_A - V_B$, at the instant when current $i = 2$ A and is increasing at a rate of 1 A per second is:



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(1) 5 volt **(2)** 6 volt
(3) 9 volt **(4)** 10 volt

11. **NEET 2023 · Q19** The magnetic energy stored in an inductor of inductance 4 μH carrying a current of 2 A is:

DIRECT Class XII Part 1 · Ch 6 · **Page 169** · U=LI^2/2 Open PDF ↗

(1) 8 μJ **(2)** 4 μJ **(3)** 4 mJ **(4)** 8 mJ

12. **NEET 2021 · Q36** Two conducting circular loops of radii R_1 and R_2 are placed in the same plane with their centres coinciding. If $R_1 \gg R_2$, the mutual inductance M between them will be directly proportional to:

DIRECT Class XII Part 1 · Ch 6 · **Page 167** · Example 6.8 Open PDF ↗

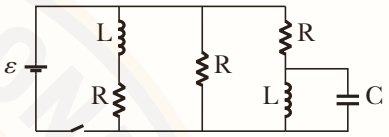
(1) $\frac{R_1}{R_2}$ **(2)** $\frac{R_2}{R_1}$ **(3)** $\frac{R_1^2}{R_2}$ **(4)** $\frac{R_2^2}{R_1}$

13. **NEET 2018 · Q21** The magnetic potential energy stored in a certain inductor is 25 mJ, when the current in the inductor is 60 mA. This inductor is of inductance

DIRECT Class XII Part 1 · Ch 6 · **Page 169** · U=LI^2/2 Open PDF ↗

(1) 1.389 H **(2)** 138.88 H
(3) 0.138 H **(4)** 13.89 H

14. **NEET 2017 · Q159** Figure shows a circuit that contains three identical resistors with resistance $R = 9.0\ \Omega$ each, two identical inductors with inductance $L = 2.0$ mH each, and an ideal battery with emf $\varepsilon = 18$ V. The current 'i' through the battery just after the switch closed is,



APPLIED Class XII Part 1 · Ch 6 · **Page 167** · inductor opposes sudden change Open PDF ↗

(1) 0.2 A **(2)** 2 A
(3) 0 ampere **(4)** 2 mA

15. **NEET 2016 · Q42** A long solenoid has 1000 turns. When a current of 4 A flows through it, the magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. The self-inductance of the solenoid is:

DIRECT Class XII Part 1 · Ch 6 · **Page 167** · N Phi = L I Open PDF ↗

(1) 4 H **(2)** 3 H **(3)** 2 H **(4)** 1 H

Electromagnetic Induction · Answer Key

1 (3)	**2** (2)	**3** (2)*	**4** (3)	**5** (4)	**6** (4)	**7** (2)	**8** (4)
9 (1)	**10** (4)	**11** (1)	**12** (4)	**13** (4)	**14** (2)*	**15** (4)	

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor. *: see note.

3. (NEET 2024 Q19) By Lenz's law, solenoid-1 (N pole receding) and solenoid-2 (S pole approaching) both oppose the change; with the winding shown this gives A to B and D to C.

14. (NEET 2017 Q159) The capacitor in the printed figure is not mentioned in the question; with only the three resistors and two inductors, just the middle resistor carries current at $t = 0$: $i = 18/9 = 2$ A.

Chapter Summary · Electromagnetic Induction

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	2	1	1	0	1	2	1	2	1	3	**15**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Magnetic flux	**1**	1	1	16 17 18 19 20 21 **22** 23 24 25 26
Faraday's law and Lenz's law	**6**	1	5	16 **17** 18 **19** 20 21 **22** 23 **24** 25 **26**
Self and mutual inductance	**8**	5	7	**16** **17** **18** 19 20 **21** 22 **23** 24 **25** **26**

Direct vs Applied

Direct from NCERT	**7**	47%
text, table, figure	6	
solved Example	1	
Exercise	0	
Applied	**8**	53%
Easy / Mod. / Hard	7 / 7 / 1	

What to focus on

- 15 questions in 11 papers, about 1.4 per paper; rank 17 of 28 chapters by number of questions.
- Most asked topic: **Self and mutual inductance** (8 questions, in 7 of the 11 years).
- 1 questions are NCERT solved Examples or Exercises: Example 6.8 (Q 12). Work these by hand.
- NCERT pages that gave three or more questions: page 168 (3 questions), page 167 (3 questions). Read these pages line by line.
- Difficulty: 7 easy, 7 moderate, 1 hard (hard: Q 7).
- 8 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 7

Alternating Current

17 questions · 3 topics · NEET 2016–2026

AC power and rms values — 2 Qs

1. NEET 2022 · Q23 The peak voltage of the ac source is equal to:

DIRECT — Class XII Part 1 · Ch 7 · **Page 180** · Vm=sqrt2 Vrms Open PDF ↗

(1) $1/\sqrt{2}$ times the rms value of the ac source
(2) the value of voltage supplied to the circuit
(3) the rms value of the ac source
(4) $\sqrt{2}$ times the rms value of the ac source

2. NEET 2018 · Q28 An inductor 20 mH, a capacitor 100 μF and a resistor 50 Ω are connected in series across a source of emf, $V = 10 \sin 314t$. The power loss in the circuit is

APPLIED — Class XII Part 1 · Ch 7 · **Page 190** Open PDF ↗

(1) 2.74 W **(2)** 0.43 W
(3) 0.79 W **(4)** 1.13 W

AC circuits (reactance, LCR, phase) — 12 Qs

3. NEET 2026 · Q30 An ac voltage $V = 220 \sin(2 \times 10^3 t)$ volt is applied to a series LCR circuit. Then the current amplitude in this circuit is: (Given: $L = 10$ mH, $C = 25\ \mu$F, $R = 100\ \Omega$)

APPLIED — Class XII Part 1 · Ch 7 · **Page 188** Open PDF ↗

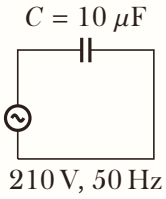
(1) 2.2 A **(2)** 5.5 A
(3) 11.0 A **(4)** 22.0 A

4. NEET 2025 · Q16 To an ac power supply of 220 V at 50 Hz, a resistor of 20 Ω, a capacitor of reactance 25 Ω and an inductor of reactance 45 Ω are connected in series. The corresponding current in the circuit and the phase angle between the current and the voltage is, respectively

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(1) 7.8 A and 30° **(2)** 7.8 A and 45°
(3) 15.6 A and 30° **(4)** 15.6 A and 45°

5. NEET 2024 · Q38 A 10 μF capacitor is connected to a 210 V, 50 Hz source as shown in figure. The peak current in the circuit is nearly ($\pi = 3.14$):



DIRECT — Class XII Part 1 · Ch 7 · **Page 185** · Example 7.4 Open PDF ↗

(1) 0.35 A **(2)** 0.58 A
(3) 0.93 A **(4)** 1.20 A

6. **NEET 2023 · Q20** In a series LCR circuit, the inductance L is 10 mH, capacitance C is 1 μF and resistance R is 100 Ω. The frequency at which resonance occurs is:

DIRECT Class XII Part 1 · Ch 7 · **Page 188** · omega0=1/sqrt(LC) Open PDF ↗

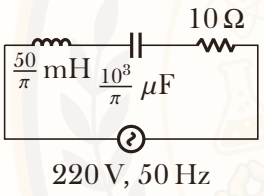
(1) 1.59 kHz **(2)** 15.9 rad/s
(3) 15.9 kHz **(4)** 1.59 rad/s

7. **NEET 2023 · Q35** An ac source is connected to a capacitor C. Due to decrease in its operating frequency:

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(1) capacitive reactance remains constant
(2) capacitive reactance decreases.
(3) displacement current increases.
(4) displacement current decreases.

8. **NEET 2023 · Q38** The net impedance of circuit (as shown in figure) will be:



APPLIED Class XII Part 1 · Ch 7 · **Page 187** Open PDF ↗

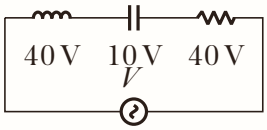
(1) 25 Ω **(2)** $10\sqrt{2}$ Ω
(3) 15 Ω **(4)** $5\sqrt{5}$ Ω

9. **NEET 2022 · Q40** A series LCR circuit with inductance 10 H, capacitance 10 μF, resistance 50 Ω is connected to an ac source of voltage, $V = 200\sin(100t)$ volt. If the resonant frequency of the LCR circuit is ν_0 and the frequency of the ac source is ν, then:

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(1) $\nu = 100$ Hz; $\nu_0 = \frac{100}{\pi}$ Hz
(2) $\nu_0 = \nu = 50$ Hz
(3) $\nu_0 = \nu = \frac{50}{\pi}$ Hz
(4) $\nu_0 = \frac{50}{\pi}$ Hz, $\nu = 50$ Hz

10. **NEET 2021 · Q20** An inductor of inductance L, a capacitor of capacitance C and a resistor of resistance 'R' are connected in series to an ac source of potential difference 'V' volts as shown in figure. Potential difference across L, C and R is 40 V, 10 V and 40 V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}$ A. The impedance of the circuit is:



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(1) $4\sqrt{2}$ Ω **(2)** $5/\sqrt{2}$ Ω
(3) 4 Ω **(4)** 5 Ω

11. **NEET 2020 · Q154** A 40 μF capacitor is connected to a 200 V, 50 Hz ac supply. The rms value of the current in the circuit is, nearly:

DIRECT Class XII Part 1 · Ch 7 · **Page 185** · Example 7.4 Open PDF ↗

(1) 1.7 A **(2)** 2.05 A
(3) 2.5 A **(4)** 25.1 A

12. **NEET 2020 · Q169** A series LCR circuit is connected to an ac voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{3}$. If instead C is removed from the circuit, the phase difference is again $\frac{\pi}{3}$ between current and voltage. The power factor of the circuit is:

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(1) zero **(2)** 0.5 **(3)** 1.0 **(4)** −1.0

13. **NEET 2016 · Q40** An inductor 20 mH, a capacitor 50 μF and a resistor 40 Ω are connected in series across a source of emf $V = 10\sin 340t$. The power loss in A.C. circuit is:

APPLIED Class XII Part 1 · Ch 7 · **Page 186** Open PDF ↗

(1) 0.51 W **(2)** 0.67 W
(3) 0.76 W **(4)** 0.89 W

14. **NEET 2016 · Q43** A small signal voltage $V(t) = V_0 \sin \omega t$ is applied across an ideal capacitor C:

DIRECT Class XII Part 1 · Ch 7 · **Page 183** · average power in a capacitor is zero Open PDF ↗

(1) Current $I(t)$, lags voltage $V(t)$ by 90°.
(2) Over a full cycle the capacitor C does not consume any energy from the voltage source.
(3) Current $I(t)$ is in phase with voltage $V(t)$.
(4) Current $I(t)$ leads voltage $V(t)$ by 180°.

Transformer and power transmission 3 Qs

15. **NEET 2024 · Q20** In an ideal transformer, the turns ratio is $\frac{N_p}{N_s} = \frac{1}{2}$. The ratio $V_s : V_p$ is equal to (the symbols carry their usual meaning):

DIRECT Class XII Part 1 · Ch 7 · **Page 195** · Vs/Vp=Ns/Np Open PDF ↗

(1) 1:4 **(2)** 1:2 **(3)** 2:1 **(4)** 1:1

16. **NEET 2023 · Q4** A 12 V, 60 W lamp is connected to the secondary of a step down transformer, whose primary is connected to ac mains of 220 V. Assuming the transformer to be ideal, what is the current in the primary winding?

DIRECT Class XII Part 1 · Ch 7 · **Page 195** · ideal transformer Open PDF ↗

(1) 0.37 A **(2)** 0.27 A
(3) 2.7 A **(4)** 3.7 A

17. **NEET 2021 · Q46** A step down transformer connected to an ac mains supply of 220 V is made to operate at 11 V, 44 W lamp. Ignoring power losses in the transformer, what is the current in the primary circuit?

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(1) 0.2 A **(2)** 0.4 A **(3)** 2 A **(4)** 4 A

Alternating Current · Answer Key

1 (4)	**2** (3)	**3** (1)	**4** (2)	**5** (3)	**6** (1)	**7** (4)	**8** (4)
9 (3)	**10** (4)*	**11** (3)	**12** (3)	**13** (1)*	**14** (2)	**15** (3)	**16** (2)
17 (1)							

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor. *: see note.

10. (NEET 2021 Q20) Official key: 5 Ω, taking the given voltages as rms values (V_{rms} = 50 V, I_{rms} = 10 A).

13. (NEET 2016 Q40) The exact value is about 0.46 W; (1) 0.51 W is the nearest option and is the intended answer.

Chapter Summary · Alternating Current

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	2	0	1	0	2	2	2	4	2	1	1	**17**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
AC power and rms values	**2**	1	2	16 17 **18** 19 20 21 **22** 23 24 25 26
AC circuits (reactance, LCR, phase)	**12**	4	8	**16** 17 18 19 **20 21 22 23 24 25 26**
Transformer and power transmission	**3**	3	3	16 17 18 19 20 **21** 22 **23 24** 25 26

Direct vs Applied

Direct from NCERT	**8**	47%
text, table, figure	6	
solved Example	2	
Exercise	0	
Applied	**9**	53%
Easy / Mod. / Hard	9 / 7 / 1	

What to focus on

- 17 questions in 11 papers, about 1.5 per paper; rank 11 of 28 chapters by number of questions.
- Most asked topic: **AC circuits (reactance, LCR, phase)** (12 questions, in 8 of the 11 years).
- Asked in 8 or more of the 11 years: AC circuits (reactance, LCR, phase).
- 2 questions are NCERT solved Examples or Exercises: Example 7.4 (Q 5), Example 7.4 (Q 11). Work these by hand.
- NCERT pages that gave three or more questions: page 188 (3 questions), page 187 (3 questions), page 185 (3 questions), page 195 (3 questions). Read these pages line by line.
- Difficulty: 9 easy, 7 moderate, 1 hard (hard: Q 2).
- 9 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 8

Electromagnetic Waves

17 questions · 3 topics · NEET 2016–2026

Displacement current and Maxwell's equations

4 Qs

1. NEET 2025 · Q26 A parallel plate capacitor made of circular plates is being charged such that the surface charge density on its plates is increasing at a constant rate with time. The magnetic field arising due to displacement current is:

DIRECT Class XII Part 1 · Ch 8 · **Page 204** · B between the plates Open PDF ↗

(1) zero at all places

(2) constant between the plates and zero outside the plates

(3) non-zero everywhere with maximum at the imaginary cylindrical surface connecting peripheries of the plates

(4) zero between the plates and non-zero outside

2. NEET 2024 · Q46 A parallel plate capacitor is charged by connecting it to a battery through a resistor. If I is the current in the circuit, then in the gap between the plates:

DIRECT Class XII Part 1 · Ch 8 · **Page 204** · i_d = i Open PDF ↗

(1) displacement current of magnitude greater than I flows but can be in any direction.

(2) there is no current.

(3) displacement current of magnitude equal to I flows in the same direction as I.

(4) displacement current of magnitude equal to I flows in a direction opposite to that of I.

3. NEET 2021 · Q2 A capacitor of capacitance 'C', is connected across an ac source of voltage V, given by $V = V_0 \sin \omega t$. The displacement current between the plates of the capacitor, would then be given by:

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(1) $I_d = V_0 \omega C \cos \omega t$ **(2)** $I_d = \dfrac{V_0}{\omega C} \cos \omega t$

(3) $I_d = \dfrac{V_0}{\omega C} \sin \omega t$ **(4)** $I_d = V_0 \omega C \sin \omega t$

4. NEET 2019 · Q43 A parallel plate capacitor of capacitance 20 μF is being charged by a voltage source whose potential is changing at the rate of 3 V/s. The conduction current through the connecting wires, and the displacement current through the plates of the capacitor, would be, respectively:

APPLIED Class XII Part 1 · Ch 8 · **Page 203** Open PDF ↗

(1) zero, 60 μA **(2)** 60 μA, 60 μA

(3) 60 μA, zero **(4)** zero, zero

Electromagnetic waves (nature, E/B = c) 10 Qs

5. **NEET 2026 · Q38** An electromagnetic wave travelling in a lossless dielectric medium having a dielectric constant $\varepsilon_r = 9$, has the electric field $E_x = E_0 \sin(kz - 2\pi \times 10^6 t)\ \text{V m}^{-1}$ where E_0 is the amplitude and k is the wave vector. Among the following options, the **incorrect** choice is

APPLIED Class XII Part 1 · Ch 8 · **Page 207** Open PDF ↗

(1) The speed of the electromagnetic wave inside the medium is $10^8\ \text{m s}^{-1}$

(2) The wavelength of the electromagnetic wave inside the medium is 300 m

(3) The magnetic field is given by the relation $B_y = \frac{B_0}{v}\sin(kz - 2\pi \times 10^6 t)$ where v is the speed of the electromagnetic wave inside the medium

(4) The direction of propagation of the electromagnetic wave is along $+z$

6. **NEET 2025 · Q9** The electric field in a plane electromagnetic wave is given by $E_z = 60\cos(5x + 1.5 \times 10^9 t)\ \text{V/m}$. Then expression for the corresponding magnetic field is (here subscripts denote the direction of the field):

DIRECT Class XII Part 1 · Ch 8 · **Page 208** · Example 8.2 Open PDF ↗

(1) $B_y = 2 \times 10^{-7}\cos(5x + 1.5 \times 10^9 t)\ \text{T}$

(2) $B_x = 2 \times 10^{-7}\cos(5x + 1.5 \times 10^9 t)\ \text{T}$

(3) $B_z = 60\cos(5x + 1.5 \times 10^9 t)\ \text{T}$

(4) $B_y = 60\sin(5x + 1.5 \times 10^9 t)\ \text{T}$

7. **NEET 2024 · Q45** The property which is not of an electromagnetic wave travelling in free space is that:

DIRECT Class XII Part 1 · Ch 8 · **Page 205** · accelerated charges radiate Open PDF ↗

(1) they originate from charges moving with uniform speed.

(2) they are transverse in nature.

(3) the energy density in electric field is equal to energy density in magnetic field.

(4) they travel with a speed equal to $\frac{1}{\sqrt{\mu_0 \varepsilon_0}}$.

8. **NEET 2023 · Q6** In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of 2.0×10^{10} Hz and amplitude $48\ \text{V m}^{-1}$. Then the amplitude of oscillating magnetic field is: (Speed of light in free space $= 3 \times 10^8\ \text{m s}^{-1}$)

DIRECT Class XII Part 1 · Ch 8 · **Page 207** · B0=E0/c Open PDF ↗

(1) $1.6 \times 10^{-6}\ \text{T}$ **(2)** $1.6 \times 10^{-9}\ \text{T}$

(3) $1.6 \times 10^{-8}\ \text{T}$ **(4)** $1.6 \times 10^{-7}\ \text{T}$

9. **NEET 2022 · Q10** When light propagates through a material medium of relative permittivity ε_r and relative permeability μ_r, the velocity of light, v is given by: (c – velocity of light in vacuum)

DIRECT Class XII Part 1 · Ch 8 · **Page 207** · v=1/sqrt(mu eps) Open PDF ↗

(1) $v = \frac{c}{\sqrt{\varepsilon_r \mu_r}}$ **(2)** $v = c$

(3) $v = \sqrt{\frac{\mu_r}{\varepsilon_r}}$ **(4)** $v = \sqrt{\frac{\varepsilon_r}{\mu_r}}$

10. **NEET 2021 · Q13** For a plane electromagnetic wave propagating in x-direction, which one of the following combination gives the correct possible directions for electric field (E) and magnetic field (B) respectively?

APPLIED Class XII Part 1 · Ch 8 · **Page 207** Open PDF ↗

(1) $\hat{j} + \hat{k},\ \hat{j} + \hat{k}$ **(2)** $-\hat{j} + \hat{k},\ -\hat{j} - \hat{k}$

(3) $\hat{j} + \hat{k},\ -\hat{j} - \hat{k}$ **(4)** $-\hat{j} + \hat{k},\ -\hat{j} + \hat{k}$

11. **NEET 2020 · Q137** The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is: (c = speed of electromagnetic waves)

APPLIED Class XII Part 1 · Ch 8 · **Page 207** · E and B carry equal energy Open PDF ↗

(1) $c : 1$ **(2)** 1:1 **(3)** $1 : c$ **(4)** $1 : c^2$

12. **NEET 2018 · Q18** An em wave is propagating in a medium with a velocity $\vec{V} = V\hat{i}$. The instantaneous oscillating electric field of this em wave is along $+y$ axis. Then the direction of oscillating magnetic field of the em wave will be along

APPLIED Class XII Part 1 · Ch 8 · **Page 207** Open PDF ↗

(1) $-y$ direction **(2)** $+z$ direction
(3) $-z$ direction **(4)** $-x$ direction

13. **NEET 2017 · Q145** In an electromagnetic wave in free space the root mean square value of the electric field is $E_{rms} = 6\,V/m$. The peak value of the magnetic field is:

APPLIED Class XII Part 1 · Ch 8 · **Page 212** · E0/B0=c Open PDF ↗

(1) $2.83 \times 10^{-8}\,T$ **(2)** $0.70 \times 10^{-8}\,T$
(3) $4.23 \times 10^{-8}\,T$ **(4)** $1.41 \times 10^{-8}\,T$

14. **NEET 2016 · Q8** Out of the following options which one can be used to produce a propagating electromagnetic wave?

DIRECT Class XII Part 1 · Ch 8 · **Page 205** · accelerated charges radiate Open PDF ↗

(1) A charge moving at constant velocity
(2) A stationary charge
(3) A chargeless particle
(4) An accelerating charge

Electromagnetic spectrum 3 Qs

15. **NEET 2026 · Q43** The following table presents the part of the electromagnetic spectrum and their corresponding major applications.

Part of the spectrum	*Application*
P. Microwave	I. For purifying the water
Q. UV rays	II. For warming the food
R. Gamma rays	III. For AM and FM communication systems
S. Radio wave	IV. For treating the cancer cells

The correct option is:

DIRECT Class XII Part 1 · Ch 8 · **Page 209** · uses listed for each band Open PDF ↗

(1) P-I, Q-II, R-III, S-IV
(2) P-I, Q-IV, R-II, S-III
(3) P-II, Q-I, R-IV, S-III
(4) P-II, Q-IV, R-III, S-I

16. **NEET 2022 · Q6** Match List-I with List-II:

List-I (EM waves)		*List-II* (λ)	
(a)	AM radio waves	(i)	10^{-10} m
(b)	Microwaves	(ii)	10^{2} m
(c)	Infrared radiations	(iii)	10^{-2} m
(d)	X-rays	(iv)	10^{-4} m

Choose the correct answer from the options given below:

DIRECT Class XII Part 1 · Ch 8 · **Page 209** · wavelength ranges Open PDF ↗

(1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(3) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
(4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

17. **NEET 2019 · Q8** Which colour of the light has the longest wavelength?

DIRECT Class XII Part 1 · Ch 8 · **Page 210** · visible 700-400 nm Open PDF ↗

(1) red **(2)** blue **(3)** green **(4)** violet

Electromagnetic Waves · Answer Key

1 (3)	**2** (3)	**3** (1)	**4** (2)	**5** (2)	**6** (1)	**7** (1)	**8** (4)
9 (1)	**10** (2)	**11** (2)	**12** (2)	**13** (1)	**14** (4)	**15** (3)	**16** (1)
17 (1)							

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Electromagnetic Waves

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	1	1	2	1	2	2	1	2	2	2	**17**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Displacement current and Maxwell's equations	**4**	2	4	16 17 18 **19** 20 **21** 22 23 **24** **25** 26
Electromagnetic waves (nature, E/B = c)	**10**	5	10	**16** **17** **18** 19 **20** **21** **22** **23** **24** **25** **26**
Electromagnetic spectrum	**3**	3	3	16 17 18 **19** 20 21 **22** 23 24 25 **26**

Direct vs Applied

Direct from NCERT	**10**	59%
text, table, figure	9	
solved Example	1	
Exercise	0	
Applied	**7**	41%
Easy / Mod. / Hard	13 / 4 / 0	

What to focus on

- 17 questions in 11 papers, about 1.5 per paper; rank 12 of 28 chapters by number of questions.
- Most asked topic: **Electromagnetic waves (nature, E/B = c)** (10 questions, in 10 of the 11 years).
- Asked in 8 or more of the 11 years: Electromagnetic waves (nature, E/B = c).
- 1 questions are NCERT solved Examples or Exercises: Example 8.2 (Q 6). Work these by hand.
- NCERT pages that gave three or more questions: page 207 (6 questions). Read these pages line by line.
- Difficulty: 13 easy, 4 moderate, 0 hard.
- 7 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 9

Ray Optics and Optical Instruments

27 questions · 5 topics · NEET 2016–2026

Reflection and spherical mirrors 3 Qs

1. **NEET 2018 · Q20** An object is placed at a distance of 40 cm from a concave mirror of focal length 15 cm. If the object is displaced through a distance of 20 cm towards the mirror, the displacement of the image will be

APPLIED Class XII Part 2 · Ch 9 · **Page 225** Open PDF ↗

(1) 30 cm towards the mirror
(2) 36 cm away from the mirror
(3) 30 cm away from the mirror
(4) 36 cm towards the mirror

2. **NEET 2017 · Q169** A beam of light from a source L is incident normally on a plane mirror fixed at a certain distance x from the source. The beam is reflected back as a spot on a scale placed just above the source L. When the mirror is rotated through a small angle θ, the spot of the light is found to move through a distance y on the scale. The angle θ is given by:

APPLIED Class XII Part 2 · Ch 9 · **Page 222** · laws of reflection Open PDF ↗

(1) $\frac{y}{x}$ **(2)** $\frac{x}{2y}$ **(3)** $\frac{x}{y}$ **(4)** $\frac{y}{2x}$

3. **NEET 2016 · Q44** Match the corresponding entries of column 1 with column 2. [Where m is the magnification produced by the mirror]

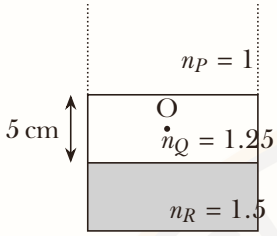
Column 1		*Column 2*	
(A)	$m = -2$	(a)	Convex mirror
(B)	$m = -\frac{1}{2}$	(b)	Concave mirror
(C)	$m = +2$	(c)	Real image
(D)	$m = +\frac{1}{2}$	(d)	Virtual image

APPLIED Class XII Part 2 · Ch 9 · **Page 226** · sign of magnification Open PDF ↗

(1) A→b and c; B→b and c; C→b and d; D→a and d
(2) A→a and c; B→a and d; C→a and b; D→c and d
(3) A→a and d; B→b and c; C→b and d; D→b and c
(4) A→c and d; B→b and d; C→b and c; D→a and d

Refraction and total internal reflection 5 Qs

4. **NEET 2026 · Q22** Consider three media P, Q and R with refractive indices 1, 1.25, and 1.5 respectively. The medium Q having a thickness of 5 cm is placed between extended media P and R as shown in the figure. An object O is placed at the centre of medium Q. If viewed from medium P near the normal direction, the apparent depth of O is h_1. For similar observation from medium R, the apparent depth is h_2. The value of $|h_1 - h_2|$, in cm, is:



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(1) 0 **(2)** 1 **(3)** 2 **(4)** 3

5. **NEET 2023 · Q8** Light travels a distance x in time t_1 in air and $10x$ in time t_2 in another denser medium. What is the critical angle for this medium?

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(1) $\sin^{-1}\left(\frac{10t_1}{t_2}\right)$ **(2)** $\sin^{-1}\left(\frac{t_2}{t_1}\right)$

(3) $\sin^{-1}\left(\frac{10t_2}{t_1}\right)$ **(4)** $\sin^{-1}\left(\frac{t_1}{10t_2}\right)$

6. **NEET 2022 · Q30** A light ray falls on a glass surface of refractive index $\sqrt{3}$, at an angle 60°. The angle between the refracted and reflected rays would be:

APPLIED Class XII Part 2 · Ch 9 · **Page 228** Open PDF ↗

(1) 120° **(2)** 30° **(3)** 60° **(4)** 90°

7. **NEET 2022 · Q38** Two transparent media A and B are separated by a plane boundary. The speed of light in those media are 1.5×10^8 m/s and 2.0×10^8 m/s, respectively. The critical angle for a ray of light for these two media is:

APPLIED Class XII Part 2 · Ch 9 · **Page 229** Open PDF ↗

(1) $\tan^{-1}(0.750)$ **(2)** $\sin^{-1}(0.500)$

(3) $\sin^{-1}(0.750)$ **(4)** $\tan^{-1}(0.500)$

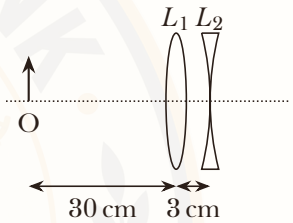
8. **NEET 2019 · Q28** In total internal reflection when the angle of incidence is equal to the critical angle for the pair of media in contact, what will be angle of refraction?

DIRECT Class XII Part 2 · Ch 9 · **Page 229** · i = ic gives r = 90 deg Open PDF ↗

(1) 180°
(2) 0°
(3) equal to angle of incidence
(4) 90°

Lenses and lens maker's formula 8 Qs

9. **NEET 2026 · Q29** The lens combination as shown in the figure, consists of two lenses, L_1 and L_2, of the focal lengths +10 cm and −10 cm, respectively. The position of the image formed is:



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(1) 20 cm to the left of the concave lens
(2) 60 cm to the left of the concave lens
(3) 30 cm to the right of the concave lens
(4) 60 cm to the right of the concave lens

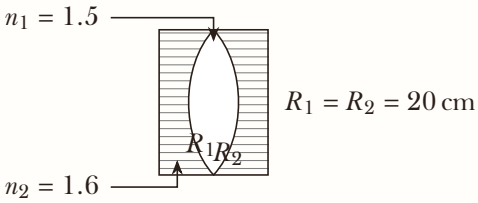
10. **NEET 2025 · Q13** In a certain camera, a combination of four similar thin convex lenses are arranged axially in contact. Then the power of the combination and the total magnification in comparison to the power (p) and magnification (m) for each lens will be, respectively

DIRECT Class XII Part 2 · Ch 9 · **Page 238** · P=sum Pi, m=m1 m2 m3 Open PDF ↗

(1) $4p$ and $4m$ **(2)** p^4 and $4m$

(3) $4p$ and m^4 **(4)** p^4 and m^4

11. **NEET 2023 · Q45** In the figure shown here, what is the equivalent focal length of the combination of lenses (Assume that all layers are thin)?



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(1) −50 cm **(2)** 40 cm
(3) −40 cm **(4)** −100 cm

12. **NEET 2023 · Q46** Two thin lenses are of same focal lengths (f), but one is convex and the other one is concave. When they are placed in contact with each other, the equivalent focal length of the combination will be:

DIRECT Class XII Part 2 · Ch 9 · **Page 237** · 1/f=1/f1+1/f2 Open PDF ↗

(1) Infinite **(2)** Zero
(3) $f/4$ **(4)** $f/2$

13. **NEET 2022 · Q8** A biconvex lens has radii of curvature, 20 cm each. If the refractive index of the material of the lens is 1.5, the power of the lens is:

DIRECT Class XII Part 2 · Ch 9 · **Page 234** · lens maker's formula Open PDF ↗

(1) infinity **(2)** +2 D
(3) +20 D **(4)** +5 D

14. **NEET 2021 · Q8** A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance 'd' in cm will be:

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(1) 25 **(2)** 15 **(3)** 50 **(4)** 30

15. **NEET 2021 · Q41** A point object is placed at a distance of 60 cm from a convex lens of focal length 30 cm. If a plane mirror were put perpendicular to the principal axis of the lens and at a distance of 40 cm from it, the final image would be formed at a distance of:

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(1) 20 cm from the lens, it would be a real image.
(2) 30 cm from the lens, it would be a real image.
(3) 30 cm from the plane mirror, it would be a virtual image.
(4) 20 cm from the plane mirror, it would be a virtual image.

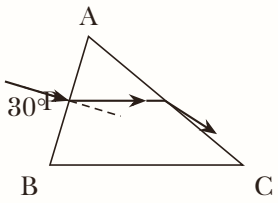
16. **NEET 2019 · Q11** Two similar thin equi-convex lenses, of focal length f each, are kept coaxially in contact with each other such that the focal length of the combination is F_1. When the space between the two lenses is filled with glycerin (which has the same refractive index ($\mu = 1.5$) as that of glass) then the equivalent focal length is F_2. The ratio $F_1 : F_2$ will be:

APPLIED Class XII Part 2 · Ch 9 · **Page 238** Open PDF ↗

(1) 2:1 **(2)** 1:2 **(3)** 2:3 **(4)** 3:4

Prism and dispersion 6 Qs

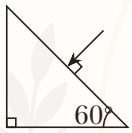
17. NEET 2024 · Q6 A light ray enters through a right angled prism at point P with the angle of incidence 30° as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC. The refractive index of the prism is:



APPLIED Class XII Part 2 · Ch 9 · Page 240 Open PDF ↗

(1) $\frac{\sqrt{3}}{2}$ (2) $\frac{\sqrt{5}}{4}$ (3) $\frac{\sqrt{5}}{2}$ (4) $\frac{\sqrt{3}}{4}$

18. NEET 2021 · Q5 Find the value of the angle of emergence from the prism. Refractive index of the glass is $\sqrt{3}$.



APPLIED Class XII Part 2 · Ch 9 · Page 239 Open PDF ↗

(1) 60° (2) 30° (3) 45° (4) 90°

19. NEET 2020 · Q140 A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ, then the angle of incidence is nearly equal to:

APPLIED Class XII Part 2 · Ch 9 · Page 239 Open PDF ↗

(1) $\frac{A}{2\mu}$ (2) $\frac{2A}{\mu}$ (3) μA (4) $\frac{\mu A}{2}$

20. NEET 2018 · Q19 The refractive index of the material of a prism is $\sqrt{2}$ and the angle of the prism is 30°. One of the two refracting surfaces of the prism is made a mirror inwards, by silver coating. A beam of monochromatic light entering the prism from the other face will retrace its path (after reflection from the silvered surface) if its angle of incidence on the prism is

APPLIED Class XII Part 2 · Ch 9 · Page 240 Open PDF ↗

(1) 30° (2) 45° (3) 60° (4) zero

21. NEET 2017 · Q152 A thin prism having refracting angle 10° is made of glass of refractive index 1.42. This prism is combined with another thin prism of glass of refractive index 1.7. This combination produces dispersion without deviation. The refracting angle of second prism should be:

APPLIED Class XII Part 2 · Ch 9 · Page 238 · thin prism delta=(mu-1)A Open PDF ↗

(1) 6° (2) 8° (3) 10° (4) 4°

22. NEET 2016 · Q22 The angle of incidence for a ray of light at a refracting surface of a prism is 45°. The angle of prism is 60°. If the ray suffers minimum deviation through the prism, the angle of minimum deviation and refractive index of the material of the prism respectively, are:

APPLIED Class XII Part 2 · Ch 9 · Page 240 · minimum deviation Open PDF ↗

(1) 45°; $\frac{1}{\sqrt{2}}$ (2) 30°; $\sqrt{2}$

(3) 45°; $\sqrt{2}$ (4) 30°; $\frac{1}{\sqrt{2}}$

Optical instruments (microscope, telescope) 5 Qs

23. NEET 2025 · Q2 A microscope has an objective of focal length 2 cm, eyepiece of focal length 4 cm and the tube length of 40 cm. If the distance of distinct vision of eye is 25 cm, the magnification in the microscope is

DIRECT Class XII Part 2 · Ch 9 · Page 244 · m=(L/fo)(D/fe) Open PDF ↗

(1) 100 (2) 125 (3) 150 (4) 250

24. **NEET 2024 · Q36** A small telescope has an objective of focal length 140 cm and an eye piece of focal length 5.0 cm. The magnifying power of telescope for viewing a distant object is:

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(1) 32 **(2)** 34 **(3)** 28 **(4)** 17

25. **NEET 2021 · Q18** A lens of large focal length and large aperture is best suited as an objective of an astronomical telescope since:

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(1) a large aperture contributes to the quality and visibility of the images.
(2) a large area of the objective ensures better light gathering power.
(3) a large aperture provides a better resolution.
(4) all of the above.

26. **NEET 2018 · Q10** An astronomical refracting telescope will have large angular magnification and high angular resolution, when it has an objective lens of

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(1) large focal length and large diameter
(2) large focal length and small diameter
(3) small focal length and large diameter
(4) small focal length and small diameter

27. **NEET 2016 · Q32** A astronomical telescope has objective and eyepiece of focal lengths 40 cm and 4 cm respectively. To view an object 200 cm away from the objective, the lenses must be separated by a distance:

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(1) 37.3 cm **(2)** 46.0 cm
(3) 50.0 cm **(4)** 54.0 cm

Ray Optics and Optical Instruments · Answer Key

1 (2)	**2** (4)	**3** (1)	**4** (2)	**5** (1)	**6** (4)	**7** (3)	**8** (4)
9 (2)	**10** (3)	**11** (4)	**12** (1)	**13** (4)	**14** (2)	**15** (4)	**16** (2)
17 (3)	**18** (1)	**19** (3)	**20** (2)	**21** (1)	**22** (2)	**23** (2)	**24** (3)
25 (4)	**26** (1)	**27** (4)					

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Ray Optics and Optical Instruments

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	3	2	3	2	1	4	3	3	2	2	2	**27**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Reflection and spherical mirrors	**3**	0	3	**16 17 18** 19 20 21 22 23 24 25 26
Refraction and total internal reflection	**5**	1	4	16 17 18 **19** 20 21 **22 23** 24 25 **26**
Lenses and lens maker's formula	**8**	3	6	16 17 18 **19** 20 **21 22 23** 24 **25 26**
Prism and dispersion	**6**	0	6	**16 17 18** 19 **20 21** 22 23 **24** 25 26
Optical instruments (microscope, telescope)	**5**	4	5	**16** 17 **18** 19 20 **21** 22 23 **24 25** 26

Direct vs Applied

Direct from NCERT	**8**	30%
text, table, figure	7	
solved Example	0	
Exercise	1	
Applied	**19**	70%
Easy / Mod. / Hard	9 / 16 / 2	

What to focus on

- 27 questions in 11 papers, about 2.5 per paper; rank 2 of 28 chapters by number of questions.
- Most asked topic: **Lenses and lens maker's formula** (8 questions, in 6 of the 11 years).
- 1 questions are NCERT solved Examples or Exercises: Exercise 9.27 (Q 24). Work these by hand.
- NCERT pages that gave three or more questions: page 238 (7 questions), page 229 (4 questions), page 240 (3 questions). Read these pages line by line.
- Difficulty: 9 easy, 16 moderate, 2 hard (hard: Q 11, 15).
- 19 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 10

Wave Optics

11 questions · 2 topics · NEET 2016–2026

Young's double-slit interference 9 Qs

1. NEET 2024 · Q35 If the monochromatic source in Young's double slit experiment is replaced by white light, then

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(1) all bright fringes will be of equal width.
(2) interference pattern will disappear.
(3) there will be a central dark fringe surrounded by a few coloured fringes.
(4) there will be a central bright white fringe surrounded by a few coloured fringes.

2. NEET 2023 · Q24 For Young's double slit experiment, two statements are given below:
Statement I: If screen is moved away from the plane of slits, angular separation of the fringes remains constant.
Statement II: If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.
In the light of the above statements, choose the *correct* answer from the options given below:

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(1) Statement I is false but Statement II is true.
(2) Both Statement I and Statement II are true.
(3) Both Statement I and Statement II are false.
(4) Statement I is true but Statement II is false.

3. NEET 2022 · Q29 In a Young's double slit experiment, a student observes 8 fringes in a certain segment of screen when a monochromatic light of 600 nm wavelength is used. If the wavelength of light is changed to 400 nm, then the number of fringes he would observe in the same region of the screen is:

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(1) 12 **(2)** 6 **(3)** 8 **(4)** 9

4. NEET 2020 · Q165 In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes:

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(1) double **(2)** half
(3) four times **(4)** one-fourth

5. NEET 2019 · Q5 In a double slit experiment, when light of wavelength 400 nm was used, the angular width of the first minima formed on a screen placed 1 m away, was found to be 0.2°. What will be the angular width of the first minima, if the entire experimental apparatus is immersed in water? ($\mu_{water} = 4/3$)

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(1) 0.266° **(2)** 0.15°
(3) 0.05° **(4)** 0.1°

6. NEET 2018 · Q9 In Young's double slit experiment the separation d between the slits is 2 mm, the wavelength λ of the light used is 5896 Å and distance D between the screen and slits is 100 cm. It is found that the angular width of the fringes is 0.20°. To increase the fringe angular width to 0.21° (with same λ and D) the separation between the slits needs to be changed to

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(1) 2.1 mm **(2)** 1.9 mm
(3) 1.8 mm **(4)** 1.7 mm

7. **NEET 2017 · Q137** Young's double slit experiment is first performed in air and then in a medium other than air. It is found that 8th bright fringe in the medium lies where 5th dark fringe lies in air. The refractive index of the medium is nearly:

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(1) 1.59 **(2)** 1.69 **(3)** 1.78 **(4)** 1.25

8. **NEET 2016 · Q6** In a diffraction pattern due to a single slit of width 'a', the first minimum is observed at an angle 30° when light of wavelength 5000 Å is incident on the slit. The first secondary maximum is observed at an angle of:

APPLIED Class XII Part 2 · Ch 10 · **Page 267** · single slit Open PDF ↗

(1) $\sin^{-1}\left(\frac{1}{4}\right)$ **(2)** $\sin^{-1}\left(\frac{2}{3}\right)$

(3) $\sin^{-1}\left(\frac{1}{2}\right)$ **(4)** $\sin^{-1}\left(\frac{3}{4}\right)$

9. **NEET 2016 · Q18** The intensity at the maximum in a Young's double slit experiment is I_0. Distance between two slits is $d = 5\lambda$, where λ is the wavelength of light used in the experiment. What will be the intensity in front of one of the slits on the screen placed at a distance $D = 10d$?

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(1) I_0 **(2)** $\frac{I_0}{4}$ **(3)** $\frac{3}{4}I_0$ **(4)** $\frac{I_0}{2}$

Polarisation and Malus' law — 2 Qs

10. **NEET 2025 · Q43** The intensity of transmitted light when a polaroid sheet, placed between two crossed polaroids at 22.5° from the polarization axis of one of the polaroid, is (I_0 is the intensity of polarised light after passing through the first polaroid):

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(1) $\frac{I_0}{2}$ **(2)** $\frac{I_0}{4}$ **(3)** $\frac{I_0}{8}$ **(4)** $\frac{I_0}{16}$

11. **NEET 2017 · Q174** Two Polaroids P_1 and P_2 are placed with their axis perpendicular to each other. Unpolarised light I_0 is incident on P_1. A third polaroid P_3 is kept in between P_1 and P_2 such that its axis makes an angle 45° with that of P_1. The intensity of transmitted light through P_2 is:

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(1) $\frac{I_0}{4}$ **(2)** $\frac{I_0}{8}$ **(3)** $\frac{I_0}{16}$ **(4)** $\frac{I_0}{2}$

Wave Optics · Answer Key

1 (4)	**2** (4)	**3** (1)	**4** (3)	**5** (2)	**6** (2)	**7** (3)	**8** (4)
9 (4)	**10** (3)	**11** (2)					

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Wave Optics

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	2	2	1	1	1	0	1	1	1	1	0	**11**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Young's double-slit interference	**9**	0	8	**16 17 18 19 20** 21 **22 23 24** 25 26
Polarisation and Malus' law	**2**	1	2	16 **17** 18 19 20 21 22 23 24 **25** 26

Direct vs Applied

Direct from NCERT	**1**	9%
text, table, figure	0	
solved Example	1	
Exercise	0	
Applied	**10**	91%
Easy / Mod. / Hard	3 / 7 / 1	

What to focus on

- 11 questions in 11 papers, about 1.0 per paper; rank 24 of 28 chapters by number of questions.
- Most asked topic: **Young's double-slit interference** (9 questions, in 8 of the 11 years).
- Asked in 8 or more of the 11 years: Young's double-slit interference.
- 1 questions are NCERT solved Examples or Exercises: Example 10.2 (Q 10). Work these by hand.
- NCERT pages that gave three or more questions: page 266 (7 questions). Read these pages line by line.
- Difficulty: 3 easy, 7 moderate, 1 hard (hard: Q 9).
- 10 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 11

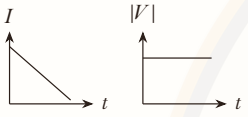
Dual Nature of Radiation and Matter

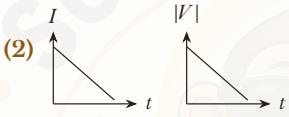
22 questions · 3 topics · NEET 2016–2026

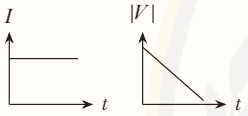
Photoelectric effect 9 Qs

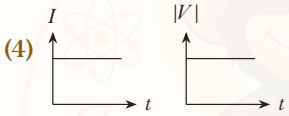
1. NEET 2026 · Q8 A beam of light falls on a metal surface such that photo-electrons are generated. If power of the light source starts to decrease linearly with time t, then variation of the photocurrent I and magnitude of the stopping potential $|V|$ with time is best represented by:

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(1) 

(2) 

(3) 

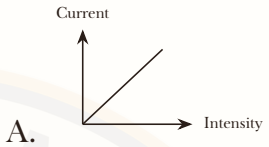
(4) 

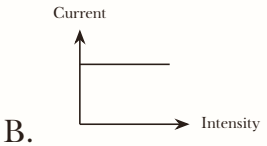
2. NEET 2026 · Q18 A ray of light with wavelength λ is incident on three different photoelectric cells namely 1, 2 and 3. The threshold wavelength of these photoelectric cells are λ_1, λ_2, and λ_3, respectively and the magnitude of stopping potentials of these cells are V_1, V_2 and V_3, respectively. The relation between λ and threshold wavelengths are $\lambda_1 < \lambda$, $\lambda_2 > \lambda$ and $\lambda_3 \gg \lambda$. The correct option is:

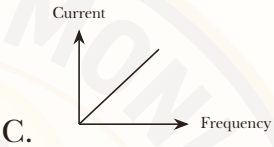
APPLIED Class XII Part 2 · Ch 11 · **Page 281** Open PDF ↗

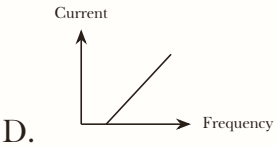
(1) $V_1 = 0, V_2 < V_3$ (2) $V_1 = 0, V_2 > V_3$
(3) $V_1 > V_2, V_3 = 0$ (4) $V_1 < V_2, V_3 = 0$

3. NEET 2025 · Q38 Which of the following options represent the variation of photoelectric current with property of light shown on the x-axis?

A. 

B. 

C. 

D. 

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(1) A only (2) A and C
(3) A and D (4) B and D

4. NEET 2023 · Q1 The work functions of Caesium (Cs), Potassium (K) and Sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV, which of these photosensitive surfaces may emit photoelectrons?

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(1) Na only (2) Cs only
(3) Both Na and K (4) K only

5. NEET 2022 · Q3 When two monochromatic lights of frequency, ν and $\frac{\nu}{2}$ are incident on a photoelectric metal, their stopping potential becomes $\frac{V_s}{2}$ and V_s respectively. The threshold frequency for this metal is:

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(1) $\frac{3}{2}\nu$ (2) 2ν (3) 3ν (4) $\frac{2}{3}\nu$

6. **NEET 2020 · Q157** Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled?

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(1) doubled **(2)** four times
(3) one-fourth **(4)** zero

7. **NEET 2018 · Q14** When the light of frequency $2\nu_0$ (where ν_0 is threshold frequency), is incident on a metal plate, the maximum velocity of electrons emitted is v_1. When the frequency of the incident radiation is increased to $5\nu_0$, the maximum velocity of electrons emitted from the same plate is v_2. The ratio of v_1 to v_2 is

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(1) 4:1 **(2)** 1:4 **(3)** 1:2 **(4)** 2:1

8. **NEET 2017 · Q158** The photoelectric threshold wavelength of silver is 3250×10^{-10} m. The velocity of the electron ejected from a silver surface by ultraviolet light of wavelength 2536×10^{-10} m is: (Given $h = 4.14\times10^{-15}$ eV s and $c = 3 \times 10^8$ m s^{-1})

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(1) $\approx 0.6 \times 10^6$ m s^{-1}
(2) $\approx 61 \times 10^3$ m s^{-1}
(3) $\approx 0.3 \times 10^6$ m s^{-1}
(4) $\approx 6 \times 10^5$ m s^{-1}

9. **NEET 2016 · Q29** When a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential is $\frac{V}{4}$. The threshold wavelength for the metallic surface is:

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(1) 4λ **(2)** 5λ **(3)** $\frac{5}{2}\lambda$ **(4)** 3λ

Photons and photon momentum 4 Qs

10. **NEET 2026 · Q19** A photon and an electron, each of 20 eV energy, move in free space. The ratio of linear momentum of electron p_e to that of photon p_{Ph}, $\frac{p_e}{p_{Ph}}$ is: [Take speed of light = 3×10^8 m s^{-1}, charge of electron = -1.6×10^{-19} C and mass of electron = 9×10^{-31} kg]

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(1) $\frac{2}{450}$ **(2)** $\frac{1}{250}$ **(3)** 225 **(4)** 275

11. **NEET 2024 · Q24** If c is the velocity of light in free space, the correct statements about photon among the following are:

A. The energy of a photon is $E = h\nu$.
B. The velocity of a photon is c.
C. The momentum of a photon, $p = \frac{h\nu}{c}$.
D. In a photon-electron collision, both total energy and total momentum are conserved.
E. Photon possesses positive charge.

Choose the correct answer from the options given below:

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(1) A, B, D and E only
(2) A and B only
(3) A, B, C and D only
(4) A, C and D only

12. **NEET 2023 · Q22** The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to:

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(1) V^2 **(2)** $\sqrt{V}$ **(3)** $\frac{1}{V}$ **(4)** $\frac{1}{\sqrt{V}}$

13. **NEET 2021 · Q31** The number of photons per second on an average emitted by the source of monochromatic light of wavelength 600 nm, when it delivers the power of 3.3×10^{-3} watt will be: ($h = 6.6 \times 10^{-34}$ J s)

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(1) 10^{18} **(2)** 10^{17} **(3)** 10^{16} **(4)** 10^{15}

de Broglie waves 9 Qs

14. **NEET 2025 · Q37** A photon and an electron (mass m) have the same energy E. The ratio ($\lambda_{photon}/\lambda_{electron}$) of their de Broglie wavelengths is: (c is the speed of light)

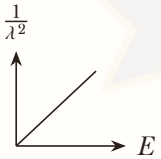
APPLIED Class XII Part 2 · Ch 11 · **Page 285** Open PDF ↗

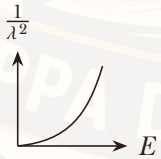
(1) $\sqrt{\dfrac{E}{2m}}$ **(2)** $c\sqrt{2mE}$

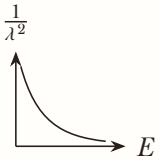
(3) $c\sqrt{\dfrac{2m}{E}}$ **(4)** $\dfrac{1}{c}\sqrt{\dfrac{E}{2m}}$

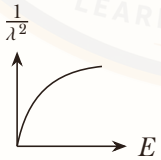
15. **NEET 2024 · Q8** The graph which shows the variation of $\left(\dfrac{1}{\lambda^2}\right)$ and its kinetic energy, E is (where λ is de Broglie wavelength of a free particle):

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(1) 

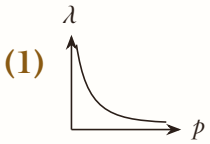
(2) 

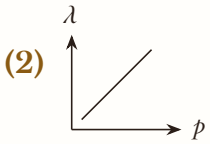
(3) 

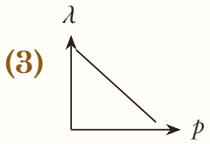
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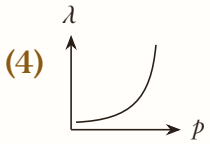
16. **NEET 2022 · Q9** The graph which shows the variation of the de Broglie wavelength (λ) of a particle and its associated momentum (p) is:

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(1) 

(2) 

(3) 

(4) 

17. **NEET 2021 · Q29** An electromagnetic wave of wavelength 'λ' is incident on a photosensitive surface of negligible work function. If 'm' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d, then:

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(1) $\lambda = \left(\dfrac{2m}{hc}\right)\lambda_d^2$ **(2)** $\lambda_d = \left(\dfrac{2mc}{h}\right)\lambda^2$

(3) $\lambda = \left(\dfrac{2mc}{h}\right)\lambda_d^2$ **(4)** $\lambda = \left(\dfrac{2h}{mc}\right)\lambda_d^2$

18. **NEET 2020 · Q175** An electron is accelerated from rest through a potential difference of V volt. If the de Broglie wavelength of the electron is 1.227×10^{-2} nm, the potential difference is:

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(1) 10 V **(2)** 10^2 V **(3)** 10^3 V **(4)** 10^4 V

19. **NEET 2019 · Q13** An electron is accelerated through a potential difference of 10,000 V. Its de Broglie wavelength is, (nearly): ($m_e = 9 \times 10^{-31}$ kg)

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(1) 12.2×10^{-13} m **(2)** 12.2×10^{-12} m

(3) 12.2×10^{-14} m **(4)** 12.2 nm

20. **NEET 2018 · Q12** An electron of mass m with an initial velocity $\vec{V} = V_0\hat{i}$ ($V_0 > 0$) enters an electric field $\vec{E} = -E_0\hat{i}$ (E_0 = constant > 0) at $t = 0$. If λ_0 is its de-Broglie wavelength initially, then its de-Broglie wavelength at time t is

APPLIED Class XII Part 2 · Ch 11 · **Page 285** Open PDF ↗

(1) $\lambda_0 t$

(2) $\lambda_0\left(1 + \dfrac{eE_0}{mV_0}t\right)$

(3) $\dfrac{\lambda_0}{\left(1 + \dfrac{eE_0}{mV_0}t\right)}$

(4) λ_0

21. **NEET 2017 · Q140** The de-Broglie wavelength of a neutron in thermal equilibrium with heavy water at a temperature T (Kelvin) and mass m, is:

APPLIED Class XII Part 2 · Ch 11 · **Page 285** · thermal KE 3kT/2 Open PDF ↗

(1) $\dfrac{h}{\sqrt{3mkT}}$

(2) $\dfrac{2h}{\sqrt{3mkT}}$

(3) $\dfrac{2h}{\sqrt{mkT}}$

(4) $\dfrac{h}{\sqrt{mkT}}$

22. **NEET 2016 · Q20** An electron of mass m and a photon have same energy E. The ratio of de-Broglie wavelengths associated with them is: (c being velocity of light)

APPLIED Class XII Part 2 · Ch 11 · **Page 285** Open PDF ↗

(1) $\dfrac{1}{c}\left(\dfrac{E}{2m}\right)^{1/2}$

(2) $\left(\dfrac{E}{2m}\right)^{1/2}$

(3) $c(2mE)^{1/2}$

(4) $\dfrac{1}{c}\left(\dfrac{2m}{E}\right)^{1/2}$

Dual Nature of Radiation and Matter · Answer Key

1 (1)	**2** (1)	**3** (1)	**4** (2)	**5** (1)	**6** (4)	**7** (3)	**8** (1)*
9 (4)	**10** (3)	**11** (3)	**12** (3)	**13** (3)	**14** (3)	**15** (1)	**16** (1)
17 (3)	**18** (4)	**19** (2)	**20** (3)	**21** (1)	**22** (1)		

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor. *: see note.

8. (NEET 2017 Q158) $K_{max} \approx 1.08$ eV gives $v \approx 6.2 \times 10^5$ m/s; options (1) and (4) are the same value.

Chapter Summary · Dual Nature of Radiation and Matter

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	2	2	2	1	2	2	2	2	2	2	3	**22**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Photoelectric effect	**9**	3	8	**16 17 18** 19 **20** 21 **22 23** 24 **25 26**
Photons and photon momentum	**4**	3	4	16 17 18 19 20 **21** 22 **23 24** 25 **26**
de Broglie waves	**9**	1	9	**16 17 18 19 20 21 22** 23 **24 25** 26

Direct vs Applied

Direct from NCERT	**7**	32%
text, table, figure	5	
solved Example	1	
Exercise	1	
Applied	**15**	68%
Easy / Mod. / Hard	8 / 14 / 0	

What to focus on

- 22 questions in 11 papers, about 2.0 per paper; rank 4 of 28 chapters by number of questions.
- Most asked topic: **de Broglie waves** (9 questions, in 9 of the 11 years).
- Asked in 8 or more of the 11 years: Photoelectric effect, de Broglie waves.
- 2 questions are NCERT solved Examples or Exercises: Example 11.1 (Q 13), Exercise 11.1 (Q 12). Work these by hand.
- NCERT pages that gave three or more questions: page 285 (7 questions), page 278 (4 questions), page 281 (4 questions). Read these pages line by line.
- Difficulty: 8 easy, 14 moderate, 0 hard.
- 15 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 12

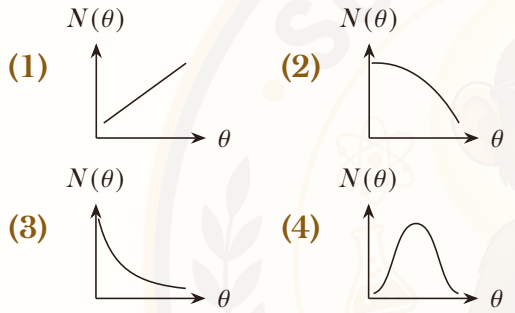
Atoms

15 questions · 2 topics · NEET 2016–2026

Alpha-particle scattering and atomic models 3 Qs

1. NEET 2026 · Q35 In Geiger–Marsden experiment, the number of scattered α-particles $N(\theta)$ is plotted as a function of scattering angle θ. Which of the following options represents the correct plot? ($N(\theta)$ on a logarithmic scale, θ from 0° to 180°.)

DIRECT Class XII Part 2 · Ch 12 · **Page 293** · Fig 12.3 Open PDF ↗



2. NEET 2024 · Q22 Given below are two statements:

Statement I: Atoms are electrically neutral as they contain equal number of positive and negative charges.

Statement II: Atoms of each element are stable and emit their characteristic spectrum.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

DIRECT Class XII Part 2 · Ch 12 · **Page 290** Open PDF ↗

(1) Statement I is incorrect but Statement II is correct.

(2) Both Statement I and Statement II are correct.

(3) Both Statement I and Statement II are incorrect.

(4) Statement I is correct but Statement II is incorrect.

3. NEET 2016 · Q23 When an α-particle of mass 'm' moving with velocity 'v' bombards on a heavy nucleus of charge 'Ze', its distance of closest approach from the nucleus depends on m as:

APPLIED Class XII Part 2 · Ch 12 · **Page 295** · distance of closest approach Open PDF ↗

(1) $\frac{1}{m}$ (2) $\frac{1}{\sqrt{m}}$ (3) $\frac{1}{m^2}$ (4) m

Bohr model of hydrogen atom 12 Qs

4. NEET 2026 · Q40 Consider that an electron is revolving in an excited state of hydrogen atom with velocity $\sqrt{25.6}\times10^5$ m s^{-1}. The radius of the orbit is $x\times10^{-9}$ m. The value of x is: [Take the mass of electron to be 9×10^{-31} kg, charge of electron $= -1.6\times10^{-19}$ C and $\frac{1}{4\pi\varepsilon_0} = 9\times10^9$ N m^{2}C^{-2}]

DIRECT Class XII Part 2 · Ch 12 · **Page 295** · r=e^2/(4 pi eps0 m v^2) Open PDF ↗

(1) 4 (2) 3 (3) 2 (4) 1

5. NEET 2025 · Q30 A particle of mass m is moving around the origin with a constant force F pulling it towards the origin. If Bohr model is used to describe its motion, the radius r of the nth orbit and the particle's speed v in the orbit depend on n as

APPLIED Class XII Part 2 · Ch 12 · **Page 299** Open PDF ↗

(1) $r \propto n^{1/3}$; $v \propto n^{1/3}$

(2) $r \propto n^{1/3}$; $v \propto n^{2/3}$

(3) $r \propto n^{2/3}$; $v \propto n^{1/3}$

(4) $r \propto n^{4/3}$; $v \propto n^{-1/3}$

6. **NEET 2025 · Q34** De-Broglie wavelength of an electron orbiting in the $n = 2$ state of hydrogen atom is close to (Given Bohr radius = 0.052 nm)

APPLIED Class XII Part 2 · Ch 12 · **Page 301** Open PDF ↗

(1) 0.067 nm **(2)** 0.67 nm
(3) 1.67 nm **(4)** 2.67 nm

7. **NEET 2024 · Q1** Match List I with List II.

List I (spectral lines of hydrogen for transitions from)		*List II* (wavelength, nm)	
A.	$n_2 = 3$ to $n_1 = 2$	I.	410.2
B.	$n_2 = 4$ to $n_1 = 2$	II.	434.1
C.	$n_2 = 5$ to $n_1 = 2$	III.	656.3
D.	$n_2 = 6$ to $n_1 = 2$	IV.	486.1

Choose the correct answer from the options given below:

APPLIED Class XII Part 2 · Ch 12 · **Page 300** · lines computed from En=-13.6/n^2 eV Open PDF ↗

(1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-III, C-I, D-II

8. **NEET 2023 · Q11** In hydrogen spectrum, the shortest wavelength in the Balmer series is λ. The shortest wavelength in the Brackett series is:

APPLIED Class XII Part 2 · Ch 12 · **Page 300** · series limits from En Open PDF ↗

(1) 16λ **(2)** 2λ **(3)** 4λ **(4)** 9λ

9. **NEET 2023 · Q36** The radius of inner most orbit of hydrogen atom is 5.3×10^{-11} m. What is the radius of third allowed orbit of hydrogen atom?

DIRECT Class XII Part 2 · Ch 12 · **Page 305** · Exercise 12.7 Open PDF ↗

(1) 4.77 Å **(2)** 0.53 Å
(3) 1.06 Å **(4)** 1.59 Å

10. **NEET 2022 · Q34** Let T_1 and T_2 be the energy of an electron in the first and second excited states of hydrogen atom, respectively. According to the Bohr's model of an atom, the ratio $T_1 : T_2$ is:

APPLIED Class XII Part 2 · Ch 12 · **Page 300** Open PDF ↗

(1) 9:4 **(2)** 1:4 **(3)** 4:1 **(4)** 4:9

11. **NEET 2020 · Q136** For which one of the following, Bohr model is **not** valid?

DIRECT Class XII Part 2 · Ch 12 · **Page 302** · Bohr model only for hydrogenic atoms Open PDF ↗

(1) Hydrogen atom
(2) Singly ionised helium atom (He^+)
(3) Deuteron atom
(4) Singly ionised neon atom (Ne^+)

12. **NEET 2019 · Q27** The total energy of an electron in an atom in an orbit is −3.4 eV. Its kinetic and potential energies are, respectively:

DIRECT Class XII Part 2 · Ch 12 · **Page 296** · K=-E, U=2E Open PDF ↗

(1) −3.4 eV, −3.4 eV **(2)** −3.4 eV, −6.8 eV
(3) 3.4 eV, −6.8 eV **(4)** 3.4 eV, 3.4 eV

13. **NEET 2018 · Q11** The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom, is

DIRECT Class XII Part 2 · Ch 12 · **Page 296** · K=-E Open PDF ↗

(1) 2 : −1 **(2)** 1 : −1 **(3)** 1:1 **(4)** 1 : −2

14. **NEET 2017 · Q168** The ratio of wavelengths of the last line of Balmer series and the last line of Lyman series is:

APPLIED Class XII Part 2 · Ch 12 · **Page 291** · series limits Open PDF ↗

(1) 1 **(2)** 4 **(3)** 0.5 **(4)** 2

15. **NEET 2016 · Q38** Given the value of Rydberg constant is $10^7\ m^{-1}$, the wave number of the last line of the Balmer series in hydrogen spectrum will be:

APPLIED

Class XII Part 2 · Ch 12 · **Page 291** · Balmer formula
Open PDF ↗

(1) $0.025 \times 10^4\ m^{-1}$ **(2)** $0.5 \times 10^7\ m^{-1}$
(3) $0.25 \times 10^7\ m^{-1}$ **(4)** $2.5 \times 10^7\ m^{-1}$

Atoms · Answer Key

1 (3)	**2** (2)*	**3** (1)	**4** (4)	**5** (3)	**6** (2)	**7** (3)	**8** (3)
9 (1)	**10** (1)	**11** (4)	**12** (3)	**13** (2)	**14** (2)	**15** (3)	

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor. *: see note.

2. (NEET 2024 Q22) NCERT (p 290) states both: atoms are neutral and each element has a characteristic spectrum.

Chapter Summary · Atoms

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	2	1	1	1	1	0	1	2	2	2	2	**15**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Alpha-particle scattering and atomic models	**3**	2	3	**16** 17 18 19 20 21 22 23 **24** 25 **26**
Bohr model of hydrogen atom	**12**	5	10	**16 17 18 19 20** 21 **22 23 24 25 26**

Direct vs Applied

Direct from NCERT	**7**	47%
text, table, figure	6	
solved Example	0	
Exercise	1	
Applied	**8**	53%
Easy / Mod. / Hard	9 / 5 / 1	

What to focus on

- 15 questions in 11 papers, about 1.4 per paper; rank 18 of 28 chapters by number of questions.
- Most asked topic: **Bohr model of hydrogen atom** (12 questions, in 10 of the 11 years).
- Asked in 8 or more of the 11 years: Bohr model of hydrogen atom.
- 1 questions are NCERT solved Examples or Exercises: Exercise 12.7 (Q 9). Work these by hand.
- NCERT pages that gave three or more questions: page 300 (3 questions). Read these pages line by line.
- Difficulty: 9 easy, 5 moderate, 1 hard (hard: Q 5).
- 8 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Class XII · Chapter 13

Nuclei

6 questions · 3 topics · NEET 2016–2026

Mass defect, binding energy, fission and fusion

4 Qs

1. NEET 2026 · Q7 Consider the following nuclear reaction

$$^{238}\mathrm{U} \longrightarrow {}^{234}\mathrm{Th} + {}^{4}\mathrm{He}$$

Take masses of ^{238}U, ^{234}Th and ^{4}He as 238.050 u, 234.043 u and 4.003 u, respectively. The Q value for the reaction, in keV, is: [Given: 1 u = 931.5 MeV c^{-2}]

DIRECT — Class XII Part 2 · Ch 13 · **Page 319** · Q=(sum mi - sum mf)c^2 Open PDF ↗

(1) 3726 **(2)** 3730 **(3)** 3736 **(4)** 3740

2. NEET 2021 · Q35 A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total gain in the Binding Energy in the process is:

DIRECT — Class XII Part 2 · Ch 13 · **Page 315** · 240 x 0.9 = 216 MeV Open PDF ↗

(1) 0.9 MeV **(2)** 9.4 MeV
(3) 804 MeV **(4)** 216 MeV

3. NEET 2020 · Q142 The energy equivalent of 0.5 g of a substance is:

DIRECT — Class XII Part 2 · Ch 13 · **Page 310** · Example 13.2 Open PDF ↗

(1) 4.5×10^{16} J **(2)** 4.5×10^{13} J
(3) 1.5×10^{13} J **(4)** 0.5×10^{13} J

4. NEET 2020 · Q158 When a uranium isotope $^{235}_{92}$U is bombarded with a neutron, it generates $^{89}_{36}$Kr, three neutrons and:

DIRECT — Class XII Part 2 · Ch 13 · **Page 315** · U-235 fission equation Open PDF ↗

(1) $^{144}_{56}$Ba **(2)** $^{91}_{40}$Zr **(3)** $^{101}_{36}$Kr **(4)** $^{103}_{36}$Kr

Radioactivity (alpha, beta, gamma)

1 Q

5. NEET 2019 · Q31 α-particle consists of:

DIRECT — Class XII Part 2 · Ch 13 · **Page 314** · alpha = helium nucleus Open PDF ↗

(1) 2 protons and 2 neutrons only
(2) 2 electrons, 2 protons and 2 neutrons
(3) 2 electrons and 4 protons only
(4) 2 protons only

Nuclear composition, size and density

1 Q

6. NEET 2022 · Q45 A nucleus of mass number 189 splits into two nuclei having mass number 125 and 64. The ratio of radius of two daughter nuclei respectively is:

DIRECT — Class XII Part 2 · Ch 13 · **Page 321** · Exercise 13.4 Open PDF ↗

(1) 25:16 **(2)** 1:1 **(3)** 4:5 **(4)** 5:4

Nuclei · Answer Key

1 (1)	**2** (4)	**3** (2)	**4** (1)	**5** (1)	**6** (4)

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor.

Chapter Summary · Nuclei

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	0	0	0	1	2	1	1	0	0	0	1	**6**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Mass defect, binding energy, fission and fusion	**4**	4	3	16 17 18 19 **20** **21** 22 23 24 25 **26**
Radioactivity (alpha, beta, gamma)	**1**	1	1	16 17 18 **19** 20 21 22 23 24 25 26
Nuclear composition, size and density	**1**	1	1	16 17 18 19 20 21 **22** 23 24 25 26

Direct vs Applied

Direct from NCERT	**6**	100%
text, table, figure	4	
solved Example	1	
Exercise	1	
Applied	**0**	0%
Easy / Mod. / Hard	6 / 0 / 0	

What to focus on

- 6 questions in 11 papers, about 0.5 per paper; rank 28 of 28 chapters by number of questions.
- Most asked topic: **Mass defect, binding energy, fission and fusion** (4 questions, in 3 of the 11 years).
- 2 questions are NCERT solved Examples or Exercises: Example 13.2 (Q 3), Exercise 13.4 (Q 6). Work these by hand.
- Difficulty: 6 easy, 0 moderate, 0 hard.

Class XII · Chapter 14

Semiconductor Electronics

11 questions · 2 topics · NEET 2016–2026

Semiconductors (intrinsic, extrinsic, bands) 2 Qs

1. **NEET 2021 · Q30** The electron concentration in an n-type semiconductor is the same as hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.

APPLIED Class XII Part 2 · Ch 14 · **Page 328** Open PDF ↗

(1) current in n-type = current in p-type.
(2) current in p-type > current in n-type.
(3) current in n-type > current in p-type.
(4) No current will flow in p-type, current will only flow in n-type.

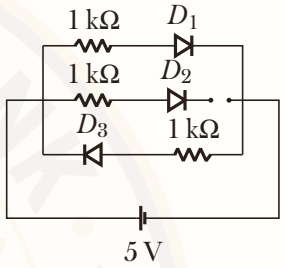
2. **NEET 2019 · Q18** For a p-type semiconductor, which of the following statements is **true**?

DIRECT Class XII Part 2 · Ch 14 · **Page 342** · Exercise 14.2 Open PDF ↗

(1) Electrons are the majority carriers and trivalent atoms are the dopants.
(2) Holes are the majority carriers and trivalent atoms are the dopants.
(3) Holes are the majority carriers and pentavalent atoms are the dopants.
(4) Electrons are the majority carriers and pentavalent atoms are the dopants.

p-n junction diode and rectifier 9 Qs

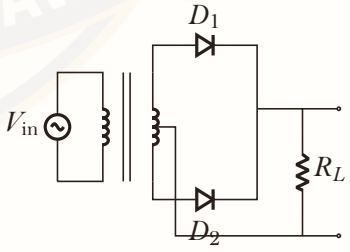
3. **NEET 2026 · Q42** Three identical p-n junction diodes D_1, D_2, and D_3 are connected across a battery as shown in the figure. If the width of the depletion regions of D_1, D_2 and D_3 are W_1, W_2 and W_3, respectively, then the correct option is:



APPLIED Class XII Part 2 · Ch 14 · **Page 335** Open PDF ↗

(1) $W_1 > W_2 > W_3$ **(2)** $W_3 = W_1 > W_2$
(3) $W_3 > W_2 > W_1$ **(4)** $W_2 > W_1 = W_3$

4. **NEET 2025 · Q40** A full wave rectifier circuit with diodes (D_1) and (D_2) is shown in the figure. If input supply voltage $V_{in} = 220\sin(100\pi t)$ volt, then at $t = 15$ ms



APPLIED Class XII Part 2 · Ch 14 · **Page 339** Open PDF ↗

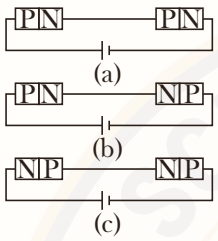
(1) D_1 is forward biased, D_2 is reverse biased
(2) D_1 is reverse biased, D_2 is forward biased
(3) D_1 and D_2 both are forward biased
(4) D_1 and D_2 both are reverse biased

5. **NEET 2023 · Q5** A full wave rectifier circuit consists of two p-n junction diodes, a centre-tapped transformer, capacitor and a load resistance. Which of these components remove the ac ripple from the rectified output?

DIRECT — Class XII Part 2 · Ch 14 · **Page 339** · capacitor filter Open PDF ↗

(1) Load resistance
(2) A centre-tapped transformer
(3) p-n junction diodes
(4) Capacitor

6. **NEET 2022 · Q26**



In the given circuits (a), (b) and (c), the potential drop across the two p-n junctions are equal in:

APPLIED — Class XII Part 2 · Ch 14 · **Page 335** Open PDF ↗

(1) Both circuits (a) and (c)
(2) Circuit (a) only
(3) Circuit (b) only
(4) Circuit (c) only

7. **NEET 2022 · Q31** In half wave rectification, if the input frequency is 60 Hz, then the output frequency would be:

DIRECT — Class XII Part 2 · Ch 14 · **Page 342** · Exercise 14.6 Open PDF ↗

(1) 120 Hz **(2)** zero
(3) 30 Hz **(4)** 60 Hz

8. **NEET 2020 · Q153** The increase in the width of the depletion region in a p-n junction diode is due to:

DIRECT — Class XII Part 2 · Ch 14 · **Page 336** · reverse bias widens depletion region Open PDF ↗

(1) forward bias only
(2) reverse bias only
(3) both forward bias and reverse bias
(4) increase in forward current

9. **NEET 2018 · Q16** In a p-n junction diode, change in temperature due to heating

APPLIED — Class XII Part 2 · Ch 14 · **Page 337** Open PDF ↗

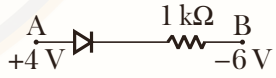
(1) does not affect resistance of p-n junction
(2) affects only forward resistance
(3) affects only reverse resistance
(4) affects the overall V–I characteristics of p-n junction

10. **NEET 2017 · Q147** Which one of the following represents forward bias diode?

APPLIED — Class XII Part 2 · Ch 14 · **Page 334** Open PDF ↗

(1) −4 V — ⊳|— −3 V
(2) −2 V — ⊳|— +2 V
(3) 3 V — ⊳|— 5 V
(4) 0 V — ⊳|— −2 V

11. **NEET 2016 · Q13** Consider the junction diode as ideal. The value of current flowing through AB is:



APPLIED — Class XII Part 2 · Ch 14 · **Page 334** · forward bias Open PDF ↗

(1) 0 A **(2)** 10^{-2} A
(3) 10^{-1} A **(4)** 10^{-3} A

Semiconductor Electronics · Answer Key

1 (3)	**2** (2)	**3** (3)	**4** (1)*	**5** (4)	**6** (1)	**7** (4)	**8** (2)
9 (4)	**10** (4)	**11** (2)					

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor. *: see note.

4. (NEET 2025 Q40) The official NTA key accepts both (1) and (2).

Chapter Summary · Semiconductor Electronics

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Questions	1	1	1	1	1	1	2	1	0	1	1	**11**

Topic-wise (in textbook order)

Topic	Qs	Direct	Years	2016 → 2026
Semiconductors (intrinsic, extrinsic, bands)	**2**	1	2	16 17 18 **19** 20 **21** 22 23 24 25 26
p-n junction diode and rectifier	**9**	3	8	**16** **17** **18** 19 **20** 21 **22** **23** 24 **25** **26**

Direct vs Applied

Direct from NCERT	4	36%
text, table, figure	2	
solved Example	0	
Exercise	2	
Applied	**7**	64%
Easy / Mod. / Hard	7 / 4 / 0	

What to focus on

- 11 questions in 11 papers, about 1.0 per paper; rank 23 of 28 chapters by number of questions.
- Most asked topic: **p-n junction diode and rectifier** (9 questions, in 8 of the 11 years).
- Asked in 8 or more of the 11 years: p-n junction diode and rectifier.
- 2 questions are NCERT solved Examples or Exercises: Exercise 14.2 (Q 2), Exercise 14.6 (Q 7). Work these by hand.
- Difficulty: 7 easy, 4 moderate, 0 hard.
- 7 applied questions: practise these after the NCERT Examples and Exercises of the chapter.

Outside the 2026-27 Syllabus

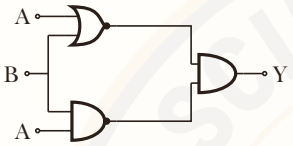
Deleted Portions

58 questions · topics removed from the rationalised NCERT

These questions were asked on topics that are no longer in the 2026-27 NCERT textbooks; 6 of them are from 2025 and 2026. They are kept here for completeness; a student following the present syllabus can skip this section. Topics are in order of how often they were asked.

Digital electronics (logic gates) — 11 Qs

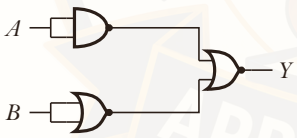
1. NEET 2025 · Q8 The output (Y) of the given logic implementation is similar to the output of an/a _____ gate.



DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) AND (2) NAND
(3) OR (4) NOR

2. NEET 2024 · Q11 The output (Y) of the given logic gate is similar to the output of an/a:



DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) AND gate (2) NAND gate
(3) NOR gate (4) OR gate

3. NEET 2024 · Q16 A logic circuit provides the output Y as per the following truth table:

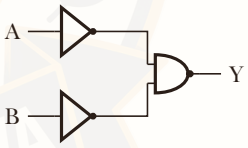
A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

The expression for the output Y is:

DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) B (2) $A \cdot B + \bar{A}$
(3) $A \cdot \bar{B} + \bar{A}$ (4) $\bar{B}$

4. NEET 2023 · Q39 For the following logic circuit, the truth table is:



DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(2)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

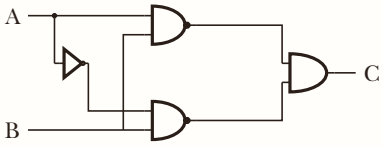
(3)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(4)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

5. **NEET 2022 · Q44**



The truth table for the given logic circuit is:

DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

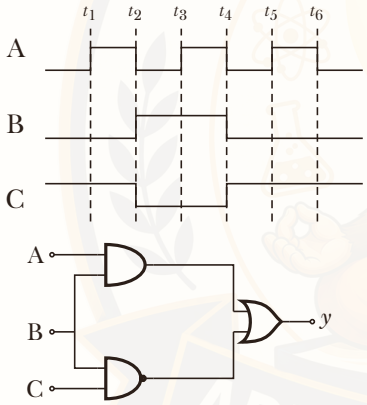
(3)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

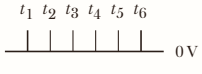
(4)

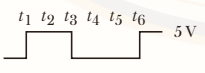
A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

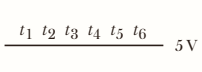
6. **NEET 2021 · Q38** For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal *y*?

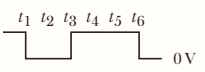


DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

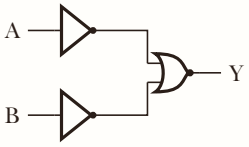
(1) 

(2) 

(3) 

(4) 

7. **NEET 2020 · Q180** For the logic circuit shown, the truth table is:



DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

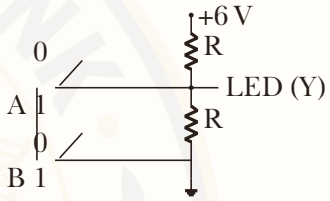
(3)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(4)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

8. **NEET 2019 · Q39**

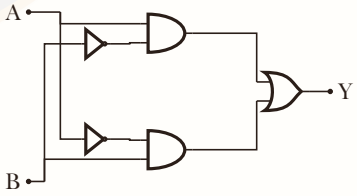


The **correct** Boolean operation represented by the circuit diagram drawn is:

DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) AND **(2)** OR
(3) NAND **(4)** NOR

9. **NEET 2018 · Q17** In the combination of the following gates the output Y can be written in terms of inputs A and B as



DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) $\overline{A \cdot B} + A \cdot B$ **(2)** $A \cdot \bar{B} + \bar{A} \cdot B$
(3) $\overline{A \cdot B}$ **(4)** $\overline{A + B}$

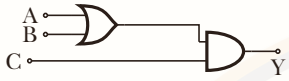
10. **NEET 2017 · Q142** The given electrical network is equivalent to:



DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) OR gate **(2)** NOR gate
(3) NOT gate **(4)** AND gate

11. **NEET 2016 · Q34** To get output 1 for the following circuit, the correct choice for the input is:



DELETED PORTION — digital electronics (logic gates) · removed from the 2026-27 NCERT; not analysed

(1) A = 0, B = 1, C = 0
(2) A = 1, B = 0, C = 0
(3) A = 1, B = 1, C = 0
(4) A = 1, B = 0, C = 1

Brewster's law — 4 Qs

12. **NEET 2025 · Q27** An unpolarized light beam travelling in air is incident on a medium of refractive index 1.73 at Brewster's angle. Then

DELETED PORTION — Brewster's law · removed from the 2026-27 NCERT; not analysed

(1) reflected light is completely polarized and the angle of reflection is close to 60°
(2) reflected light is partially polarized and the angle of reflection is close to 30°
(3) both reflected and transmitted light are perfectly polarized with angles of reflection and refraction close to 60° and 30°, respectively
(4) transmitted light is completely polarized with angle of refraction close to 30°

13. **NEET 2024 · Q4** An unpolarised light beam strikes a glass surface at Brewster's angle. Then

DELETED PORTION — Brewster's law · removed from the 2026-27 NCERT; not analysed

(1) the reflected light will be completely polarised but the refracted light will be partially polarised.
(2) the reflected light will be partially polarised.
(3) the refracted light will be completely polarised.
(4) both the reflected and refracted light will be completely polarised.

14. **NEET 2020 · Q138** The Brewster's angle i_b for an interface should be:

DELETED PORTION — Brewster's law · removed from the 2026-27 NCERT; not analysed

(1) $0° < i_b < 30°$ **(2)** $30° < i_b < 45°$
(3) $45° < i_b < 90°$ **(4)** $i_b = 90°$

15. **NEET 2018 · Q8** Unpolarised light is incident from air on a plane surface of a material of refractive index 'μ'. At a particular angle of incidence 'i', it is found that the reflected and refracted rays are perpendicular to each other. Which of the following options is correct for this situation?

DELETED PORTION — Brewster's law · removed from the 2026-27 NCERT; not analysed

(1) $i = \sin^{-1}\left(\frac{1}{\mu}\right)$

(2) Reflected light is polarised with its electric vector perpendicular to the plane of incidence

(3) Reflected light is polarised with its electric vector parallel to the plane of incidence

(4) $i = \tan^{-1}\left(\frac{1}{\mu}\right)$

Radioactive decay law (half-life, mean life, activity) — 4 Qs

16. **NEET 2023 · Q28** The half life of a radioactive substance is 20 minutes. In how much time, the activity of substance drops to $\left(\frac{1}{16}\right)^{\text{th}}$ of its initial value?

DELETED PORTION — radioactive decay law (half-life) · removed from the 2026-27 NCERT; not analysed

(1) 80 minutes **(2)** 20 minutes
(3) 40 minutes **(4)** 60 minutes

17. **NEET 2021 · Q34** The half-life of a radioactive nuclide is 100 hours. The fraction of original activity that will remain after 150 hours would be:

DELETED PORTION — radioactive decay law (half-life) · removed from the 2026-27 NCERT; not analysed

(1) 1/2 **(2)** $\frac{1}{2\sqrt{2}}$ **(3)** $\frac{2}{3}$ **(4)** $\frac{2}{3\sqrt{2}}$

18. **NEET 2018 · Q13** For a radioactive material, half-life is 10 minutes. If initially there are 600 number of nuclei, the time taken (in minutes) for the disintegration of 450 nuclei is

DELETED PORTION — radioactive decay law (half-life) · removed from the 2026-27 NCERT; not analysed

(1) 30 **(2)** 10 **(3)** 20 **(4)** 15

19. **NEET 2017 · Q141** Radioactive material 'A' has decay constant '8λ' and material 'B' has decay constant 'λ'. Initially they have same number of nuclei. After what time, the ratio of number of nuclei of material 'B' to that 'A' will be $\frac{1}{e}$?

DELETED PORTION — radioactive decay law (decay constant) · removed from the 2026-27 NCERT; not analysed

(1) $\frac{1}{7\lambda}$ **(2)** $\frac{1}{8\lambda}$ **(3)** $\frac{1}{9\lambda}$ **(4)** $\frac{1}{\lambda}$

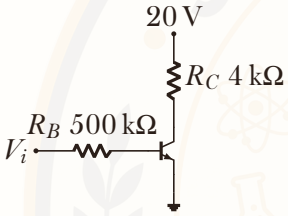
Transistor (incl. amplifier) — 4 Qs

20. NEET 2020 · Q156 For transistor action, which of the following statements is **correct**?

DELETED PORTION — transistor · removed from the 2026-27 NCERT; not analysed

(1) Base, emitter and collector regions should have same doping concentrations.
(2) Base, emitter and collector regions should have same size.
(3) Both emitter junction as well as the collector junction are forward biased.
(4) The base region must be very thin and lightly doped.

21. NEET 2018 · Q15 In the circuit shown in the figure, the input voltage V_i is 20 V, $V_{BE} = 0$ and $V_{CE} = 0$. The values of I_B, I_C and β are given by



DELETED PORTION — transistor · removed from the 2026-27 NCERT; not analysed

(1) $I_B = 20\ \mu A$, $I_C = 5$ mA, $\beta = 250$
(2) $I_B = 25\ \mu A$, $I_C = 5$ mA, $\beta = 200$
(3) $I_B = 40\ \mu A$, $I_C = 10$ mA, $\beta = 250$
(4) $I_B = 40\ \mu A$, $I_C = 5$ mA, $\beta = 125$

22. NEET 2017 · Q138 In a common emitter transistor amplifier the audio signal voltage across the collector is 3 V. The resistance of collector is 3 kΩ. If current gain is 100 and the base resistance is 2 kΩ, the voltage and power gain of the amplifier is:

DELETED PORTION — transistor (CE amplifier) · removed from the 2026-27 NCERT; not analysed

(1) 15 and 200 (2) 150 and 15000
(3) 20 and 2000 (4) 200 and 1000

23. NEET 2016 · Q17 A npn transistor is connected in common emitter configuration in a given amplifier. A load resistance of 800 Ω is connected in the collector circuit and the voltage drop across it is 0.8 V. If the current amplification factor is 0.96 and the input resistance of the circuit is 192 Ω, the voltage gain and the power gain of the amplifier will respectively be:

DELETED PORTION — transistor (CE amplifier) · removed from the 2026-27 NCERT; not analysed

(1) 4, 3.84 (2) 3.69, 3.84
(3) 4, 4 (4) 4, 3.69

Alpha- and beta-decay — 3 Qs

24. NEET 2024 · Q28 $^{290}_{82}X \xrightarrow{\alpha} Y \xrightarrow{e^+} Z \xrightarrow{\beta^-} P \xrightarrow{e^-} Q$. In the nuclear emission stated above, the mass number and atomic number of the product Q respectively, are:

DELETED PORTION — alpha- and beta-decay · removed from the 2026-27 NCERT; not analysed

(1) 286, 81 (2) 280, 81
(3) 286, 80 (4) 288, 82

25. NEET 2022 · Q11 In the given nuclear reaction, the element X is:

$$^{22}_{11}\text{Na} \longrightarrow \text{X} + e^+ + \nu$$

DELETED PORTION — alpha- and beta-decay · removed from the 2026-27 NCERT; not analysed

(1) $^{22}_{12}$Mg (2) $^{23}_{11}$Na (3) $^{23}_{10}$Ne (4) $^{22}_{10}$Ne

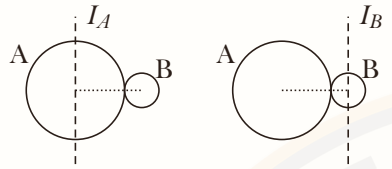
26. NEET 2021 · Q19 A radioactive nucleus $^{A}_{Z}$X undergoes spontaneous decay in the sequence $^{A}_{Z}\text{X} \rightarrow {}_{Z-1}\text{B} \rightarrow {}_{Z-3}\text{C} \rightarrow {}_{Z-2}\text{D}$, where Z is the atomic number of element X. The possible decay particles in the sequence are:

DELETED PORTION — alpha- and beta-decay · removed from the 2026-27 NCERT; not analysed

(1) α, β^-, β^+ (2) α, β^+, β^-
(3) β^+, α, β^- (4) β^-, α, β^+

Parallel axis theorem 3 Qs

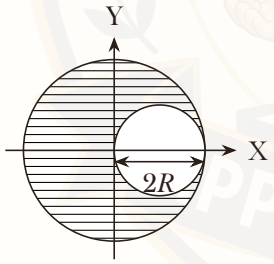
27. **NEET 2026 · Q27** A solid sphere A of radius R and mass M is attached at a point to a smaller solid sphere B of radius $r < R$ and mass $m < M$. Assume that the line joining their centres lies along the horizontal. The moment of inertia of the system calculated about a vertical axis passing through the centre of A is I_A and that calculated about a vertical axis passing through the centre of B is I_B. The difference $I_A - I_B$ is:



DELETED PORTION parallel axis theorem · removed from the 2026-27 NCERT; not analysed

(1) $(M-m)(R+r)^2$ **(2)** $(m-M)(R+r)^2$
(3) $(m-M)(R-r)^2$ **(4)** 0

28. **NEET 2025 · Q39** A sphere of radius R is cut from a larger solid sphere of radius $2R$ as shown in the figure. The ratio of the moment of inertia of the smaller sphere to that of the rest part of the sphere about the Y-axis is:



DELETED PORTION parallel axis theorem · removed from the 2026-27 NCERT; not analysed

(1) $\frac{7}{8}$ **(2)** $\frac{7}{40}$ **(3)** $\frac{7}{57}$ **(4)** $\frac{7}{64}$

29. **NEET 2016 · Q1** From a disc of radius R and mass M, a circular hole of diameter R, whose rim passes through the centre is cut. What is the moment of inertia of the remaining part of the disc about a perpendicular axis, passing through the centre?

DELETED PORTION parallel axis theorem · removed from the 2026-27 NCERT; not analysed

(1) $15MR^2/32$ **(2)** $13MR^2/32$
(3) $11MR^2/32$ **(4)** $9MR^2/32$

Potentiometer 3 Qs

30. **NEET 2021 · Q27** In a potentiometer circuit a cell of EMF 1.5 V gives balance point at 36 cm length of wire. If another cell of EMF 2.5 V replaces the first cell, then at what length of the wire, the balance point occurs?

DELETED PORTION potentiometer · removed from the 2026-27 NCERT; not analysed

(1) 60 cm **(2)** 21.6 cm
(3) 64 cm **(4)** 62 cm

31. **NEET 2017 · Q150** A potentiometer is an accurate and versatile device to make electrical measurements of E.M.F. because the method involves:

DELETED PORTION potentiometer · removed from the 2026-27 NCERT; not analysed

(1) potential gradients
(2) a condition of no current flow through the galvanometer
(3) a combination of cells, galvanometer and resistances
(4) cells

32. **NEET 2016 · Q31** A potentiometer wire is 100 cm long and a constant potential difference is maintained across it. Two cells are connected in series first to support one another and then in opposite direction. The balance points are obtained at 50 cm and 10 cm from the positive end of the wire in the two cases. The ratio of emf's is:

DELETED PORTION potentiometer · removed from the 2026-27 NCERT; not analysed

(1) 5:1 **(2)** 5:4 **(3)** 3:4 **(4)** 3:2

Resistor colour code — 3 Qs

33. NEET 2023 · Q18 Resistance of a carbon resistor determined from colour codes is (22000 ± 5%) Ω. The colour of third band must be:

DELETED PORTION — resistor colour code · removed from the 2026-27 NCERT; not analysed

(1) Yellow (2) Red
(3) Green (4) Orange

34. NEET 2020 · Q164 The color code of a resistance is given below:



The values of resistance and tolerance, respectively, are:

DELETED PORTION — resistor colour code · removed from the 2026-27 NCERT; not analysed

(1) 470 kΩ, 5% (2) 47 kΩ, 10%
(3) 4.7 kΩ, 5% (4) 470 Ω, 5%

35. NEET 2018 · Q5 A carbon resistor of (47 ± 4.7) kΩ is to be marked with rings of different colours for its identification. The colour code sequence will be

DELETED PORTION — resistor colour code · removed from the 2026-27 NCERT; not analysed

(1) Yellow – Green – Violet – Gold
(2) Yellow – Violet – Orange – Silver
(3) Violet – Yellow – Orange – Silver
(4) Green – Orange – Violet – Gold

Rolling motion — 3 Qs

36. NEET 2019 · Q9 A disc of radius 2 m and mass 100 kg rolls on a horizontal floor. Its centre of mass has speed of 20 cm/s. How much work is needed to stop it?

DELETED PORTION — kinetic energy of rolling · removed from the 2026-27 NCERT; not analysed

(1) 3 J (2) 30 kJ (3) 2 J (4) 1 J

37. NEET 2018 · Q41 A solid sphere is in rolling motion. In rolling motion a body possesses translational kinetic energy (K_t) as well as rotational kinetic energy (K_r) simultaneously. The ratio $K_t : (K_t + K_r)$ for the sphere is

DELETED PORTION — kinetic energy of rolling · removed from the 2026-27 NCERT; not analysed

(1) 10:7 (2) 5:7 (3) 7:10 (4) 2:5

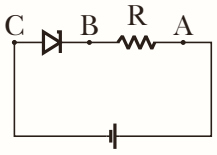
38. NEET 2016 · Q21 A disk and a sphere of same radius but different masses roll off on two inclined planes of the same altitude and length. Which one of the two objects gets to the bottom of the plane first?

DELETED PORTION — kinetic energy of rolling · removed from the 2026-27 NCERT; not analysed

(1) Disk
(2) Sphere
(3) Both reach at the same time
(4) Depends on their masses

Zener diode — 3 Qs

39. NEET 2026 · Q16 An ideal Zener diode with breakdown voltage of −3 V is reverse biased with a negative input voltage $V_i = -5$ V. The magnitude of voltage difference between point B and A is:



DELETED PORTION — Zener diode · removed from the 2026-27 NCERT; not analysed

(1) 3 V (2) 2 V (3) 1 V (4) 0 V

40. NEET 2023 · Q16 Given below are two statements:
Statement I: Photovoltaic devices can convert optical radiation into electricity.
Statement II: Zener diode is designed to operate under reverse bias in breakdown region.
In the light of the above statements, choose the *most appropriate* answer from the options given below:

DELETED PORTION — Zener diode · removed from the 2026-27 NCERT; not analysed

(1) Statement I is incorrect but Statement II is correct.
(2) Both Statement I and Statement II are correct.
(3) Both Statement I and Statement II are incorrect.
(4) Statement I is correct but Statement II is incorrect.

41. NEET 2021 · Q1 Consider the following statements (A) and (B) and identify the correct answer.
(A) A zener diode is connected in reverse bias, when used as a voltage regulator.
(B) The potential barrier of p-n junction lies between 0.1 V to 0.3 V.

DELETED PORTION — Zener diode · removed from the 2026-27 NCERT; not analysed

(1) (A) and (B) both are correct.
(2) (A) and (B) both are incorrect.
(3) (A) is correct and (B) is incorrect.
(4) (A) is incorrect but (B) is correct.

Doppler effect — 2 Qs

42. NEET 2017 · Q144 Two cars moving in opposite directions approach each other with speed of 22 m/s and 16.5 m/s respectively. The driver of the first car blows a horn having a frequency 400 Hz. The frequency heard by the driver of the second car is [velocity of sound 340 m/s]:

DELETED PORTION — Doppler effect · removed from the 2026-27 NCERT; not analysed

(1) 361 Hz **(2)** 411 Hz
(3) 448 Hz **(4)** 350 Hz

43. NEET 2016 · Q4 A siren emitting a sound of frequency 800 Hz moves away from an observer towards a cliff at a speed of 15 $m\,s^{-1}$. Then, the frequency of sound that the observer hears in the echo reflected from the cliff is: (Take velocity of sound in air = 330 $m\,s^{-1}$)

DELETED PORTION — Doppler effect · removed from the 2026-27 NCERT; not analysed

(1) 765 Hz **(2)** 800 Hz
(3) 838 Hz **(4)** 885 Hz

Earth's magnetism (incl. angle of dip) 2 Qs

44. NEET 2019 · Q26 At a point A on the earth's surface the angle of dip, $\delta = +25°$. At a point B on the earth's surface the angle of dip, $\delta = -25°$. We can interpret that:

DELETED PORTION — Earth's magnetism (angle of dip) · removed from the 2026-27 NCERT; not analysed

(1) A and B are both located in the northern hemisphere.
(2) A is located in the southern hemisphere and B is located in the northern hemisphere.
(3) A is located in the northern hemisphere and B is located in the southern hemisphere.
(4) A and B are both located in the southern hemisphere.

45. NEET 2017 · Q149 If θ_1 and θ_2 be the apparent angles of dip observed in two vertical planes at right angles to each other, then the true angle of dip θ is given by:

DELETED PORTION — Earth's magnetism (angle of dip) · removed from the 2026-27 NCERT; not analysed

(1) $\tan^2\theta = \tan^2\theta_1 + \tan^2\theta_2$
(2) $\cot^2\theta = \cot^2\theta_1 - \cot^2\theta_2$
(3) $\tan^2\theta = \tan^2\theta_1 - \tan^2\theta_2$
(4) $\cot^2\theta = \cot^2\theta_1 + \cot^2\theta_2$

Resolving power (telescope, microscope) 2 Qs

46. NEET 2020 · Q168 Assume that light of wavelength 600 nm is coming from a star. The limit of resolution of telescope whose objective has a diameter of 2 m is:

DELETED PORTION — resolving power of a telescope · removed from the 2026-27 NCERT; not analysed

(1) 3.66×10^{-7} rad (2) 1.83×10^{-7} rad
(3) 7.32×10^{-7} rad (4) 6.00×10^{-7} rad

47. NEET 2017 · Q151 The ratio of resolving powers of an optical microscope for two wavelengths $\lambda_1 = 4000$ Å and $\lambda_2 = 6000$ Å is:

DELETED PORTION — resolving power of microscope · removed from the 2026-27 NCERT; not analysed

(1) 9:4 (2) 3:2 (3) 16:81 (4) 8:27

Coefficient of restitution 1 Q

48. NEET 2018 · Q32 A moving block having mass m, collides with another stationary block having mass $4m$. The lighter block comes to rest after collision. When the initial velocity of the lighter block is v, then the value of coefficient of restitution (e) will be

DELETED PORTION — coefficient of restitution · removed from the 2026-27 NCERT; not analysed

(1) 0.8 (2) 0.25 (3) 0.5 (4) 0.4

Eddy currents 1 Q

49. NEET 2019 · Q6 In which of the following devices, the eddy current effect is **not** used?

DELETED PORTION — eddy currents · removed from the 2026-27 NCERT; not analysed

(1) induction furnace
(2) magnetic braking in train
(3) electromagnet
(4) electric heater

Force between capacitor plates 1 Q

50. NEET 2018 · Q23 The electrostatic force between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A, is

DELETED PORTION — force between capacitor plates · removed from the 2026-27 NCERT; not analysed

(1) proportional to the square root of the distance between the plates.
(2) linearly proportional to the distance between the plates.
(3) independent of the distance between the plates.
(4) inversely proportional to the distance between the plates.

Intensity of EM waves 1 Q

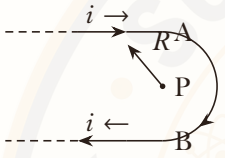
51. NEET 2020 · Q150 Light with an average flux of $20\ \mathrm{W/cm^2}$ falls on a non-reflecting surface at normal incidence having surface area $20\ \mathrm{cm^2}$. The energy received by the surface during time span of 1 minute is:

DELETED PORTION — intensity of EM waves · removed from the 2026-27 NCERT; not analysed

(1) $10 \times 10^3\ \mathrm{J}$ **(2)** $12 \times 10^3\ \mathrm{J}$
(3) $24 \times 10^3\ \mathrm{J}$ **(4)** $48 \times 10^3\ \mathrm{J}$

Magnetic field of a finite straight wire 1 Q

52. NEET 2023 · Q44 A very long conducting wire is bent in a semi-circular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by:

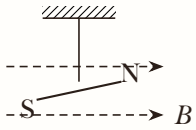


DELETED PORTION — magnetic field of a finite straight wire · removed from the 2026-27 NCERT; not analysed

(1) $\dfrac{\mu_0 i}{4R}\left[1-\dfrac{2}{\pi}\right]$ pointed into the page
(2) $\dfrac{\mu_0 i}{4R}$ pointed into the page
(3) $\dfrac{\mu_0 i}{4R}$ pointed away from the page
(4) $\dfrac{\mu_0 i}{4R}\left[1-\dfrac{2}{\pi}\right]$ pointed away from page

Oscillation period of a magnetic needle 1 Q

53. NEET 2024 · Q3 In a uniform magnetic field of 0.049 T, a magnetic needle performs 20 complete oscillations in 5 seconds as shown. The moment of inertia of the needle is $9.8 \times 10^{-6}\ \mathrm{kg\,m^2}$. If the magnitude of magnetic moment of the needle is $x \times 10^{-5}\ \mathrm{A\,m^2}$, then the value of x is:



DELETED PORTION — oscillation period of a magnetic needle · removed from the 2026-27 NCERT; not analysed

(1) $1280\pi^2$ **(2)** $5\pi^2$
(3) $128\pi^2$ **(4)** $50\pi^2$

Rainbow 1 Q

54. NEET 2019 · Q15 Pick the **wrong** answer in the context with rainbow.

DELETED PORTION — rainbow · removed from the 2026-27 NCERT; not analysed

(1) When the light rays undergo two internal reflections in a water drop, a secondary rainbow is formed.
(2) The order of colours is reversed in the secondary rainbow.
(3) An observer can see a rainbow when his front is towards the sun.
(4) Rainbow is a combined effect of dispersion, refraction and reflection of sunlight.

Sharpness of resonance (Q-factor) 1 Q

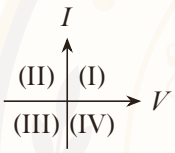
55. **NEET 2021 · Q39** A series LCR circuit containing 5.0 H inductor, 80 μF capacitor and 40 Ω resistor is connected to 230 V variable frequency ac source. The angular frequencies of the source at which power transferred to the circuit is half the power at the resonant angular frequency are likely to be:

DELETED PORTION — sharpness of resonance (half-power frequencies) · removed from the 2026-27 NCERT; not analysed

(1) 25 rad/s and 75 rad/s
(2) 50 rad/s and 25 rad/s
(3) 46 rad/s and 54 rad/s
(4) 42 rad/s and 58 rad/s

Solar cell 1 Q

56. **NEET 2024 · Q5** Consider the following statements A and B and identify the correct answer:



A. For a solar-cell, the I-V characteristics lies in the IV quadrant of the given graph.
B. In a reverse biased pn junction diode, the current measured in (μA), is due to majority charge carriers.

DELETED PORTION — solar cell (special purpose diodes) · removed from the 2026-27 NCERT; not analysed

(1) Both A and B are incorrect.
(2) A is correct but B is incorrect.
(3) A is incorrect but B is correct.
(4) Both A and B are correct.

Types of errors 1 Q

57. **NEET 2023 · Q23** The errors in the measurement which arise due to unpredictable fluctuations in temperature and voltage supply are:

DELETED PORTION — types of errors · removed from the 2026-27 NCERT; not analysed

(1) Random errors
(2) Instrumental errors
(3) Personal errors
(4) Least count errors

Velocity selector 1 Q

58. **NEET 2025 · Q3** An electron (mass 9×10^{-31} kg and charge 1.6×10^{-19} C) moving with speed $c/100$ (c = speed of light) is injected into a magnetic field $\vec{B}$ of magnitude 9×10^{-4} T perpendicular to its direction of motion. We wish to apply a uniform electric field $\vec{E}$ together with the magnetic field so that the electron does not deflect from its path. Then (speed of light $c = 3\times10^{8}\ \mathrm{m\,s^{-1}}$)

DELETED PORTION — velocity selector (crossed E and B) · removed from the 2026-27 NCERT; not analysed

(1) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is $27\times10^{4}\ \mathrm{V\,m^{-1}}$
(2) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is $27\times10^{2}\ \mathrm{V\,m^{-1}}$
(3) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is $27\times10^{2}\ \mathrm{V\,m^{-1}}$
(4) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is $27\times10^{4}\ \mathrm{V\,m^{-1}}$

Deleted Portions · Answer Key

1 (4)	**2** (1)	**3** (4)	**4** (3)	**5** (4)	**6** (3)	**7** (1)	**8** (3)
9 (2)	**10** (2)	**11** (4)	**12** (1)	**13** (1)	**14** (3)	**15** (2)	**16** (1)
17 (2)	**18** (3)	**19** (1)*	**20** (4)	**21** (4)	**22** (2)	**23** (1)	**24** (1)
25 (4)	**26** (3)	**27** (2)	**28** (3)	**29** (2)	**30** (1)	**31** (2)	**32** (4)
33 (4)	**34** (4)	**35** (2)	**36** (1)	**37** (2)	**38** (2)	**39** (2)	**40** (2)
41 (3)	**42** (3)	**43** (3)	**44** (3)	**45** (4)	**46** (1)	**47** (2)	**48** (2)
49 (4)	**50** (3)	**51** (3)	**52** (1)	**53** (1)	**54** (3)	**55** (3)	**56** (2)
57 (1)	**58** (2)						

Answers of 2021, 2022, 2023, 2025 and 2026 are from the official NTA final keys (checked by the editor); answers of 2016–2020 and 2024 were solved by the editor. *: see note.

19. (NEET 2017 Q141) $N_B/N_A = e^{7\lambda t}$, so the ratio is e (not $1/e$) at $t = 1/(7\lambda)$; the intended answer is (1).

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	Total
Deleted-portion questions	6	7	8	5	6	6	2	6	6	4	2	**58**

Chapter-wise Weightage, 2016–2026

Number of in-syllabus questions from each 2026-27 NCERT chapter in every paper. Dir %: share of the chapter's questions that are direct from the NCERT textbook.

Ch	Chapter	16	17	18	19	20	21	22	23	24	25	26	Total	Dir
XI.1	Units and Measurements	0	1	1	1	2	3	3	1	3	3	3	**21**	29%
XI.2	Motion in a Straight Line	1	0	1	1	1	1	2	3	1	1	1	**13**	31%
XI.3	Motion in a Plane	3	2	0	2	0	2	1	1	1	1	0	**13**	23%
XI.4	Laws of Motion	1	2	2	2	1	1	1	2	2	2	2	**18**	39%
XI.5	Work, Energy and Power	2	1	1	3	0	2	2	1	2	2	2	**18**	44%
XI.6	System of Particles and Rotational Motion	1	3	3	1	2	2	3	2	2	2	1	**22**	45%
XI.7	Gravitation	2	2	2	2	1	2	1	2	2	2	2	**20**	45%
XI.8	Mechanical Properties of Solids	0	1	1	1	2	0	1	1	1	0	0	**8**	62%
XI.9	Mechanical Properties of Fluids	1	1	1	2	1	1	2	2	1	1	2	**15**	20%
XI.10	Thermal Properties of Matter	3	2	1	2	1	1	0	0	1	1	1	**13**	31%
XI.11	Thermodynamics	2	2	3	1	1	0	1	1	1	1	2	**15**	40%
XI.12	Kinetic Theory	1	1	1	1	3	1	1	1	1	2	2	**15**	40%
XI.13	Oscillations	0	2	1	3	1	2	1	1	2	2	2	**17**	53%
XI.14	Waves	2	1	2	0	1	0	1	1	0	1	1	**10**	40%
XII.1	Electric Charges and Fields	1	1	1	3	1	2	1	2	0	1	0	**13**	54%
XII.2	Electrostatic Potential and Capacitance	1	2	0	0	4	4	3	2	4	2	4	**26**	42%
XII.3	Current Electricity	1	1	2	3	4	3	4	4	4	3	1	**30**	23%
XII.4	Moving Charges and Magnetism	2	1	2	2	1	4	4	1	1	2	2	**22**	55%
XII.5	Magnetism and Matter	1	1	1	0	1	0	0	1	2	0	0	**7**	57%
XII.6	Electromagnetic Induction	1	2	1	1	0	1	2	1	2	1	3	**15**	47%
XII.7	Alternating Current	2	0	1	0	2	2	2	4	2	1	1	**17**	47%
XII.8	Electromagnetic Waves	1	1	1	2	1	2	2	1	2	2	2	**17**	59%
XII.9	Ray Optics and Optical Instruments	3	2	3	2	1	4	3	3	2	2	2	**27**	30%
XII.10	Wave Optics	2	2	1	1	1	0	1	1	1	1	0	**11**	9%
XII.11	Dual Nature of Radiation and Matter	2	2	2	1	2	2	2	2	2	2	3	**22**	32%
XII.12	Atoms	2	1	1	1	1	0	1	2	2	2	2	**15**	47%
XII.13	Nuclei	0	0	0	1	2	1	1	0	0	0	1	**6**	100%
XII.14	Semiconductor Electronics	1	1	1	1	1	1	2	1	0	1	1	**11**	36%
	Class XI total												**218**	
	Class XII total												**239**	

Class XII gives 52% of the in-syllabus questions and Class XI 48%. The six heaviest chapters, Current Electricity (30), Ray Optics and Optical Instruments (27), Electrostatic Potential and Capacitance (26), System of Particles and Rotational Motion (22), Moving Charges and Magnetism (22), Dual Nature of Radiation and Matter (22), together give 149 questions, 33% of the total. The pages that follow give the year-wise summary, the most repeated topics and the preparation plan.

NEET 2016–2026 · Summary

Every question of the eleven papers was checked one by one against the four 2026-27 NCERT Physics books (Class XI Part 1 & 2, Class XII Part 1 & 2). Direct: the answer is stated in the textbook (text, summary, table or figure), or the question is an NCERT solved Example or Exercise (same or changed numbers), or it needs only one printed formula. Applied: the concept is in the textbook but the question needs several steps or a new situation. Deleted portion: the topic was removed from the rationalised syllabus; such questions are not analysed. Percentages are of the analysed questions.

Year	Direct				Applied	Analysed	Deleted	Direct
	Total	Text	Example	Exercise			portion	%
2026	**16**	15	0	1	27	43	2	37%
2025	**11**	9	2	0	30	41	4	27%
2024	**21**	18	2	1	23	44	6	48%
2023	**23**	19	2	2	21	44	6	52%
2022	**29**	22	5	2	19	48	2	60%
2021	**14**	9	5	0	30	44	6	32%
2020	**26**	21	3	2	13	39	6	67%
2019	**20**	17	1	2	20	40	5	50%
2018	**11**	8	2	1	26	37	8	30%
2017	**5**	2	1	2	33	38	7	13%
2016	**7**	5	2	0	32	39	6	18%
Total	**183**	145	25	13	274	457	**58**	40%

Of the 515 questions, **457** are within the present syllabus. Of these, **183** (40%) are direct from the NCERT textbook and 274 (60%) are applications of NCERT concepts. **58** questions (11% of all 515) are from deleted portions; 38 of them are in the papers of 2016–2021, set before the rationalised syllabus, and only 6 in 2025 and 2026.

Deleted portions asked, 2016–2026 (number of questions):

Digital electronics (logic gates) (11);
Brewster's law (4);
Radioactive decay law (half-life, mean life, activity) (4);
Transistor (incl. amplifier) (4);
Alpha- and beta-decay (3);
Parallel axis theorem (3);
Potentiometer (3);
Resistor colour code (3);
Rolling motion (3);
Zener diode (3);
Doppler effect (2);
Earth's magnetism (incl. angle of dip) (2);
Resolving power (telescope, microscope) (2);
Coefficient of restitution (1);
Eddy currents (1);
Force between capacitor plates (1);
Intensity of EM waves (1);
Magnetic field of a finite straight wire (1);
Oscillation period of a magnetic needle (1);
Rainbow (1);
Sharpness of resonance (Q-factor) (1);
Solar cell (1);
Types of errors (1);
Velocity selector (1).

The 15 Most Repeated Topics

Every in-syllabus question of 2016–2026 was given one topic label while the answer keys were solved. The table counts the questions on each topic over the eleven papers (deleted portions left out; ties are ordered by the number of years in which the topic appeared). Years in bold are the years the topic was asked.

#	Topic	NCERT chapter	Qs	Years	2016 → 2026
1	Simple harmonic motion	Oscillations (XI)	**17**	10	16 **17 18 19 20 21 22 23 24 25 26**
2	Kinetic theory of gases	Kinetic Theory (XI)	**15**	11	**16 17 18 19 20 21 22 23 24 25 26**
3	Rotational dynamics (torque, moment of inertia)	Rotational Motion (XI)	**15**	10	**16 17 18 19 20 21 22 23 24 25** 26
4	Kinematics in one dimension	Motion in a Straight Line (XI)	**13**	10	**16** 17 **18 19 20 21 22 23 24 25 26**
5	Bohr model of hydrogen atom	Atoms (XII)	**12**	10	**16 17 18 19 20** 21 **22 23 24 25 26**
6	Measurement and errors	Units & Measurement (XI)	**12**	9	16 17 **18 19 20 21 22 23 24 25 26**
7	AC circuits (reactance, LCR, phase)	Alternating Current (XII)	**12**	8	**16** 17 18 19 **20 21 22 23 24 25 26**
8	Circular motion	Laws of Motion (XI)	**12**	7	**16 17 18 19** 20 21 22 23 **24 25 26**
9	Work-energy theorem and power	Work, Energy & Power (XI)	**11**	9	**16 17** 18 **19** 20 **21 22 23 24 25 26**
10	Electromagnetic waves (nature, E/B = c)	Electromagnetic Waves (XII)	**10**	10	**16 17 18** 19 **20 21 22 23 24 25 26**
11	Dimensional analysis	Units & Measurement (XI)	**10**	6	16 **17** 18 19 20 **21 22** 23 **24 25 26**
12	de Broglie waves	Dual Nature (XII)	**9**	9	**16 17 18 19 20 21 22** 23 **24 25** 26
13	Photoelectric effect	Dual Nature (XII)	**9**	8	**16 17 18** 19 **20** 21 **22 23** 24 **25 26**
14	Thermodynamic processes	Thermodynamics (XI)	**9**	8	**16 17 18 19 20** 21 **22** 23 **24** 25 **26**
15	Young's double-slit interference	Wave Optics (XII)	**9**	8	**16 17 18 19 20** 21 **22 23 24** 25 26

These 15 topics carry **175** of the 457 in-syllabus questions (38%), about 16 questions in every paper. Nine of them are Class XI topics and 6 are Class XII topics: NEET weighs the two years almost equally.

Questions asked again, almost word for word (in the present syllabus)

2016 Q26, 2019 Q45, 2021 Q26, 2022 Q42	field inside and outside a thick wire (Example 4.7)
2019 Q16, 2020 Q143, 2025 Q32	weight of a body inside or above the earth (Exercises 7.15, 7.16)
2016 Q38, 2017 Q168, 2023 Q11	series limits of the hydrogen spectrum
2021 Q24 ≈ 2022 Q13	Galileo's law of odd numbers (Example 2.5)
2018 Q10 ≈ 2021 Q18	telescope objective: large focal length and large aperture
2018 Q11 ≈ 2019 Q27	kinetic and total energy in a Bohr orbit ($K = -E$)
2020 Q148 ≈ 2022 Q14	centre of mass of two masses on a light rod
2020 Q154 ≈ 2024 Q38	current through a capacitor on 50 Hz ac (Example 7.4)
2020 Q155 ≈ 2026 Q2	mean free path $l = 1/(\sqrt{2}\, n\pi d^2)$

How to Prepare for NEET Physics

Every point below comes from the question-by-question analysis of the eleven papers in this book.

1. What the papers show

- **40%** of the in-syllabus questions (183 of 457) are direct from the NCERT textbook; the rest (274) apply NCERT concepts.
- 25 questions are NCERT solved Examples and 13 are NCERT Exercises, often with the same numbers.
- Class XII gives **52%** of the questions, Class XI 48%.
- Heaviest chapters (questions in 11 years): Current Electricity (30), Ray Optics (27), Electrostatic Potential & Capacitance (26), Dual Nature (22), Moving Charges & Magnetism (22), Rotational Motion (22).
- Difficulty: 230 easy, 212 moderate, only 15 hard. NEET rewards clear concepts and careful arithmetic more than tricks.
- 58 old questions are on deleted topics, but only 6 in 2025 and 2026. The rationalised NCERT is the syllabus now.
- Questions do come back (see the previous page), so the old papers are worth solving fully.

2. What to study from

1. **NCERT Physics 2026-27** (XI Parts 1–2, XII Parts 1–2): the main book. Read every line, including the boxes, tables, figures, Summary and Points to Ponder.
2. **NCERT solved Examples and Exercises**: work all of them by hand. They are the single biggest source of numerical questions.
3. **NCERT Exemplar Physics**: for the applied questions; the MCQ and short-answer sections are enough.
4. **NCERT Lab Manual**: NEET 2020, 2021, 2025 and 2026 asked screw-gauge, vernier and optical-bench questions from it.
5. **NEET papers 2016–2026** (this book), solved under time, with the answer keys.
6. One standard reference for extra MCQ practice only after NCERT is complete. Do not start a new book in the last month.

3. How to work

- Make a one-page formula and fact sheet for each chapter while reading NCERT. Revise from these sheets.
- Keep an error notebook: every wrong or guessed question, the reason, and the right idea. Read it every week.
- Practise the graph, circuit and ray-diagram questions; about one question in 5 has a figure (104 of 515).
- Learn units, dimensions, standard values (e, h, c, m_e, 13.6 eV, 1 u = 931.5 MeV) and the ratio and proportion short cuts; they save time in nearly every paper.
- Check every old question against the present syllabus; skip the ones marked Deleted Portion.

4. Plan of action

Phase	Work
1. Foundation	NCERT chapter by chapter, Class XII (about half the marks): text, Examples, Exercises, formula sheet.
2. Class XI	The same for Class XI (the other half), with more time on Units and Measurement, Rotational Motion, Thermodynamics, Kinetic Theory and Oscillations.
3. Topic drills	Solve this book topic by topic; start with the 15 most repeated topics. Then Exemplar MCQs on the same topics.
4. Full papers	One complete NEET Physics section every two days: 45 questions in 50 minutes. Check with the key and update the error notebook.
5. Last 2 weeks	Only revision: formula sheets, NCERT Summaries, error notebook, and a paper on alternate days. No new material.

5. In the examination hall

- NEET gives about one minute per question over all three subjects; aim to finish Physics in 50 minutes. First pass: answer every direct NCERT question at once (about a third of the paper).
- Second pass: the numericals; leave any question that takes more than two minutes and come back.
- Each correct answer gives 4 marks and each wrong answer costs 1. Answer a doubtful question only when two options can be ruled out.
- Fill the OMR as you go, not at the end.

6. A weekly rhythm

Five days of NCERT and practice, one day of a timed paper, one day of revision and the error notebook. A steady 2–3 hours of Physics a day, kept up for months, beats long sessions before the exam.